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# DEPARTMENT OF DEFENSE HANDBOOK

## RESISTORS, SELECTION AND USE OF

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# FOREWORD

1. This handbook is approved for use by all Departments and Agencies of the Department of Defense.

2. This handbook provides selected standard resistors for use in the design of Department of Defense equipment. This handbook is for guidance only. This handbook cannot be cited as a requirement. If it is, the contractor does not have to comply.

    a. The application information and performance characteristics contained in this handbook are offered for guidance and are not to be considered as mandatory. Additional application information will be added when coordinated with the Department of Defense.
    b. Additional resistor types of this handbook will be developed as standard resistors of a given specification family are selected and coordinated with the Department of Defense.

3. Comments, suggestions, or questions on this document should be addressed to: DLA Land and Maritime, ATTN: VAT, Post Office Box 3990, Columbus, Ohio 43218-3990 or by email Resistor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.



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## 1. SCOPE

1.1 Scope. This handbook is for guidance only. This handbook cannot be cited as a requirement. If it is, the contractor does not have to comply. This handbook consists of the following:

a. Selected standard resistor types, for use in the design and manufacturer of Department of Defense equipment under the jurisdiction of the Department of Defense.
b. Guides for the choice and application of resistors for use in Department of Defense equipment.

Requirements for resistors listed in this handbook are covered in the applicable specification (see 2.1). When it has been determined that circuit requirements cannot be met by using resistor styles or characteristics listed in the applicable specifications, the design engineer should, with the approval of the cognizant activity, select from the applicable resistor specification styles or characteristics not listed herein.

1.2 Purpose of handbook.

a. To provide the equipment designer with a selection of standard resistors for use in most Department of Defense applications.
b. To control and minimize the variety of resistors used in Department of Defense equipment in order to facilitate logistic support of equipment in the field.
c. To outline criteria pertaining to the use, choice, and application of resistors in Department of Defense equipment.

1.3 DLA Land and Maritime SpecFinder. DLA Land and Maritime SpecFinder is an online tool that can be used in conjunction with this handbook. The SpecFinder is a search tool to help you find standardization documents by selecting device characteristics. If you require certain performance features but don't know which documents might meet your needs, the Specification Finder will help you narrow your search down to a few candidate documents. The tool can be found at: https://landandmaritimeapps.dla.mil/Programs/SpecFind/default.aspx.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation.

DEPARTMENT OF DEFENSE SPECIFICATIONS

| | | |
|---|---|---|
| MIL-PRF-19 | - | Resistor, Variable, Wirewound, Low operating Temperature, General Specification For. (Inactive for New Design) |
| MIL-PRF-22 | - | Resistor, Variable, Wirewound, Power Type, General Specification For. |
| MIL-PRF-26 | - | Resistor, Fixed, Wirewound, Power Type, General Specification For. |
| MIL-PRF-29 | - | Resistor, Fixed, Meter Multiplier, External, High Voltage, Ferrule Terminal Type, General Specification For |
| MIL-PRF-93 | - | Resistor, Fixed, Wirewound, Accurate, General Specification For (Inactive for New Design) |



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DEPARTMENT OF DEFENSE SPECIFICATIONS (continued)

| | | |
|---|---|---|
| MIL-PRF-94 | - | Resistor, Variable, Composition, General Specification For |
| MIL-PRF-914 | - | Resistor Networks, Fixed, Film, Surface Mount, Nonestablished Reliability, and Established Reliability, General Specification For |
| MIL-PRF-12934 | - | Resistor, Variable, Precision, General Specification For |
| MIL-PRF-18546 | - | Resistor, Fixed, Wirewound, Power Type, Chassis Mounted, General Specification For. |
| MIL-PRF-22097 | - | Resistor, Variable, Nonwirewound, Adjustment Type, General Specification For |
| MIL-PRF-22684 | - | Resistor, Fixed, Film, Insulated, General Specification For |
| MIL-PRF-23648 | - | Resistor, Thermal (Thermistor) Insulated, General Specification For |
| MIL-PRF-27208 | - | Resistor, Variable, Wirewound, Nonprecision, General Specification For |
| MIL-PRF-32159 | - | Resistor, Chip, Fixed, Film, Zero Ohm, Industrial, High Reliability, Space Level, General Specification For |
| MIL-PRF-32192 | - | Resistor, Chip, Thermal (Thermistor), General Specification For |
| MIL-PRF-32663 | - | Resistor, Chip, Fixed, Bulk Metal Foil, Established Reliability, Space Level |
| MIL-PRF-39005 | - | Resistor, Fixed, Wirewound, Accurate, Nonestablished Reliability, Established Reliability, General Specification For |
| MIL-PRF-39007 | - | Resistor, Fixed, Wirewound, Power Type, Nonestablished Reliability, Established Reliability, and Space Level, General Specification For |
| MIL-PRF-39009 | - | Resistor, Fixed, Wirewound, Power Type, Chassis Mounted, Nonestablished Reliability, and Established Reliability, General Specification For |
| MIL-PRF-39015 | - | Resistor, Variable, Wirewound, Nonestablished Reliability, and Established Reliability, General Specification For |
| MIL-PRF-39017 | - | Resistor, Fixed, Film, Insulated, Nonestablished Reliability, and Established Reliability, General Specification For |
| MIL-PRF-39023 | - | Resistor, Variable, Nonwirewound, Precision, General Specification For |
| MIL-PRF-39035 | - | Resistor, Variable, Nonwirewound, Adjustment Type, Nonestablished Reliability, and Established Reliability, General Specification For |
| MIL-PRF-49462 | - | Resistor, Fixed, Film, High Voltage, General Specification For |
| MIL-PRF-49465 | - | Resistor, Fixed, Metal Element, Power Type, Very Low Resistance Values, General Specification For |
| MIL-PRF-55182 | - | Resistor, Fixed, Film, Nonestablished Reliability, Established Reliability, and Space Level, General Specification For |
| MIL-PRF-55342 | - | Resistor, Chip, Fixed, Film, Nonestablished Reliability, Established Reliability, Space Level, General Specification For |
| MIL-PRF-83401 | - | Resistor Network, Fixed, Film, and Capacitor-Resistor Networks, Ceramic Capacitor and Fixed Film Resistor, General Specification For |



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DEPARTMENT OF DEFENSE SPECIFICATIONS (continued)

| | | |
|---|---|---|
| MIL-PRF-83530 | - | Resistor, Voltage Sensitive, Varistor, Metal-Oxide, General Specification For (Inactive for New Design) |
| MIL-R-10509 | - | Resistor, Fixed, Film, High Stability, General Specification For (Inactive for New Design) |

(Copies of these documents are available online at https://quicksearch.dla.mil).

2.3 Order of precedence Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence unless otherwise noted. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Rating and design application terms. A list of common terms used in rating and design application of resistors is as follows:

a. Ambient operating temperature. The temperature of the air surrounding an object, neglecting small localized variations.

b. Contact resistance variation. The apparent resistance seen between the wiper and the resistance element when the wiper is energized with a specified current and moved over the adjustment travel in either direction at a constant speed. The output variations are measured over a specified frequency bandwidth, exclusive of the effects due to roll-on or roll-off of the terminations and is expressed in ohms or percent of total nominal resistance.

c. Critical value of resistance. For a given voltage rating and a given power rating, there is only one value of resistance that will dissipate full rated power at rated voltage. This value of resistance is commonly referred to as the “critical value of resistance.” For values of resistance below the critical value, the maximum (element) voltage is never reached and, for values of resistance above critical value, the power dissipated becomes lower than rated. Figure 1 shows this relationship.

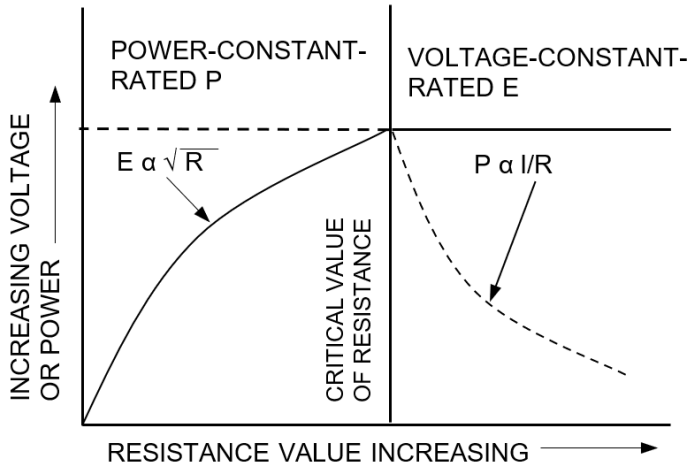



FIGURE 1. Maximum working voltage and critical value of resistance.



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d. Resistance value deviations. From its nominal value; it is the percent change from normal resistor tolerance + percent change from TCR + percent change from self-heating.

e. Dielectric strength. The ultimate breakdown voltage of the dielectric or insulation of the resistor when the voltage is applied between the case and all terminals tied together. Dielectric strength is usually specified at sea level and simulated high altitude air pressures.

f. Hot-spot temperature. As defined in Department of Defense specifications, the maximum temperature measured on the resistor due to both internal heating and the ambient operating temperature. Maximum hot-spot temperature is predicated on thermal limits of the materials and the design. The hot-spot temperature is also usually established as the top temperature on the derating curve at which the resistor is derated to zero power.

g. Insulation resistance. The dc resistance measured between all terminals connected together and the case, exterior insulation, or external hardware.

h. Maximum (element) working voltage($E = \sqrt{PR}$). The maximum voltage stress (dc or rms) that may be applied to the resistor (resistance element) is a function of (1) the materials used, (2) the required performance, and (3) the physical dimensions. (See figure 1)

i. Noise. An unwanted voltage fluctuation generated within the resistor. Total noise of a resistor always includes Johnson noise 1/ which is dependent only on the resistance value and temperature of the resistance element. Depending on the type of element and construction, total noise may also include noise caused by current flow, and noise caused by cracked bodies and loose end caps or leads. For variable resistors, noise may also be caused by jumping of contact over turns of wire and by an imperfect electrical path between the contact and resistance element.

j. Resistance temperature characteristic (temperature coefficient). The magnitude of change in resistance due to temperature, usually expressed in percent per degree Celsius or parts per million per degree Celsius (ppm/$^{O}$C). If the changes are linear over the operating temperature range, the parameter is known as “temperature coefficient”.

k. Resistance tolerance. The permissible deviation of the manufactured resistance value (expressed in percent) from the specified nominal resistance value at standard (or stated) environmental conditions.

l. Stability. The overall ability of a resistor to maintain its initial resistance value over extended periods of time when subjected to any combination of environmental conditions and electrical stresses.

## 4. GENERAL REQUIREMENTS

4.1 Choice of resistor types. The variety of resistor types used in any particular equipment should be the minimum necessary to obtain satisfactory performance. Where more than one type of resistor may be used in a given application (such as, fixed, film, insulated versus fixed, film, insulated (high stability)), consideration should be given to cost and availability (use of strategic materials, multiple sources). The resistors identified in this handbook meet all the criteria for standard types (see 1.1 and 4.4).

4.1.1 Reliability. Where quantitative reliability requirements specified as part of the equipment requirements are such that the use of parts with established reliability is dictated, such parts should be selected from the established reliability specification.

4.1.2 Qualified sources. After a preliminary selection of the desired resistor has been made, reference should be made to the applicable qualified products list for listing of qualified sources.

4.2 Item identification. A type designation for any resistor referenced herein may be constructed as indicated in the example given in the applicable section. The Part Identification Number (PIN) designations are depicted in the applicable specification.

4.3 Conflict of requirements. This handbook provides selected standard resistors for use in the design of Department of Defense equipment. This handbook is for guidance only. This handbook cannot be sited as a requirement.

1/ Johnson, J. B., “Thermal Agitation of Electricity in Conductors,” Physical Review, volume 32 (July, 1928, 97-109).



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4.4 Criteria for inclusion in this handbook. The criteria for the inclusion of resistor types in this handbook are as follows:

a. The resistor should be the best type available for general use in military equipment.
b. Coordinated Department of Defense specifications should be available (see 2.1).
c. Resistors should be in production, or should have been in production.

5. DETAILED REQUIREMENTS

5.1 Detailed requirements. The detailed requirements for standard resistor types are contained in the applicable specification of this handbook.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. General application notes are as indicated in the appendix.

6.2 Subject term (key word) listing.

| | |
|---|---|
| Adjustable | Space level |
| Chip | Surface mount |
| Established reliability | Thermistor |
| Film | Trimmer |
| Fixed | Variable |
| Lead-screw | Varistor |
| Network | Wirewound |
| Nonestablished reliability | |
| Nonwirewound | |

6.3 Tin whisker growth. The use of alloys with tin content greater than 97 percent, by mass, may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions, on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead, by mass, have shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM-B545 (Standard Specification for Electrodeposited Coatings of Tin).

6.4 Change from previous issue. The margins of this handbook are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations.



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# APPENDIX A

# GENERAL APPLICATION INFORMATION

## A.1. SCOPE

A.1.1 Scope. The application information in this handbook is designed to help in the selection of specified resistors (application information pertaining to specific resistors types is contained in the applicable sections). As with other types of components, the most important thing a user must decide is which of the numerous types of resistors will be best for use in the military equipment being designed. Proper selection in its broadest sense is the first step in building reliable equipment. To properly select the resistors to be used, the user must know as much as possible about the types from which to choose. The advantages and disadvantages should be known, as well as their behavior under various environmental conditions, their construction, and their effect on circuits and the effect of circuits on them, and a knowledge of what makes resistors fail. This appendix is not a mandatory part of the handbook. The information contained herein is intended for guidance only.

A.1.1.1 Resistor types. All variable and fixed resistors, of the types widely used in electronic equipment, can be grouped into one of three basic types. They are “composition” types, “film” types, or “wirewound” types. As the name indicates, the “composition” type is made of a mixture of resistive material and a binder which are molded into the proper shape and resistance value. The “film” type is composed of a resistive film deposited on, or inside of, an insulating cylinder or filament. The “wirewound” type is made up of resistance wire, wound on an insulated form. These basic types differ from each other in size, cost, resistance range, power rating, and general characteristics. Some are better than others for particular purposes; no one type has all of the best characteristics. The choice among them, therefore, depends on the requirements, both initial and long-term; the environment in which they must exist; and numerous other factors which the designer must understand. The designer must realize that the summaries of the requirements of a particular application must be taken into consideration and compared with the advantages and drawbacks of each of the several types, before a final choice is made. Tables A-I, A-II, A-III, A-IV, A-V and A-VI provide a selection guide for fixed and variable resistors included in this handbook.

The Department of Defense resistor specification categories are shown in figure A-1.

A.1.1.2 Resistance Tolerances. Resistance tolerances will be coded as follows:

| Code | Tolerance | Code | Tolerance | Code | Tolerance |
|---|---|---|---|---|---|
| V | 0.0005 | B | 0.1 | H | 3.0 |
| T | 0.01 | D | 0.5 | J | 5.0 |
| Q | 0.02 | F | 1.0 | K | 10.0 |
| A | 0.05 | G | 2.0 | | |

## A.2. APPLICABLE DOCUMENTS.

A.2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

A.2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B545 - Standard Specification for Electrodeposited Coatings of Tin

(Copies of this document are available from at https://www.astm.org/)



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APPENDIX A

Resistor Fixed

| Film/Foil | | |
|---|---|---|
| Precision | Semi-precision | Special |
| MIL-PRF-32663 (ER)<br>MIL-PRF-55182 (ER) | MIL-PRF-22684<br>MIL-PRF-39017 (ER)<br>MIL-PRF-55342 (ER) | MIL-PRF-32159<br>MIL-PRF-49462<br>MIL-PRF-49465 |

| Wirewound | |
|---|---|
| Accurate | Power |
| MIL-PRF-39005 (ER) | MIL-PRF-26<br>MIL-PRF-18546<br>MIL-PRF-39007 (ER)<br>MIL-PRF-39009 (ER) |

| Network |
|---|
| MIL-PRF-914 (ER)<br>MIL-PRF-83401 |

Resistor Variable

| General purpose. | |
|---|---|
| Wirewound | Nonwirewound |
| MIL-PRF-22 | MIL-PRF-94<br>(COMPOSITION) |

| Precision | |
|---|---|
| Wirewound | Nonwirewound |
| MIL-PRF-12934 | MIL-PRF-39023 |

| Semiprecision |
|---|
| Wirewound |
| N/A |

| Adjustable Trimmer | |
|---|---|
| Wirewound | Nonwirewound |
| MIL-PRF-27208<br>MIL-PRF-39015 (ER) | MIL-PRF-22097<br>MIL-PRF-39035(ER) |

Resistor - Special

| Thermal | Fixed | Varistor |
|---|---|---|
| MIL-PRF-23648<br>MIL-PRF-32192 | MIL-PRF-29 | N/A |

FIGURE A-1. Department of Defense resistor specification categories.



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APPENDIX A

A.3. GENERAL CHARACTERISTICS OF RESISTORS

A.3.1 General characteristics of fixed resistors.

A.3.1.1 Fixed, Film Resistors. See table A-I.

A.3.1.1.1 MIL-PRF-22684, RL42...TX, Resistor, Fixed, Film, Insulated. These fixed, film, resistors have semi-precision characteristics and small sizes. The sizes and wattage rating are comparable to those of MIL-R-39008 (cancelled) and stability is between MIL-R-39008 and MIL-PRF-55182. Design parameter tolerances are looser than those of MIL-PRF-55182 but good stability makes them desirable in most electronic circuits. These resistors are capable of full load operation at an ambient temperature of +70$^{O}$C and have a resistance-temperature characteristic of ±200 parts per million per degree Celsius (ppm/$^{O}$C). See MIL-PRF-39017, RLR, Resistor, Fixed, Film, Insulated, Nonestablished Reliability and Established Reliability.

NOTE: MIL-PRF-22684 designers are cautioned in using these resistors in pulse application. The usage requirements for the resistor must be evaluated for each application. See Appendix E.

A.3.1.1.2 MIL-PRF-32159, RCZ, Resistor, Chip, Fixed, Film, Zero Ohm, Industrial, High Reliability, Space Level. These fixed, film, chip resistors are primarily intended for incorporation into surface mount applications. This specification has three product levels, a space level “T” part number with 100 percent burn-in and screening, a high reliability 100 percent burn-in screened “M” part number level, and a part level “C” using the manufacturer’s inspection system to validate conformance.

NOTE:
1. FR level is in percent per 1000 hours.
2. MIL-PRF-32159 designers are cautioned in using these resistors in pulse application. The usage requirements for the resistor must be evaluated for each application. See Appendix E.

A.3.1.1.3 MIL-PRF-32663, RFB or RFG Resistor, Chip, Fixed, Bulk Metal Foil, Established Reliability, Space Level. These fixed, bulk metal foil (BMF), ultra-precision chip resistors are primarily intended for incorporation into surface mount applications. These resistors have a high degree of stability with respect to time under normal conditions. ER resistors covered by this specification have life failure rates (FR) ranging from 1 percent to 0.001 percent per 1,000 hours.

NOTE:
1. FR level is in percent per 1000 hours.
2. MIL-PRF-32663 designers are cautioned in using these resistors in high power pulse application. The usage requirements for the resistor must be evaluated for each application. See Appendix E.

A.3.1.1.4 MIL-PRF-39017, RLR, Resistor, Fixed, Film, Insulated, Nonestablished Reliability and Established Reliability. These fixed, film, resistors have semi-precision characteristics and small sizes. The sizes and wattage rating are comparable to those of MIL-R-39008 (cancelled) and stability is between MIL-R-39008 and MIL-PRF-55182. Design parameter tolerances are looser than those of MIL-PRF-55182 but good stability makes them desirable in most electronic circuits. These resistors are capable of full load operation at an ambient temperature of +70$^{O}$C and have a resistance-temperature characteristic of ±100 ppm/$^{O}$C and ±350 ppm/$^{O}$C. The resistors have product levels ranging from Non-ER, and a life failure rate (FR) 1.0 percent to 0.001 percent per 1,000 hours. The FR levels are established at a 60 percent confidence level on basis of life tests.
Replaces MIL-PRF-22684 RL42...TX, Resistor, Fixed, Film, Insulated inactive specification sheets.

NOTE:
1. FR level is in percent per 1000 hours.
2. MIL-PRF-39017 designers are cautioned in using these resistors in pulse application. The usage requirements for the resistor must be evaluated for each application. See Appendix E.



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A.3.1.1.5 MIL-PRF-49462, RHV, Resistor, Fixed, Film, High Voltage. These resistors are intended for use in electronic circuits where high voltages and high resistance values are used.

NOTE: MIL-PRF-49462 designers are cautioned in using these resistors in pulse application. The usage requirements for the resistor must be evaluated for each application. See Appendix E.

A.3.1.1.6 MIL-PRF-49465, RLV, Resistor, Fixed, Metal Element, Power Type, Very Low Resistance Value. These are power type, very low resistance values (1 ohm and below), fixed resistors (2 terminal and 4 terminal) for use in electrical, electronic, communications, and associated equipment. Included are precision resistors of 1, 3, and 5 percent initial resistance tolerances with power ratings ranging from 3 watts to 5 watts at +25$^{O}$C derated to 0 watts at +275$^{O}$C

NOTE: MIL-PRF-49465 designers are cautioned in using these resistors in pulse application. The usage requirements for the resistor must be evaluated for each application. See Appendix E.

A.3.1.1.7 MIL-PRF-55182, RNC, RNN, or RNR, Resistor, Fixed, Film, Nonestablished Reliability, Established Reliability, and Space Level. These fixed, film resistors including both hermetically and nonhermetically sealed types possess a high degree of stability, with respect to time, under severe environmental conditions, with an established reliability. Use in circuits requiring higher stability than provided by composition resistors or film, insulated, resistors, and where ac frequency requirements are critical. Operation is satisfactory from dc to 100 megahertz (MHz). Metal films are characterized by low temperature coefficients and are usable for ambient temperatures of +125$^{O}$C or higher with small degradation. The resistors have product levels ranging from Non-ER (C), a life failure rate (FR) 1.0 (M) percent to 0.001 (S) percent per 1,000 hours and space level (T). The FR levels are established at a 60 percent confidence level on basis of life tests. Replaces MIL-R-10509 RN, Resistor, Fixed, Film (High Stability) inactive specification sheets.

NOTE:
1. FR level is in percent per 1000 hours.
2. MIL-PRF-55182 designers are cautioned in using these resistors in pulse application. The usage requirements for the resistor must be evaluated for each application. See Appendix E.

A.3.1.1.8 MIL-PRF-55342, RM, Resistor, Chip, Fixed, Film, Nonestablished Reliability, Established Reliability, and Space Level. These chip resistors are primarily intended for incorporation into hybrid microelectronic circuits. They are designed for use in critical circuitry where stability, long life, reliable operation, and accuracy are of prime importance. These resistors are uncased, leadless chip devices and possess a high degree of stability with respect to time, under severe environmental conditions. The resistors have product levels ranging from Non-ER (C), a life failure rate (FR) 1.0 (M) percent to 0.001 (S) or (V) and space level (T). The FR levels are established at a 60 percent confidence level on basis of life tests.

NOTE:
1. FR level is in percent per 1000 hours.
2. MIL-PRF-55342 designers are cautioned in using these resistors in pulse application. The usage requirements for the resistor must be evaluated for each application. See Appendix E.

TABLE A-I. Fixed film resistor selection guidance table.

| MIL-PRF-22684 - Resistor, Fixed, Film, Insulated | | | | | | | |
|---|---|---|---|---|---|---|---|
| Specification sheet | Styles available | Power and max voltage ratings | Resistance tolerance (± percent) (see A.1.1.2) | Resistance range (ohms) | Resistance temperature coefficient (ppm/$^{O}$C) | Maximum body size (inches) and Configuration (see figure A-2) | |
| MIL-PRF-22684/8 | RL42…TX | 2 W / 500 V | G and J | 10 to 1.5 M | ±200 | 0.728 x 0.336 | A |



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TABLE A-I. Fixed film resistor selection guidance table. Continued

| MIL-PRF-32159 - Resistor, Chip, Fixed, Film, Zero Ohm, Industrial, High Reliability, Space Level | | | | | | | |
|---|---|---|---|---|---|---|---|
| Specification sheet | Styles available | Power ratings | Termination | Current rating (in amps) | Max resistance value (in ohms) | Maximum body size (inches) and Configuration (see figure A-2) | |
| MIL-PRF-32159/1 | RCZ0502 | 0.050 | B,G,W<br>C,D<br>U,T | 1.30<br>0.91<br>0.58 | 0.030<br>0.060<br>0.150 | 0.061 X 0.030 | |
| MIL-PRF-32159/2 | RCZ0505 | 0.100 | B,G,W<br>C,D<br>U,T | 2.20<br>1.80<br>1.00 | 0.020<br>0.030<br>0.100 | 0.061 X 0.055 | |
| MIL-PRF-32159/3 | RCZ1005 | 0.200 | B,G,W<br>C,D<br>U,T | 2.80<br>2.20<br>1.30 | 0.025<br>0.040<br>0.125 | 0.112 X 0.055 | |
| MIL-PRF-32159/4 | RCZ1505 | 0.150 | B,G,W<br>C,D<br>U,T | 2.10<br>1.70<br>0.93 | 0.035<br>0.050<br>0.175 | 0.162 X 0.055 | |
| MIL-PRF-32159/5 | RCZ2208 | 0.225 | B,G,W<br>C,D<br>U,T | 2.50<br>2.10<br>1.10 | 0.035<br>0.050<br>0.175 | 0.237 X 0.080 | |
| MIL-PRF-32159/6 | RCZ0705 | 0.150 | B,G,W<br>C,D<br>U,T | 2.70<br>2.20<br>1.20 | 0.020<br>0.030<br>0.100 | 0.086 X 0.055 | |
| MIL-PRF-32159/7 | RCZ1206 | 0.225 | B,G,W<br>C,D<br>U,T | 3.20<br>2.50<br>1.40 | 0.025<br>0.040<br>0.125 | 0.134 X 0.068 | T |
| MIL-PRF-32159/8 | RCZ2010 | 0.800 | B,G,W<br>C,D<br>U,T | 5.70<br>4.50<br>2.50 | 0.025<br>0.040<br>0.125 | 0.218 X 0.103 | |
| MIL-PRF-32159/9 | RCZ2512 | 1.000 | B,G,W<br>C,D<br>U,T | 6.30<br>5.00<br>2.80 | 0.025<br>0.040<br>0.125 | 0.268 X 0.129 | |
| MIL-PRF-32159/10 | RCZ1010 | 0.500 | B,G,W<br>C,D<br>U,T | 5.00<br>4.10<br>2.20 | 0.020<br>0.030<br>0.100 | 0.112 X 0.105 | |
| MIL-PRF-32159/11 | RCZ0402 | 0.040 | B,G,W<br>C,D<br>U,T | 1.20<br>0.82<br>0.52 | 0.030<br>0.060<br>0.150 | 0.050 X 0.027 | |
| MIL-PRF-32159/12 | RCZ0603 | 0.070 | B,G,W<br>C,D<br>U,T | 1.50<br>1.10<br>0.68 | 0.030<br>0.060<br>0.150 | 0.070 X 0.037 | |
| MIL-PRF-32159/13 | RCZ0302 | 0.035 | B,G,W<br>C,D<br>U,T | 1.10<br>0.76<br>0.48 | 0.030<br>0.060<br>0.150 | 0.038 X 0.027 | |
| MIL-PRF-32159/14 | RCZ0201 | 0.05 | B,G<br>C,D<br>U,T | 0.50<br>0.35<br>0.22 | 0.005<br>0.060<br>0.150 | 0.022/0.024<br>x 0.012 | |



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TABLE A-I. Fixed film resistor selection guidance table - Continued.

MIL-PRF-32663, Resistor, Chip, Fixed, Bulk Metal Foil, Established Reliability, Space Level

| Specification sheet | Styles available | Power and max voltage ratings | Resistance tolerance (± percent) (see A.1.1.2) | Resistance range (ohms) 1/ | Resistance temperature coefficient (ppm/°C) 2/ | Maximum body size (inches) and Configuration (see figure A-2) | |
|---|---|---|---|---|---|---|---|
| MIL-PRF-32663/1 | RFB0603 | 75 mW 25 V | ± 0.01 ± 0.02 ± 0.05 ± 0.1 ± 0.25 ± 0.5 ± 1.0 | 100 to 3.2K | ± 3 ± 5 ± 10 | 0.063 x 0.032 x 0.025 | T |
| MIL-PRF-32663/2 | RFB0805 | 150 mW 40 V | | 10 to 7K | | 0.080 x 0.050 x 0.025 | |
| MIL-PRF-32663/3 | RFB1206 | 250 mW 90 V | | 10 to 20K | | 0.126 x 0.062 x 0.025 | |
| MIL-PRF-32663/4 | RFB1506 | 300 mW 110 V | | 10 to 25K | | 0.150 x 0.062 x 0.025 | |
| MIL-PRF-32663/5 | RFB2010 | 400 mW 190 V | | 10 to 55K | | 0.198 x 0.097 x 0.025 | |
| MIL-PRF-32663/6 | RFB2512 | 500 mW 306 V | | 10 to 100K | | 0.249 x 0.127 x 0.025 | |
| MIL-PRF-32663/7 | RFG0603 | 100 / 75 mW 3/ 25 V | | 100 to 3.2K | | 0.063 x 0.032 x 0.025 | |
| MIL-PRF-32663/8 | RFG0805 | 225 / 150 mW 3/ 40 V | | 10 to 7K | | 0.080 x 0.050 x 0.025 | |
| MIL-PRF-32663/9 | RFG1206 | 400 / 250 mW 3/ 90 V | | 10 to 20K | | 0.126 x 0.062 x 0.025 | |
| MIL-PRF-32663/10 | RFG1506 | 450 / 250 mW 3/ 110 V | | 10 to 25K | | 0.150 x 0.062 x 0.025 | |
| MIL-PRF-32663/11 | RFG2010 | 600 / 400 mW 3/ 190 V | | 10 to 55K | | 0.198 x 0.097 x 0.025 | |
| MIL-PRF-32663/12 | RFG2512 | 800 / 500 mW 3/ 306 V | | 10 to 100K | | 0.249 x 0.127 x 0.025 | |

1/ Resistance range based on the characteristic and resistance tolerance, see applicable specification sheets for available combinations.

2/ Based on Characteristic. see applicable specification sheet.

3/ Based on mounting (ceramic/poly substrates vs. FR4), see applicable specification sheet.

MIL-PRF-39017 - Resistor, Fixed, Film, Insulated, Nonestablished Reliability and Established Reliability

| Specification sheet | Styles available | Power and max voltage ratings | Resistance tolerance (± percent) (see A.1.1.2) | Resistance range (ohms) | Resistance temperature coefficient (ppm/°C) | Maximum body size (inches) and Configuration (see figure A-2) | |
|---|---|---|---|---|---|---|---|
| MIL-PRF-39017/1 | RLR07 | 0.25 W 250 V | F, G, J and K | 1 to 10 M 11 M to 22 M | ±100 ±350 | 0.281 x 0.98 | A |
| MIL-PRF-39017/2 | RLR20 | 0.5 W 350 V | | 1 to 3.01 M 3.3 M to 22 M | | 0.416 x 0.161 | |
| MIL-PRF-39017/3 | RLR32 | 1 W 500 V | | 1 to 2.7 M 3.0 M to 22 M | | 0.593 x 0.205 | |
| MIL-PRF-39017/5 | RLR05 | 0.125 W 200 V | | 2.7 to 1 M 1.1 M to 22 M | | 0.170 x 0.074 | |



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TABLE A-I. Fixed film resistor selection guidance table - Continued.

| MIL-PRF-49462 - Resistors, Fixed, Film, High Voltage | | | | | | | |
|---|---|---|---|---|---|---|---|
| Specification sheet | Styles available | Power and max voltage ratings | Resistance tolerance (± percent) (see A.1.1.2) | Resistance range (ohms) | Resistance temperature coefficient (ppm/°C) | Maximum body size (inches) and Configuration (see figure A-2) | |
| MIL-PRF-49462/3 | RHV30<br>RHV31<br>RHV32<br>RHV33<br>RHV34<br>RHV35 | 0.25 W / 750 V<br>0.5 W / 1.5 kV<br>1.0 W / 3.0 kV<br>2.0 W / 5.0 kV<br>3.0 W / 10.0 kV<br>5.0 W / 20.0 kV | F, G, J and K | 100 k to 100 M<br>100 k to 392 M<br>49.9 k to 499 M<br>100 k to 499 M<br>200 k to 1 G<br>330 k to 1 G | R < 500 MΩ<br>RTC ≤ 200<br>R ≥ 500 MΩ<br>RTC ≤ 500 | 0.306 x 0.098<br>0.431 x 0.154<br>0.752 x 0.328<br>1.124 x 0.328<br>2.124 x 0.328<br>3.124 x 0.328 | A |

| MIL-PRF-49465 - Resistor, Fixed, Metal Element, Power Type, Very Low Resistance Values | | | | | | | |
|---|---|---|---|---|---|---|---|
| Specification sheet | Styles available | Power ratings | Resistance tolerance (± percent) (see A.1.1.2) | Resistance range (ohms) | Resistance temperature coefficient (ppm/°C) | Maximum body size (inches) and Configuration (see figure A-2) | |
| MIL-PRF-49465/1 | RLV10 | 5 W | | 0.01 to 0.0249<br>0.025 to 0.0499<br>0.05 to 0.0749<br>0.075 to 0.099<br>0.1 to 0.5 | ±150<br>±125<br>±100<br>±50<br>±50 | 0.999 x 0.406 | O |

| MIL-PRF-55182 - Resistor, Fixed, Film, Nonestablished Reliability, Established Reliability, and Space Level | | | | | | | |
|---|---|---|---|---|---|---|---|
| Specification sheet | Styles available | Power ratings and max voltage rating | Resistance tolerance (± percent) (see A.1.1.2) | Resistance range (ohms) | Resistance temperature coefficient (ppm/°C) | Maximum body size (inches) and Configuration (see figure A-2) | |
| MIL-PRF-55182/1 | RNR/N/C50 | 0.100 W<br>200 V | B, D, and F | 10 to 2.0 M | ±25,<br>±50,<br>±100 | 0.250 x 0.109 | A |
| MIL-PRF-55182/3 | RNR/N/C55 | 0.125 W<br>250 V | | 10 to 0.2 M | | 0.281 x 0.155 | |
| MIL-PRF-55182/5 | RNR/N/C60 | 0.250 W<br>300 V | | 1.0 to 8.06 M | | 0.625 x 0.188 | |
| MIL-PRF-55182/6 | RNR/N/C65 | 0.50 W<br>350 V | | 1.0 to 15 M | | 0.750 x 0.250 | |
| MIL-PRF-55182/7 | RNR/N/C70 | 0.05 W<br>200 V | | 10 to 0.796 M | | 0.180 x 0.065 | |
| MIL-PRF-55182/9 | RNC90 | 0.3 W<br>300 V | V, T, A, B, D, and F | 4.99 to 200K | ±5, ±10 | 0.310 x 0.336 x 0.110 | K |
| MIL-PRF-55182/10 | RNR/C75 | 1.0 W<br>750 V | B, D, and F | 10 to 20 M | ±25, ±50, ±100 | 1.062 x 0.375 | A |



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TABLE A-I. Fixed film resistor selection guidance table - Continued.

| MIL-PRF-55342 - Resistor, Chip, Fixed, Film, Nonestablished Reliability, Established Reliability, Space Level | | | | | | | |
|---|---|---|---|---|---|---|---|
| Specification sheet | Styles available | Power ratings and max voltage rating | Resistance tolerance (± percent) (see A.1.1.2) | Resistance range (ohms) | Resistance temperature coefficient (ppm/°C) | Body size (inches) and Configuration (see figure A-2) | |
| MIL-PRF-55342/1 | RM0502 | 0.05 W 40 V | B, F, G, J and K | 1 to 22M | ±25, ±50, ±100, ±200, ±300 | 0.050 x 0.025 x 0.010/0.030 | T |
| MIL-PRF-55342/2 | RM0505 | 0.125W 40 V | | | | 0.05 x 0.05 x 0.012/0.033 | |
| MIL-PRF-55342/3 | RM1005 | 0.2 W 50 V | | | | 0.10 x 0.05 x 0.015/0.033 | |
| MIL-PRF-55342/4 | RM1505 | 0.150 W 40 V | | | | 0.15 x 0.05 x 0.015/0.033 | |
| MIL-PRF-55342/5 | RM2208 | 0.225 W 40 V | | | | 0.230 x 0.085 x 0.015/0.033 | |
| MIL-PRF-55342/6 | RM0705 | 0.150 W 50V | | | | 0.075 x 0.05 x 0.015/0.033 | |
| MIL-PRF-55342/7 | RM1206 | 0.25 W 100V | | | | 0.126 x 0.063 x 0.015/0.033 | |
| MIL-PRF-55342/8 | RM2010 | 0.8 W 150 V | | | | 0.206 x 0.098 x 0.015/0.033 | |
| MIL-PRF-55342/9 | RM2512 | 1 W 200 V | | | | 0.248 x 0.124 x 0.015/ .033 | |
| MIL-PRF-55342/10 | RM1010 | 0.5 W 75 V | | | | 0.100 x 0.100 x 0.015/0.033 | |
| MIL-PRF-55342/11 | RM0402 | 0.050 W 25 V | | | | 0.041 x 0.022 x 0.010/0.033 | |
| MIL-PRF-55342/12 | RM0603 | 0.1 W 50 V | | | | 0.060 x 0.032 x 0.010/0.033 | |
| MIL-PRF-55342/13 | RM0302 | 0.040 W 15 V | | | | 0.032 x 0.022 x 0.010/0.025 | |
| MIL-PRF-55342/14 | RM0201 | 0.050W 30V | | | | 0.012 x 0.009 X0.022/0.024 | |



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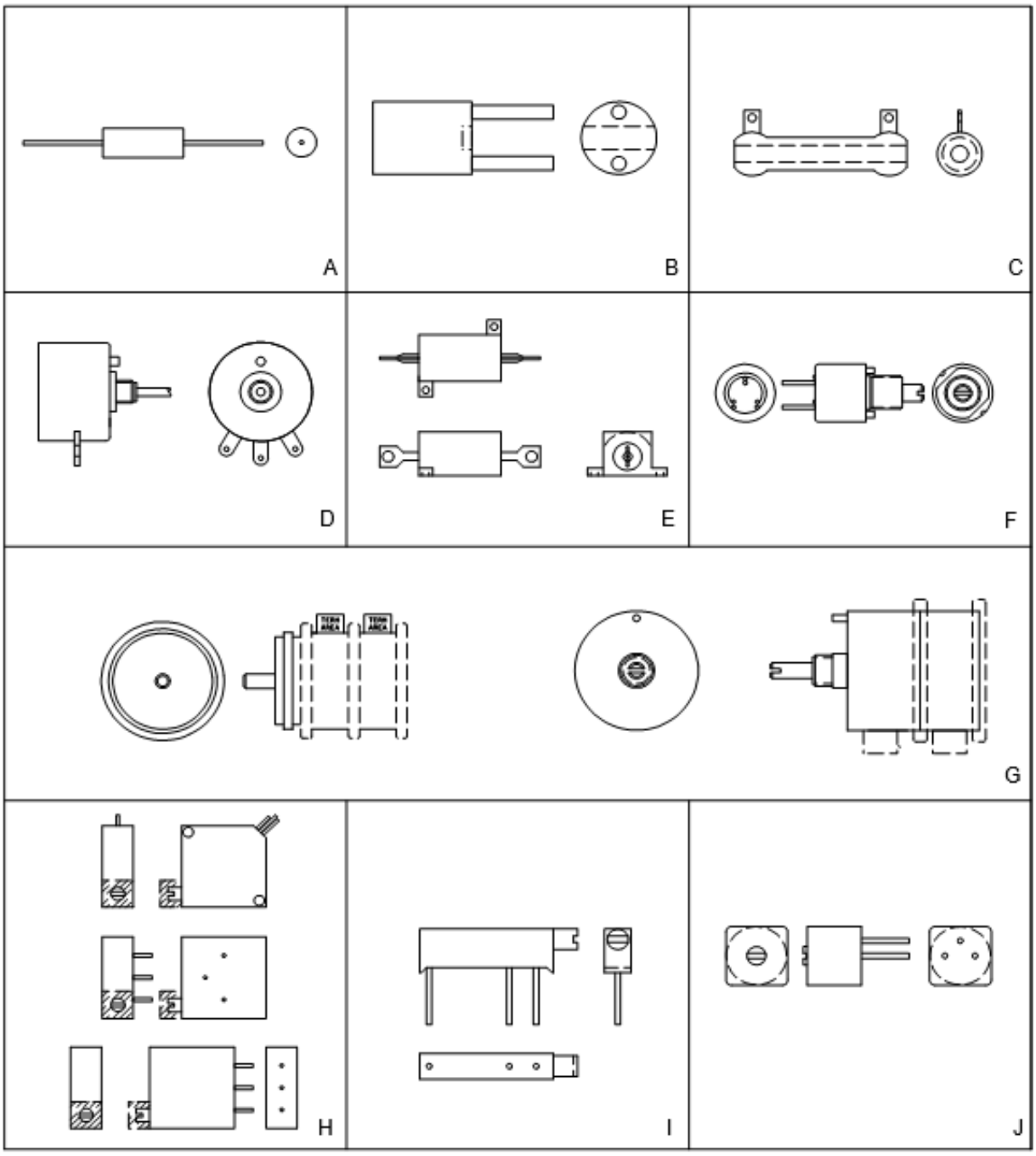



FIGURE A-2. Configurations.



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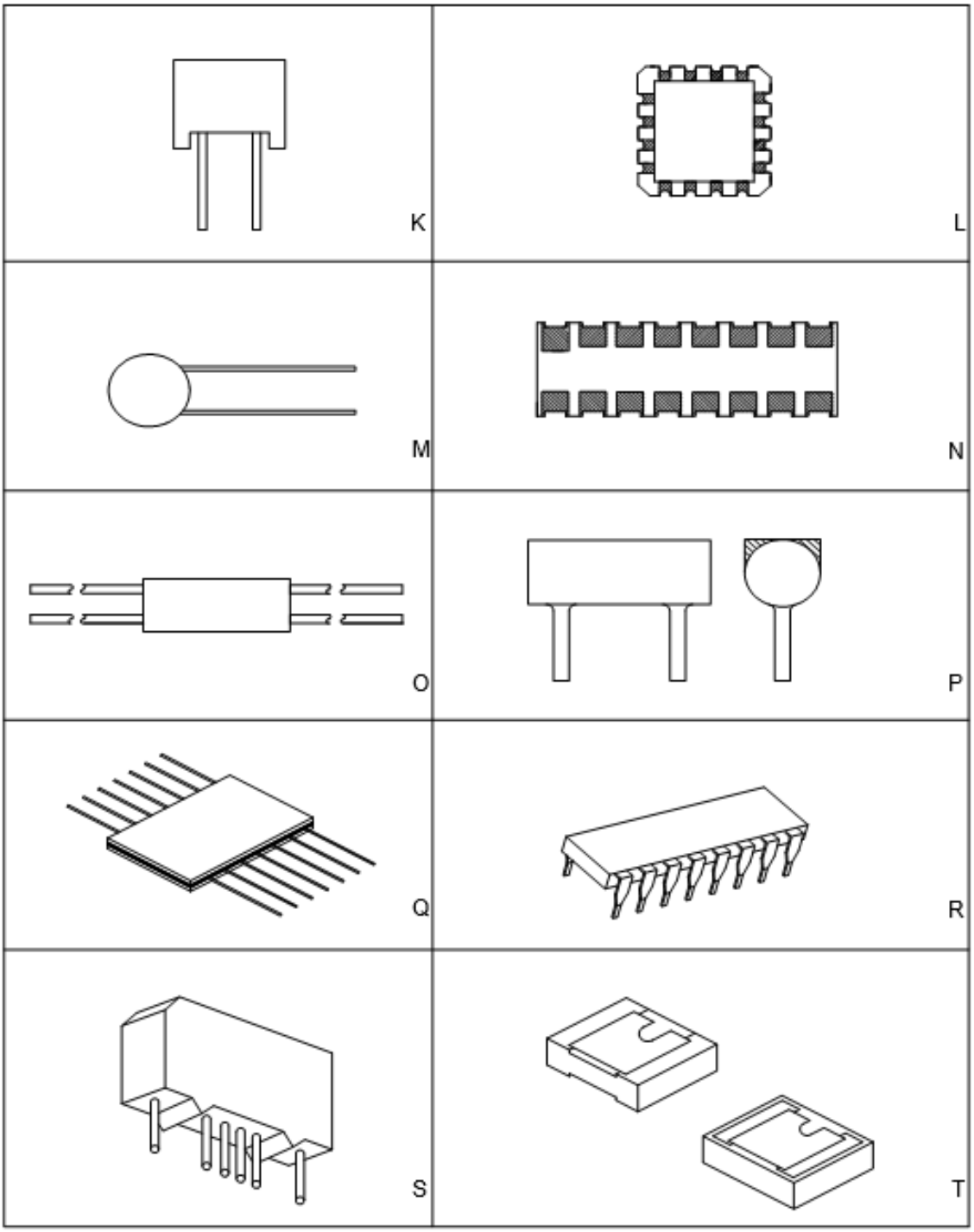



FIGURE A-2. Configurations. - Continued.



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A.3.1.2 Fixed Wirewound Resistors. See table A-II.

A.3.1.2.1 MIL-PRF-26, RW, Resistor, Fixed, Wirewound, Power Type. These power type, wirewound fixed resistors are for use in electrical, communication and associated equipment. Included are general purpose styles of 5 percent initial resistance tolerance with power ratings ranging from 3 watts to 240 watts at +25$^{O}$C, derated to 0 power at 350$^{O}$C and precision axial lead types of 0.1 percent, 0.5 percent, and 1 percent initial resistance tolerance with power ranging from 1 watt to 10 watts at +25$^{O}$C, derated to 0 power at +275$^{O}$C. Use where large power dissipation is required and where ac performance is relatively unimportant (such as, when used as voltage divider or bleeder resistors in dc power supplies, or for series dropping). They are generally satisfactory for use at frequencies up to 20 kilohertz (kHz) even though the ac characteristics are controlled. Neither the wattage rating nor the rated continuous working voltage may be exceeded. See MIL-PRF-39007, RWR, Resistor Fixed, Wirewound, Power Type, Nonestablished Reliability, Established Reliability, and Space Level.

A.3.1.2.2 MIL-PRF-18546, RE, Resistor, Fixed, Wirewound, Power Type, Chassis Mounted. These fixed, wirewound resistors are established reliability chassis mounted, power type units. They utilize the principle of heat dissipation through a metal mounting surface. These resistors are capable of full load operation at an ambient temperature of +25$^{O}$C when mounted on a specified chassis area. Use where power tolerance and relatively large power dissipation is required for a given unit size than is provided by MIL-PRF-26 resistors, where ac performance is noncritical (such as, voltage divider or bleeder resistors in dc power supplies or series-dropping circuits). See MIL-PRF-39009, RER, Resistor, Fixed, Wirewound, Power Type, Chassis Mounted, Nonestablished Reliability, Established Reliability.

A.3.1.2.3 MIL-PRF-39005, RBR, Resistor, Fixed, Wirewound, Accurate, Nonestablished Reliability, Established Reliability. These nonestablished reliability, established reliability, accurate, wirewound fixed resistors that have a maximum resistance tolerance of 1 percent and a high degree of stability with respect to time under specified environmental conditions. These resistors are suitable for continuous full load (or voltage) operation at any ambient temperature up to +125$^{O}$C, and when properly derated to +145$^{O}$C. Use in circuits requiring higher stability than provided by composition or film resistors, and where ac frequency performance is not critical. Operation is satisfactory from dc to 50 kHz. The resistors have product levels ranging from Non-ER, and a life failure rate (FR) 1.0 percent to 0.001 percent per 1,000 hours. The FR levels are established at a 60 percent confidence level on basis of life tests. Replaces MIL-PRF-93, RB Resistors, Fixed, Wire-Wound (Accurate) inactive specification sheets.

NOTE:
1. FR level is in percent per 1000 hours.

A.3.1.2.4 MIL-PRF-39007, RWR, Resistor Fixed, Wirewound, Power Type, Nonestablished Reliability, Established Reliability, and Space Level. These nonestablished reliability, established reliability, or space level, axial leaded precision power type, wirewound fixed resistors having a +25$^{O}$C ambient operating temperature derated to 0 load at +250$^{O}$C. Use where large power dissipation is required and where ac performance is relatively unimportant (such as, when used as voltage divider or bleeder resistors in dc power supplies, or for series dropping). They are generally satisfactory for use at frequencies up to 20 kilohertz (kHz) even though the ac characteristics are controlled. Neither the wattage rating nor the rated continuous working voltage may be exceeded. The resistors have product levels ranging from Non-ER (C), a life failure rate (FR) 1.0 (M) percent to 0.001 (S) percent per 1,000 hours and space level (T) The FR levels are established at a 60 percent confidence level on basis of life tests. Replaces MIL-PRF-26, RW, Resistor, Fixed, Wirewound, Power Type inactive specification sheets.

NOTE:
1. FR level is in percent per 1000 hours.



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A.3.1.2.5 MIL-PRF-39009, RER, Resistor, Fixed, Wirewound, Power Type, Chassis Mounted, Nonestablished Reliability, Established Reliability. These fixed, wirewound resistors are established reliability chassis mounted, power type units. They utilize the principle of heat dissipation through a metal mounting surface with full rated wattage at +25$^{O}$C. These resistors should not be used in circuits where their ac performance is of critical importance, although provision have been made, in particular styles, to minimize inductance. The resistors have product levels ranging from Non-ER, and a life failure rate (FR) 1.0 percent to 0.001 percent per 1,000 hours. The FR levels are established at a 60 percent confidence level on basis of life tests. Replaces MIL-PRF-18546, RE, Resistor, Fixed, Wirewound, Power Type, Chassis Mounted inactive specification sheets.

NOTE:
1. FR level is in percent per 1000 hours.

TABLE A-II. Fixed wirewound resistor selection guidance table.

MIL-PRF-26 - Resistor, Fixed, Wirewound, Power Type

| Specification sheet | Styles available | Power ratings and max voltage rating | Resistance tolerance (± percent) (see A.1.1.2) | Resistance range (ohms) | Resistance temperature coefficient (ppm/$^{O}$C) 1/ | Maximum body size (inches) and Configuration (see figure A-2) | |
|---|---|---|---|---|---|---|---|
| MIL-PRF-26/3 | RW29 | 11 | J and K | 0.1 to 5.6K | ±650 | 1.812 x 0.500 | C |
| | RW31 | 14 | | 0.1 to 6.8K | | 1.562 x 0.594 | |
| | RW33 | 26 | | 0.1 to 18K | | 3.062 x 0.594 | |
| | RW35 | 55 | | 0.1 to 43K | ±500 | 4.062 x 0.906 | |
| | RW37 | 113 | | 0.1 to 91K | ±400 | 6.062 x 1.312 | |
| | RW38 | 159 | | 0.1 to 150K | ±10 | 8.062 x 1.312 | |
| | RW47 | 210 | | 0.1 to 180K | | 10.562 x 1.312 | |
| MIL-PRF-26/4 | RW56 | 14 | | 0.1 to 9.1K | ±400 / ±260 | 2.094 x .563 | A |

1/ Based on resistance range, see specification sheet for ranges.

MIL-PRF-18546 - Resistor, Fixed, Wire-Wound, Power Type, Chassis Mounted

| Specification sheet | Styles available | Power ratings and max voltage rating | Resistance tolerance (± percent) (see A.1.1.2) | Resistance range (ohms) | Resistance temperature coefficient (ppm/$^{O}$C) | Maximum body size (inches) and Configuration (see figure A-2) | |
|---|---|---|---|---|---|---|---|
| MIL-PRF-18546/2 | RE77 | 75 | F | 0.05 to 29.4K | ±30 (R ≥ 2k) | 3.594 x 1.781 x 2.843 | E |
| | RE80 | 120 | | 0.1 to 35.7K | ±50 (R < 2k) | 4.594 x 2.219 x 3.031 | |



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TABLE A-II. Fixed wirewound resistor selection guidance table - Continued.

MIL-PRF-39005 - Resistor, Fixed, Wirewound, Accurate, Nonestablished Reliability, Established Reliability

| Specification sheet | Styles available | Power ratings and max voltage rating | Resistance tolerance (± percent) (see A.1.1.2) | Resistance range (ohms) | Resistance temperature coefficient (ppm/°C) | Maximum body size (inches) and Configuration (see figure A-2) | |
|---|---|---|---|---|---|---|---|
| MIL-PRF-39005/1 | RBR52 1/ | 0.5 W 600 V | T, Q, A, B and F | 0.1 to 1.21 M | ±90 (R < 1) ±30 (1 ≤ R < 10) ±15 (10 ≤ R < 100) ±10 (R ≥ 100) | 1.020 x 0.390 | A |
| MIL-PRF-39005/2 | RBR53 2/ | 0.33 W 300 V | | 0.1 to 1.1 M | | 0.770 x 0.390 | |
| MIL-PRF-39005/3 | RBR54 3/ | 0.25 W 300 V | | 0.1 to .562 M | | 0.770 x 0.265 | |
| MIL-PRF-39005/4 | RBR55 4/ | 0.15 W 200 V | | 0.1 to .332 M | | 0.520 x 0.265 | |
| MIL-PRF-39005/5 | RBR56 5/ | 0.125 W 150 V | | 0.1 to .220 M | | 0.364 x 0.265 | |
| MIL-PRF-39005/6 | RBR71 6/ | 0.125 W 150 V | | 0.1 to .15 M | | 0.343 x 0.281 | B |
| MIL-PRF-39005/11 | RBR80 7/ RBR81 8/ | 0.1 W 100 V | | 10 to .120 M 10 to .250 M | ±5 (R ≥ 100) ±10 (R < 100) | 0.325 x 0.160 | P |

1/ Suggested replacement use DLA Land and Maritime drawing 14006 (Non ER)
2/ Suggested replacement use DLA Land and Maritime drawing 15002 (Non ER)
3/ Suggested replacement use DLA Land and Maritime drawing 14007 (Non ER)
4/ Suggested replacement use DLA Land and Maritime drawing 15003 (Non ER)
5/ Suggested replacement use DLA Land and Maritime drawing 14008 (Non ER)
6/ Suggested replacement use DLA Land and Maritime drawing 15004 (Non ER)
7/ Suggested replacement use DLA Land and Maritime drawing 15007 (Non ER)
8/ Suggested replacement use DLA Land and Maritime drawing 09008 (Non ER)

MIL-PRF-39007 - Resistor, Fixed, Wirewound, Power Type, Nonestablished Reliability, Established Reliability, and Space Level

| Specification sheet | Styles available | Power ratings and max voltage rating | Resistance tolerance (± percent) (see A.1.1.2) | Resistance range (ohms) | Resistance temperature coefficient (ppm/°C) | Maximum body size (inches) and Configuration (see figure A-2) | |
|---|---|---|---|---|---|---|---|
| MIL-PRF-39007/7 | RWR78 | 10 | B, D, and F | 0.1 to 39.2 k | | 1.842 x 0.406 | A |
| MIL-PRF-39007/8 | RWR80 | 2 | | 0.1 to 3.16 k | ±650 (.1 ≤ R ≤ .499) | 0.437 x 0.125 | |
| MIL-PRF-39007/9 | RWR81 | 1 | | 0.1 to 1 k | ±400 (.499 < R ≤ 1) | 0.281 x 0.105 | |
| MIL-PRF-39007/10 | RWR84 | 7 | | 0.1 to 12.4 k | ± 20 (R >10) | 0.937 x 0.343 | |
| MIL-PRF-39007/11 | RWR89 | 3 | | 0.1 to 4.12 k | | 0.622 x 0.218 | |
| MIL-PRF-39007/12 | RWR82 | 1.5 | | 0.1 to 1.3 k | ± 50 (1 < R ≤ 10) | 0.328 x 0.101 | |



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TABLE A-II. Fixed wirewound resistor selection guidance table - Continued.

| MIL-PRF-39009 - Resistor, Fixed, Wirewound, Power Type, Chassis Mounted, Nonestablished Reliability, and Established Reliability | | | | | | | |
|---|---|---|---|---|---|---|---|
| Specification sheet | Styles available | Power ratings and max voltage rating | Resistance tolerance (± percent) (see A.1.1.2) | Resistance range (ohms) | Resistance temperature coefficient (ppm/°C) | Maximum body size (inches) and Configuration (see figure A-2) | |
| MIL-PRF-39009/1 | RER60 | 5 | F | 0.1 to 3.32K | ±100 (R < 1)<br>± 50 (1 ≤ R < 19.6)<br>± 30 (R ≥ 20) | 0.662 x 0 .677 x 0.351 | E |
| | RER65 | 10 | | 0.1 to 5.62K | | 0.812 x 0.843 x 0.437 | |
| | RER70 | 20 | | 0.1 to 12.1K | | 1.124 x 1.125 x 0.593 | |
| | RER75 | 30 | | 0.1 to 39.2K | | 2.000 x 1.187x x 0.656 | |
| MIL-PRF-39009/2 | RER40 | 5 | | 1 to 1.65K | | 0.662 x 0.677 x 0.351 | |
| | RER45 | 10 | | 1 to 2.80K | | 0.812 x 0.843 x 0.437 | |
| | RER50 | 20 | | 1 to 6.04K | | 1.124 x 1.125 x 0.593 | |
| | RER55 | 30 | | 1 to 19.6K | | 2.000 x 1.187 x 0.656 | |

A.3.1.3 Fixed Film Networks. See table A-III.

A.3.1.3.1 MIL-PRF-914, RNS, Resistor, Network, Fixed, Film, Surface Mount, Nonestablished Reliability, and Established Reliability. These networks are primarily intended for use in surface mount applications where space is a major concern. These resistors can either be hermetically or nonhermetically sealed and consist entirely of fixed, film resistors. The resistors have product levels ranging from Non-ER, and a life failure rate (FR) 1.0 percent to 0.001 percent per 1,000 hours. The FR levels are established at a 60 percent confidence level on basis of life tests.

NOTE:
1. FR level is in percent per 1000 hours.
2. MIL-PRF-914 designers are cautioned in using these resistors in pulse application. The usage requirements for the resistor must be evaluated for each application. See Appendix E.

A.3.1.3.2 MIL-PRF-83401, RZ, Resistor Network, Fixed Film, and Capacitor-Resistor Networks, Ceramic Capacitor and Fixed Film Resistor. These networks are designed for use in critical circuitry where stability, long life, reliable operation, and accuracy are of prime importance. They are particularly desirable for use where miniaturization is important and ease of assembly is desired. They are useful where a number of resistors of the same resistance value are required in the circuit.

NOTE: MIL-PRF-83401 designers are cautioned in using these resistors in pulse application. The usage requirements for the resistor must be evaluated for each application. See Appendix E.



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APPENDIX A

TABLE A-III. Fixed film network resistor selection guidance table.

| MIL-PRF-914 - Resistor Network, Fixed, Film, Surface Mount, Nonestablished Reliability, and Established Reliability | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Specification sheet | Styles available | Schematic | Power ratings and Characteristics | | Resistance tolerance (± percent) (see A.1.1.2) | Resistance range (ohms) | Resistance temperature coefficient (ppm/°C) | Maximum body size (inches) and Configuration (see figure A-2) | |
| | | | K and M | C, R, H, and V | | | | | |
| MIL-PRF-914/3 | RNS030 | W | 0.10/0.80 | 0.050/0.40 | B, D, F, G and J | 10 to 2.2 M | ±25, ±50, ±100, ±300 | 0.300 x 0.300 x 0.050 | L |
| | | M | 0.05/0.75 | 0.025/0.40 | | | | 0.300 x 0.300 x 0.085 | |
| | | E | 0.10/0.80 | 0.050/0.40 | | | | 0.300 x 0.300 x 0.090 | |
| | | S | 0.05/0.70 | 0.025/0.35 | | | | 0.300 x 0.300 x 0.050 | |
| MIL-PRF-914/4 | RNS040 | P | 0.10/1.0 | 0.050/0.50 | | | | 0.350 x 0.350 x 0.050 | |
| | | M | 0.05/0.95 | 0.025/0.475 | | | | 0.350 x 0.035 x .085 | |
| | | E | 0.10/1.0 | 0.050/0.50 | | | | 0.350 x 0.350 x 0.090 | |
| | | S | 0.05/1.0 | 0.025/0.50 | | | | 0.350 x 0.350 x 0.050 | |
| MIL-PRF-914/5 | RNS050 | A | 0.10/0.80 | 0.050/0.40 | | | | 0.150 x 0.410 x 0.035 | N |
| | | B | 0.055/0.80 | 0.025/0.375 | | | | | |
| | | C | 0.050/0.80 | 0.100/0.40 | | | | | |
| | | J | 0.030/0.80 | 0.015/0.40 | | | | | |



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TABLE A-III. Fixed film network resistor selection guidance table. Continued

| MIL-PRF-83401 - Resistor Network, Fixed, Film, and Capacitor-Resistor Networks, Ceramic Capacitor and Fixed Film Resistor | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Specification sheet (MIL-PRF-) | Styles available | Schematic | Power ratings and Characteristics | | Resistance tolerance (± percent) (see A.1.1.2) | Resistance range (ohms) | Resistance temperature coefficient (ppm/°C) | Maximum body size (inches) and Configuration (see figure A-2) | |
| | | | H, K and M | C and V | | | | | |
| 83401/1 | RZ010 | A<br>B<br>J | 0.2/1.4<br>0.1/1.3<br>0.050/1.2 | 0.1/0.7<br>0.05/0.65<br>0.025/0.6 | | | | 0.785 x 0.305 x 0.200 | R |
| 83401/2 | RZ020 | A<br>B<br>J | 0.2/1.6<br>0.1/1.5<br>0.050/1.4 | 0.1/0.8<br>0.05/0.75<br>0.025/0.7 | B, D, F, G and J | | | 0.876 x 0.305 x 0.200 | |
| 83401/3 | RZ030 | A<br>B<br>J | 0.05/0.35 | 0.025/0.325<br>0.015/0.35 | | | | 0.385 x 0.305 x 0.075 | Q |
| 83401/4 | RZ040 | C<br>G<br>H | 0.2/1.0<br>0.2/0.6<br>0.11/0.88 | 0.1/0.5<br>0.1/0.3<br>0.06/0.48 | | | | 0.598 x 0.098 x 0.350 | |
| 83401/5 | RZ050 | C<br>G<br>H | 0.2/1.4<br>0.2/0.8<br>0.11/1.32 | 0.1/0.7<br>0.1/0.4<br>0.06/0.72 | F, G and J | | | 0.798 x 0.098 x 0.350 | |
| 83401/6 | RZ060 | C<br>G<br>H | 0.2/1.8<br>0.2/1.0<br>0.11/1.8 | 0.1/0.9<br>0.1/0.5<br>0.06/0.9 | | | | 0.998 x 0.098 x 0.352 | S |
| 83401/7 | RZ070 | C<br>G<br>H | 0.12/0.6<br>0.12/0.36<br>0.07/0.6 | 0.06/0.3<br>0.06/0.18<br>0.04/0.3 | | 10 to 1 M | | 0.598 x 0.098 x 0.197 | |
| 83401/8 | RZ080 | C<br>G<br>H | 0.12/0.84<br>0.12/0.48<br>0.07/0.84 | 0.06/0.42<br>0.06/0.24<br>0.04/0.42 | | | ±50,<br>±100,<br>±300 | 0.798 x 0.098 x 0.197 | |
| 83401/9 | RZ090 | C<br>G<br>H | 0.12/1.08<br>0.12/0.60<br>0.07/1.08 | 0.06/0.54<br>0.06/0.30<br>0.04/0.54 | | | | 0.798 x 0.098 x 0.197 | |
| 83401/10 | RZ100 | A<br>B<br>J | 0.05/0.4<br>0.025/0.375<br>0.015/0.42 | | B, D, F, G and J | | | 0.410 x 0.305 x 0.075 | Q |
| 83401/13 | RZ130 | A<br>B<br>J | 0.2/1.4<br>0.1/1.3<br>0.050/1.2 | 0.1/0.7<br>0.05/0.65<br>0.025/0.6 | | | | 0.785 x 0.305 x 0.200 | R |
| 83401/14 | RZ140 | A<br>B<br>J | 0.2/1.6<br>0.1/1.5<br>0.050/1.4 | 0.1/0.8<br>0.05/0.75<br>0.025/0.7 | | | | 0.876 x 0.305 x 0.200 | |
| 83401/15 | RZ150 | A<br>B<br>J | 0.05/0.35<br>0.025/0.325<br>0.015/0.35 | | | | | 0.385 x 0.305 x 0.075 | Q |
| 83401/18 | RZ180 | A001<br>A002 | 0.10 | 0.40<br>0.50 | G | See schematics per specification sheet | | 0.598 x 0.098 x 0.350 | S |
| 83401/19 | RZ190 | 1/<br>2/<br>3/ | 0.10 4/ | 0.06<br>0.05<br>0.04 | | | | 0.798 x 0.098 x 0.197 | |
| 83401/21 | RZ210 | C<br>G<br>H | 0.12/0.6<br>0.12/0.36<br>0.07/0.6 | 0.06/0.3<br>0.06/0.18<br>0.04/0.3 | B, D, F, G and J | | | 0.598 x 0.098 x 0.197 | |
| 83401/22 | RZ220 | C<br>G<br>H | 0.12/0.84<br>0.12/0.48<br>0.07/0.84 | 0.06/0.42<br>0.06/0.24<br>0.04/0.42 | | | | 0.798 x0 .098 x 0.197 | |
| 83401/23 | RZ230 | C<br>G<br>H | 0.12/1.08<br>0.12/0.60<br>0.07/1.08 | 0.06/0.54<br>0.06/0.30<br>0.04/0.54 | | | | 0.998 x 0.098 x 0.197 | |
| 83401/24 | RZ240 | C<br>G<br>H | 0.12/1.08<br>0.12/0.60<br>0.07/1.08 | 0.06/0.54<br>0.06/0.30<br>0.04/0.54 | | | | 0.998 x 0.098 x 0.197 | |

1/ A001-A006, A011-A012
2/ A007-A009
3/ A010
4/ A001-A0012



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A.3.1.4 Variable Wirewound Resistors. (See table A-IV)

A.3.1.4.1 MIL-PRF-22, RP, Resistor, Variable, Wirewound, Power Type. These resistors have a resistance element of wire, wound linearly on an insulted strip shaped in an arc, such that a contact bears uniformly on the resistance element when adjusted by a control shaft. The power ratings cover a range from 6.25 watts to 1,000 watts, inclusive. Use in such applications as motor speed control, generator field control, lamp dimming, heater and oven control, potentiometer uses, and applications where variations of voltage and current are expected.

A.3.1.4.2 MIL-PRF-12934, RR, Resistor, Variable, Wirewound, Precision. These precision, wirewound, variable resistors whose electrical output (in terms of percent of applied voltage) are linear or nonlinear with respects to the angular position of the shaft. These resistors are capable of full-load operation at maximum ambient temperatures of +70$^{O}$C and +85$^{O}$C and are suitable for continuous operation, when properly derated, to maximum temperatures of +125$^{O}$C and +150$^{O}$C. Use in servo mounting applications requiring precise electrical and mechanical output and performance. Used in computer, antenna, flight control, and bomb navigation systems.

A.3.1.4.3 MIL-PRF-27208, RT, Resistor, Variable, Wirewound, Nonprecision. These wirewound variable resistors with a contact bearing uniformly over the entire surface of the entire resistive element, wound linearly, when positioned by an multiturn lead screw actuator. These resistors are capable of full-load operation at maximum ambient temperatures of +85$^{O}$C and are suitable for continuous operation, when properly derated, to maximum temperatures of +150$^{O}$C. Use for matching, balancing, and adjusting circuit variables in computers, telemetering equipment, and other critical applications. See MIL-PRF-39015, RTR, Resistor, Variable, Wirewound, Lead Screw Actuated, Nonestablished Reliability and Established Reliability.

A.3.1.4.4 MIL-PRF-39015, RTR, Resistor, Variable, Wirewound, Lead Screw Actuated, Nonestablished Reliability and Established Reliability. These nonestablished and established reliability lead screw actuated, wirewound variable resistors with a contact bearing uniformly over the surface of the entire resistive element, wound linearly, when position by a multiturn lead screw actuator. These resistors are capable of full load operation (when the maximum resistance is engaged), at maximum ambient temperature of +85$^{O}$C and are suitable for continuous operation, when properly derated, at a maximum ambient temperature of +150$^{O}$C. Use for matching, balancing, and adjusting circuit variables in computers, telemetering equipment, and other critical applications. The resistors have product levels ranging from Non-ER, and a life failure rate (FR) 1.0 percent to 0.001 percent per 1,000 hours. The FR levels are established at a 60 percent confidence level on basis of life tests. Replaces MIL-PRF-27208, RT, Resistor, Variable, Wirewound, Nonprecision inactive specification sheets.

NOTE:
1. FR level is in percent per 1000 hours.



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TABLE A-IV. Variable wirewound resistor selection guidance table.

| MIL-PRF-22 - Resistor, Variable, Wirewound, Power Type | | | | | |
|---|---|---|---|---|---|
| Specification sheet | Styles available | Power ratings | Resistance range (ohms) | Maximum body size (inches) and Configuration (see figure A-2) | |
| MIL-PRF-22/1 | RP06 | 12.5 | 1 to 3.5k | 0.906 x 0.751 | |
| MIL-PRF-22/2 | RP07 | 6.25 | 1 to 3.5k | 1.094 x 1.126 | |
| MIL-PRF-22/3 | RP10 | 25 | 1 to 5k | 1.680 x 1.410 | |
| MIL-PRF-22/4 | RP11 | 12.5 | 2 to 5k | 1.880 x 1.750 | |
| MIL-PRF-22/5 | RP15 | 50 | 1 to 10k | 2.410 x 1.440 | |
| MIL-PRF-22/6 | RP16 | 25 | 1 to 10k | 2.750 x 1.750 | |
| MIL-PRF-22/7 | RP20 | 75 | 2 to 10k | 2.810 x 1.780 | |
| MIL-PRF-22/8 | RP25 | 100 | 2 to 10k | 3.190 x 1.780 | D |
| MIL-PRF-22/9 | RP30 | 150 | 2 to 10k | 4.060 x 2.030 | |
| MIL-PRF-22/10 | RP35 | 225 | 2 to 2.5k | 5.090 x 2.160 | |
| MIL-PRF-22/11 | RP40 | 300 | 2 to 2.5k | 6.090 x 2.410 | |
| MIL-PRF-22/12 | RP45 | 500 | 2 to 2.5k | 8.090 x 2.250 | |
| MIL-PRF-22/14 | RP55 | 1,000 | 2 to 2.5k | 12.310 x 3.250 | |
| MIL-PRF-22/15 | RP05 | 5 | 10 to 5k | 0.525 x 0.687 | |



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TABLE A-IV. Variable wirewound resistor selection guidance table. Continued

| MIL-PRF-12934 - Resistor, Variable, Wirewound, Precision | | | | | |
|---|---|---|---|---|---|
| Specification sheet | Styles available | Power ratings | Resistance range (ohms) | Maximum body size (inches) and Configuration (see figure A-2) | |
| MIL-PRF-12934/1 | RR0900 | 1.25 | 100 to 10K | 0.880 x 0.812 | |
| MIL-PRF-12934/2 | RR1000 | 2.0 | 100 to 50K | 0.880 x 1.625 | |
| MIL-PRF-12934/4 | RR1100 | 1.5 | 100 to 20K | 1.067 x 0.812 | |
| MIL-PRF-12934/5 | RR1300 | 2.0 | 100 to 40K | 1.468 x 1.062 | |
| MIL-PRF-12934/6 | RR1400 | 3.0 | 200 to 200K | 1.468 x 2.250 | |
| MIL-PRF-12934/9 | RR2000 | 4.0 | 100 to 60K | 2.031 x 1.312 | |
| MIL-PRF-12934/10 | RR2100 | 5.0 | 200 to 250K | 2.031 x 2.250 | |
| MIL-PRF-12934/15 | RR3000 | 6.0 | 200 to 100K | 3.031 x 1.312 | |
| MIL-PRF-12934/16 | RR3100 | 1.25 | 100 to 10K | 0.906 x 0.750 | |
| MIL-PRF-12934/17 | RR3200 | 1.5 | 100 to 20K | 1.093 x 0.750 | |
| MIL-PRF-12934/18 | RR3300 | 2.0 | 100 to 40K | 1.468 x 1.062 | |
| MIL-PRF-12934/19 | RR3400 | 4.0 | 100 to 60K | 2.031 x 1.156 | G |
| MIL-PRF-12934/20 | RR3500 | 6.0 | 200 to 100K | 3.031 x 1.156 | |
| MIL-PRF-12934/27 | RR3600 | 1.5 | 100 to 50K | 0.906 x 1.076 | |
| MIL-PRF-12934/28 | RR3700 | 1.5 | 100 to 50K | 0.906 x 1.076 | |
| MIL-PRF-12934/29 | RR3800 | 1.5 | 100 to 100K | 0.906 x 1.219 | |
| MIL-PRF-12934/30 | RR3900 | 1.5 | 100 to 100K | 0.906 x 1.219 | |
| MIL-PRF-12934/31 | RR4000 | 2.0 | 100 to 50K | 0.890 x 1.500 | |
| MIL-PRF-12934/32 | RR4100 | 5.0 | 200 to 250K | 1.844 x 2.094 | |
| MIL-PRF-12934/33 | RR2002 | 4.0 | 20K | 2.005 x 1.781 | |
| MIL-PRF-12934/34 | RR1004 | 2.0 | 100K | 0.880 x 1.580 | |
| MIL-PRF-12934/35 | 2RR2104 | 5.0 | 20K / 20K | 2.005 x 4.311 | |
| MIL-PRF-12934/36 | RR3601 | 1.5 | 10K | 0.922 x 0.875 | |
| MIL-PRF-12934/37 | 2RR3100 | 0.25mW | 665 / 665 | 1.062 x 0.765 | |

| MIL-PRF-27208 - Resistor, Variable, Wirewound, Nonprecision | | | | | |
|---|---|---|---|---|---|
| Specification sheet | Styles available | Power ratings | Resistance range (ohms) | Maximum body size (inches) and Configuration (see figure A-2) | |
| MIL-PRF-27208/8 | RT12 | 0.75 | 10 to 20K | 1.250 x 0.315 x 0.190 | I |
| MIL-PRF-27208/10 | RT26 | 0.25 | 10 to 5K | 0.250 x 0.250 x 0.165 | H |



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TABLE A-IV. Variable wirewound resistor selection guidance table. Continued

| MIL-PRF-39015 - Resistor, Variable, Wirewound, Lead Screw Actuated, Nonestablished Reliability and Established Reliability | | | | | |
|---|---|---|---|---|---|
| Specification sheet | Styles available | Power ratings | Resistance range (ohms) | Maximum body size (inches) and Configuration (see figure A-2) | |
| MIL-PRF-39015/2 | RTR22 | 0.75 | 10 to 20K | 0.500 x 0.500 x 0.235 | H |
| MIL-PRF-39015/3 | RTR24 | | 10 to 10K | 0.375 x 0.375 x 0.195 | |

A.3.1.5 Variable Nonwirewound Resistors. See table A-V.

A.3.1.5.1 MIL-PRF-94, RV, Resistor, Variable Composition. These variable resistors have a composition resistance element shaped in an arc, and a contact bearing uniformly thereon, so that a change in resistance is produced between the terminal of the contact and the terminal of either end of the resistance element when the operating shaft is turned. These resistors are capable of full load operation (where maximum resistance is engaged) at a maximum ambient temperature of +70$^{O}$C, and suitable for continuous operation when properly derated, at a maximum temperature of +120$^{O}$C. Use where initial setting stability is not critical and long-term stability needs to be no better than ±20 percent.

A.3.1.5.2 MIL-PRF-22097, RJ, Resistor, Variable, Nonwirewound, Adjustment Type. These multiturn lead screw actuated and single-turn nonwirewound variable resistors with a contact bearing uniformly over the entire surface of the entire resistive element, when positioned by the actuator. These resistors are capable of full load operation (where maximum resistance is engaged) at a maximum ambient temperature of +70$^{O}$C and +85$^{O}$C, and are suitable for continuous operation when properly derated, at a maximum temperature of +120$^{O}$C and +150$^{O}$C, respectively. Use for matching, balancing, and adjusting circuit variables in computers, telemetering equipment, and other critical applications. See MIL-PRF-39035, RJR, Resistor, Variable, Nonwirewound, Adjustment Type, Nonestablished Reliability and Established Reliability.

A.3.1.5.3 MIL-PRF-39023, RQ, Resistor, Variable, Nonwirewound Precision. These precision, nonwirewound, variable resistors whose electrical output (in terms of percent of applied voltage) are linear or nonlinear with respect to the angular position of the operating shaft. These resistors have a resistance tolerance of ±10 percent. These resistors are capable of full-load operation at maximum ambient temperatures of +70$^{O}$C and are suitable for continuous operation, when properly derated, to maximum temperatures of +125$^{O}$C. Use in servo mounting application requiring precise electrical and mechanical output and performance. Used in computer, antenna, flight control, and bomb navigation systems.

A.3.1.5.4 MIL-PRF-39035, RJR, Resistor, Variable, Nonwirewound, Adjustment Type, Nonestablished Reliability and Established Reliability. These nonestablished and established reliability multiturn lead screw actuated, and single-turn nonwirewound variable resistors with a contact bearing uniformly over the surface of the entire resistive element, when position by an actuator. These resistors are capable of full load operation (when the maximum resistance is engaged), at maximum ambient temperature of +85$^{O}$C and are suitable for continuous operation, when properly derated, at a maximum ambient temperature of +150$^{O}$C. Use for matching, balancing, and adjusting circuit variables in computers, telemetering equipment, and other critical applications. The resistors have product levels ranging from Non-ER, and a life failure rate (FR) 1.0 percent to 0.001 percent per 1,000 hours. The FR levels are established at a 60 percent confidence level on basis of life tests. Replaces MIL-PRF-22097, RJ, Resistor, Variable, Nonwirewound, Adjustment Type, inactive specification sheets.

NOTE:
1. FR level is in percent per 1000 hours.



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TABLE A-V. Variable nonwirewound resistor selection guidance table.

MIL-PRF-94 - Resistor, Variable Composition

| Specification sheet | Styles available | Power ratings and max voltage rating | Resistance tolerance (± percent) (see A.1.1.2) | Resistance range (ohms) | Maximum body size (inches) and Configuration (see figure A-2) | |
|---|---|---|---|---|---|---|
| MIL-PRF-94/2 | RV5 | 0.5/0.25* | ±10/±20 | 250 to 2.5M | 0.375 x 0.750 | F |
| MIL-PRF-94/3 | RV6 | 0.5/0.25* | | 100 to 5M | 0.453 x 0.500 | |
| MIL-PRF-94/4 | RV2 | 1.0/0.5* | | 100 to 2.5M | 0.453 x 0.906 | |
| MIL-PRF-94/5 | RV4 | 2.0/1.0* | | 50 to 2.5M | 0.609 x 1.094 | |
| MIL-PRF-94/6 | 2RV7 | various | | 50 to 5M | 1.266 x 1.094 | |
| MIL-PRF-94/7 | RV8 | 0.5/0.25* | | 100 to 5M | 0.581 x 0.500 | |

* - A taper/C and F taper

MIL-PRF-22097 - Resistor, Variable, Nonwirewound, Adjustment Type

| Specification sheet | Styles available | Power ratings and max voltage rating | Resistance tolerance (± percent) (see A.1.1.2) | Resistance range (ohms) | Maximum body size (inches) and Configuration (see figure A-2) | |
|---|---|---|---|---|---|---|
| MIL-PRF-22097/2 | RJ12 | 0.75 | ±10 | 10 to 1M | 1.250 x 0.190 x 0.315 | I |
| MIL-PRF-22097/4 | RJ24 | 0.50 | | | 0.375 x 0.150 x 0.375 | H |

MIL-PRF-39023 - Resistor, Variable, Nonwirewound Precision

| Specification sheet | Styles available | Power ratings and max voltage rating | Resistance tolerance (± percent) (see A.1.1.2) | Resistance range (ohms) | Maximum body size (inches) and Configuration (see figure A-2) | |
|---|---|---|---|---|---|---|
| MIL-PRF-39023/1 | RQ090 | 1.00 | ±10 | 100 to 1M | 0.906 x 0.810 | H |
| MIL-PRF-39023/2 | RQ110 | 1.25 | | 100 to 1M | 1.125 x 0.810 | |
| MIL-PRF-39023/3 | RQ150 | 1.50 | | 100 to 1M | 1.500 x 1.060 | |
| MIL-PRF-39023/4 | RQ200 | 2.00 | | 100 to 1M | 2.031 x 1.310 | |
| MIL-PRF-39023/5 | RQ300 | 3.00 | | 100 to 1M | 3.031 x 1.310 | |
| MIL-PRF-39023/6 | RQ100 | 2.50 | | 1000 to 1M | 0.906 x 1.880 | |
| MIL-PRF-39023/7 | RQ160 | 3.50 | | 1000 to 3M | 1.468 x 2.500 | |
| MIL-PRF-39023/8 | RQ210 | 4.50 | | 1000 to 3M | 2.031 x 2.900 | |
| MIL-PRF-39023/9 | RQ091 | 1.00 | | 100 to 1M | 0.906 x 0.750 | |
| MIL-PRF-39023/10 | RQ051 | 0.50 | | 5000 | 0.9644 x 0.550 | |

MIL-PRF-39035 - Resistor, Variable, Nonwirewound, Adjustment Type, Nonestablished Reliability and Established Reliability

| Specification sheet | Styles available | Power ratings and max voltage rating | Resistance tolerance (± percent) (see A.1.1.2) | Resistance range (ohms) | Maximum body size (inches) and Configuration (see figure A-2) | |
|---|---|---|---|---|---|---|
| MIL-PRF-39035/2 | RJR24 | 0.50 | ±10 | 10 to 1M | 0.375 x 0.150 x 0.375 | H |
| MIL-PRF-39035/3 | RJR26 | 0.25 | | | 0.270 x 0.165 x 0.250 | |
| MIL-PRF-39035/4 | RJR50 | | | | 0.25 1/ x 0.25 | J |

1/ Diameter



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A.3.1.6 General terms for variable resistors.

a. Precision potentiometer. A mechanical-electrical transducer dependent upon the relative position of a moving contact (wiper) and a resistance element for its operation. It delivers to a high degree of accuracy a voltage output that is some specified function of applied voltage and shaft position, (see figure A-3).

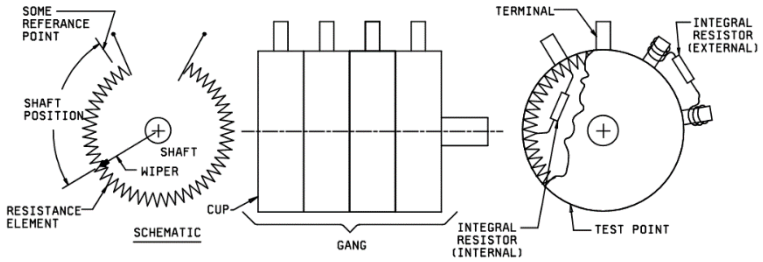



FIGURE A-3. Precision potentiometer.

b. Cup. A single mechanical section of a potentiometer which may contain one or more electrical resistance elements.

c. Gang. An assembly of two or more cups on a common operating shaft.

d. Shaft. The mechanical input element of the potentiometer.

e. Shaft position: An indication of the position of the wiper relative to a reference point.

f. Terminal. An external member that provides electrical access to the potentiometer resistance element and wiper.

g. Integral resistor. An internal or external resistor preconnected to the electrical element and forming an integral part of the cup assembly to provide a desired electrical characteristic. The resistor may be a separate entity, a part of the wirewound or nonwirewound resistance element, or a layer type resistor formed on the same insulating substrate as the resistance element.

h. Test point. An additional terminal used only to facilitate measurement.

I. Tap. An electrical connection fixed to the resistance element and capable of carrying rated element current.

NOTE: Certain types of taps (minimum width) essential for tighter conformities may require lower current limits. These taps are referred to as voltage taps.

j. Dead space. A region of the resistance element constructed to provide a discontinuity in the output. The dead space is usually found between the input terminals of a continuous rotation single turn potentiometer.



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k. Total applied voltage (E). The total voltage applied between the designated input terminals, (see figure A-4).

NOTE: When plus (+) and minus (-) voltages are applied to the potentiometer, the total applied voltage (commonly called peak-to-peak applied voltage) is equal to the sum of the two voltages. Each individual voltage is referred to as zero-to-peak applied voltage

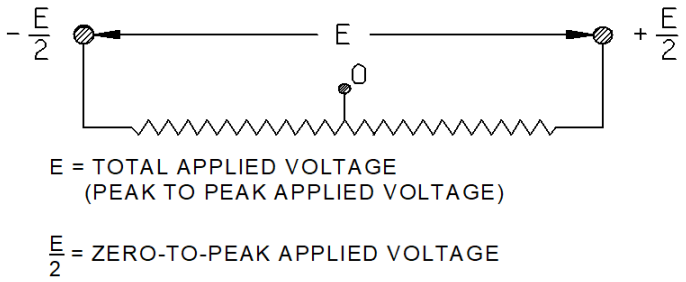



FIGURE A-4. Total applied voltage.

l. Output voltage. The voltage between the wiper and the designated reference point. Unless otherwise specified, the designated reference point is the CCW terminal.

m. Output ratio. The ratio of the output voltage to the designated input reference voltage is the total applied voltage.

n. Total variable output. The difference between the maximum and minimum output ratios. These ratios correspond to the minimum voltages at each input terminal. (see figure A-5).

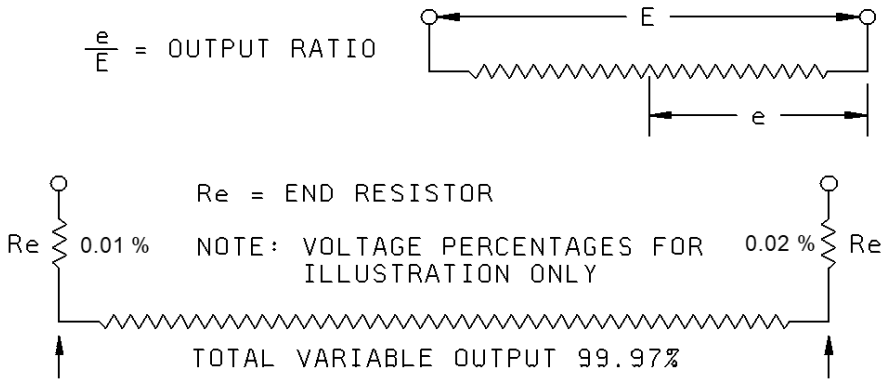



FIGURE A-5. Total variable output.

o. End voltage. The voltage between the wiper terminal and an end terminal when the shaft is positioned at the corresponding theoretical end point. End voltage is expressed as a percent of the total applied voltage.



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p. Minimum voltage. The smallest or lowest voltage between the wiper terminal and an end terminal when the shaft is positioned near the corresponding end of electrical continuity travel. Minimum voltage is expressed as a percent of the total applied voltage.

q. Shorted segment. A portion of the resistance element over which the output ratio remains constant within specified limits as the wiper traverses the segment with a specified load resistance.

r. Output slope. The ratio between the rate of change of output ratio and the rate of change of shaft travel.

$$\text{MATHEMATICALLY:} \quad A = \frac{\Delta \frac{e}{E}}{\Delta \frac{\theta}{\theta_T}}$$

$\theta_A$ may be substituted for $\theta_T$ where applicable.

NOTE: The theoretical output slope is the first derivative of the normalized theoretical function characteristic.

$$\text{MATHEMATICALLY:} \quad A = \frac{df\left[\frac{\theta}{\theta_T}\right]}{d\left[\frac{\theta}{\theta_T}\right]} = \frac{d\left[\frac{e}{E}\right]}{d\left[\frac{\theta}{\theta_T}\right]}$$

s. Slope ratio. The ratio of the largest to the smallest output slopes of a monotonic theoretical function characteristic.

t. Gradient. The rate of change of output ratio relative to shaft travel.

$$\text{MATHEMATICALLY: } G = \frac{d\left[\frac{e}{E}\right]}{d\theta}$$

u. Load resistance (RL). The external resistance as seen by the output voltage; (connected between the wiper and the designated reference point).

NOTE: No load means an infinite load resistance.

v. Loading error. The difference between the output ratio with an infinite load resistance and the output ratio with a specified finite load resistance, at any shaft position as long as it is the same position for both output ratio measurements, (see figure A-6).

NOTE: Elimination of loading error, by compensating the resistance element to give the desired output with a specified load resistance, is referred to as “load compensation.”



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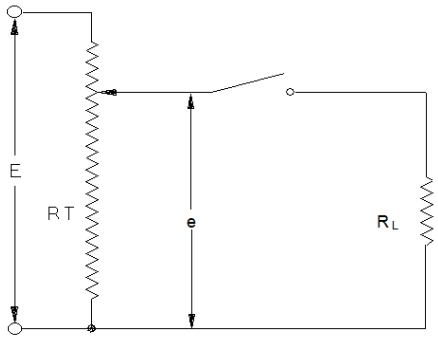
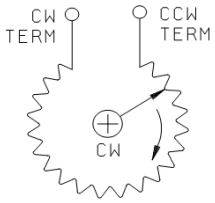



FIGURE A-6. Loading error.

w. Direction of travel. For rotary potentiometers, CW or CCW when viewing the specified mounting end of the potentiometer. The designation of terminals in the figure corresponds to the direction of shaft travel. Unless otherwise specified (see figure A-7), the output ratio and shaft position increase with CW (or extending) direction of travel.




View of shaft and element
from specified mounting end

FIGURE A-7. Direction of travel.

x. Total mechanical travel. The total travel of the shaft between integral stops, under specified stop load. In potentiometers without stops, the mechanical travel is continuous.

y. Mechanical overtravel. The shaft travel between each theoretical end point and its adjacent corresponding limit of total mechanical travel, (see figure A-8).

NOTE: The relationship of the electrical travels to each other and to the input terminals shown above is given for illustration only and may vary from one potentiometer to another.



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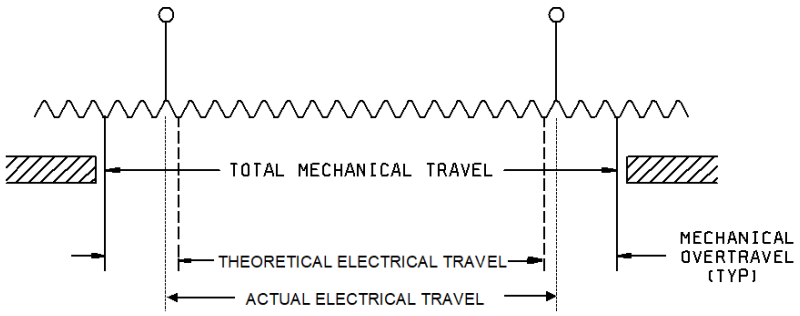



FIGURE A-8. Mechanical overtravel.

z. Backlash. the maximum difference in shaft position that occurs when the shaft is moved to the same actual output ratio point from opposite directions. (see figure A-9)

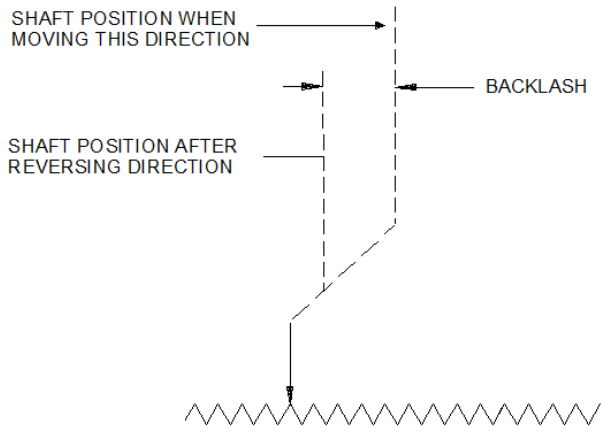



FIGURE A-9. Backlash.

aa. Theoretical end point. The shaft positions corresponding to the end of the theoretical electrical travel as determined from the index point.

ab. Index point. A point of reference fixing the relationship between a specified shaft position and the output ratio. It is used to establish a shaft position reference, (see figure A-10).



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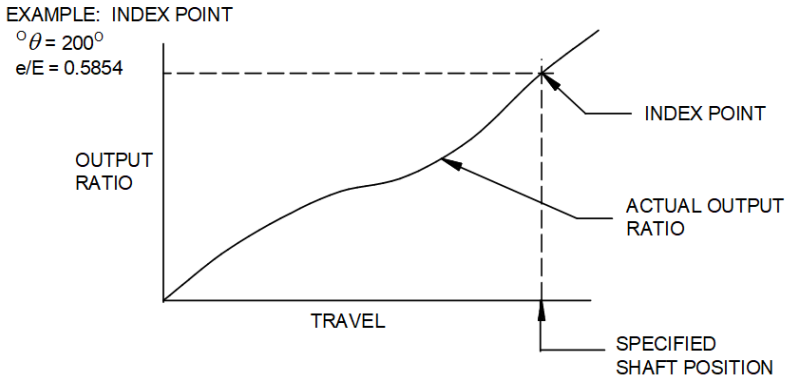



FIGURE A-10. Index point.

ac. Theoretical electrical travel. The specified shaft travel over which the theoretical function characteristic extends between defined output ratio limits, as determined from the index point. (see figure A-11)

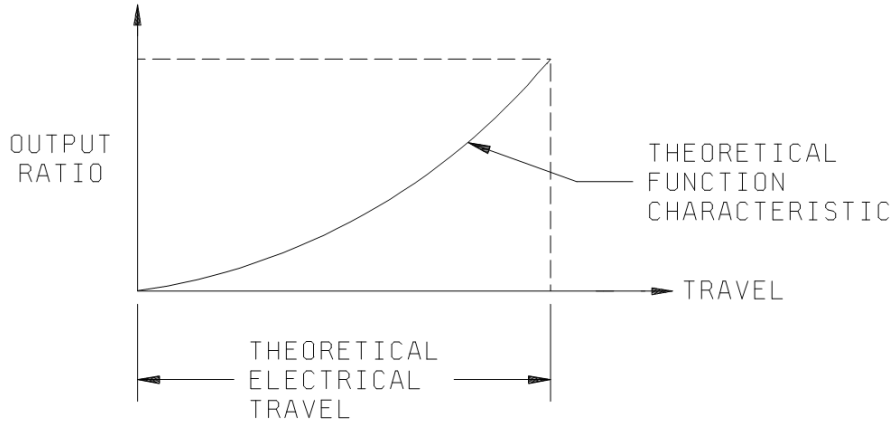



FIGURE A-11. Theoretical electrical travel.

ad. Electrical overtravel. The shaft travel over which there is continuity between the wiper terminal and the resistance element beyond each end of the theoretical electrical travel. (In cases where absolute linearity or absolute conformity is specified, theoretical electrical travel should be substituted for actual electrical travel in this definition.)

ae. Electrical continuity travel. The total travel of the shaft over which electrical continuity is maintained between the wiper and the resistance element. (see figure A-12)



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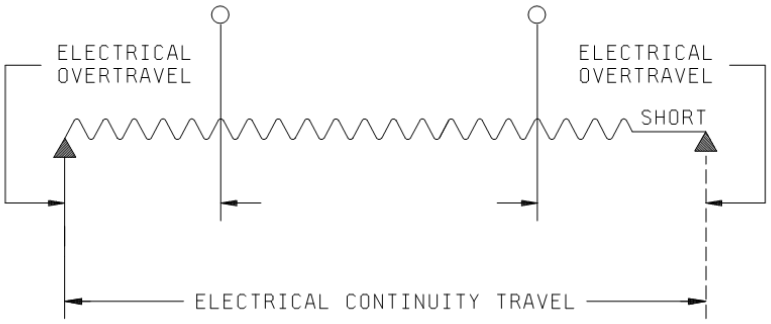



FIGURE A-12. Electrical continuity travel.

af. Tap location. The position of a tap relative to some reference. This is commonly expressed in terms of an output ratio and a shaft position. When a shaft position is specified, the tap location is the center of the effective tap width.

ag. Effective tap width. The travel of the shaft during which the voltage at the wiper terminal and the tap terminal are the same, as the wiper is moved past the tap in one direction. (see figure A-13)

NOTE: In some instances, the tap width may be essentially zero (i.e., no flat zone), but the tap may have a significant effect on conformity. In these cases the term “effective tap width” should not be applied. Instead, the effect of the tap on the output characteristics should be considered in terms of conformity.

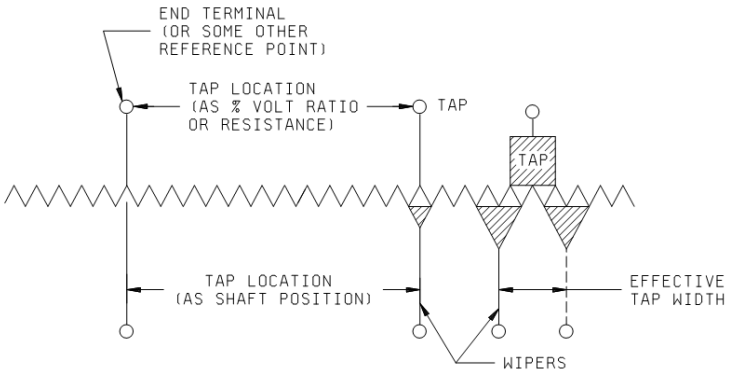



FIGURE A-13. Effective tap width.

ah. Phasing. The relative alignment of the phasing points of each cup of a gang potentiometer.

NOTE: Unless otherwise specified, phasing requirements apply to a single specified phasing point in each cup, and all cups are aligned to the phasing point of the first cup.

ai. Total resistance (dc input impedance). The dc resistance between the input terminals with the shaft positioned so as to give a maximum resistance value.



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aj. DC output impedance. The maximum dc resistance between the wiper and either end terminal with the input shorted.

ak. Minimum resistance. Refer to tap resistance or minimum voltage for applicable definition

al. End resistance. Refer to end voltage for applicable definition.

am. Tap resistance. The minimum resistance obtainable between a tap terminal and a wiper position on the resistance element, measured without drawing wiper current.

an. Temperature coefficient of resistance. The temperature coefficient of resistance is the unit change in resistance per degree centigrade change from a reference temperature and expressed in parts per million per degree centigrade as follows:

$$TC = \frac{R_2 - R_1}{R_1(T_2 - T_1)} \times 10^6$$

Where:

$R_1$ = Resistance at reference temperature in ohms.

$R_2$ = Resistance at test temperature in ohms

$T_1$ = Reference temperature in degrees centigrade ($^{O}C$)

$T_2$ = Test temperature in degrees centigrade ($^{O}C$)

ao. Apparent contact resistance. Refer to output smoothness.

ap. Equivalent noise resistance. Refer to output smoothness.

aq. Function characteristic. The relationship between the output ratio and the shaft position.

$$\text{MATHEMATICALLY: } \frac{e}{E} = f(\theta)$$

ar. Conformity. The fidelity of the relationship between the actual function characteristic and the theoretical function characteristic. (see figure A-14)

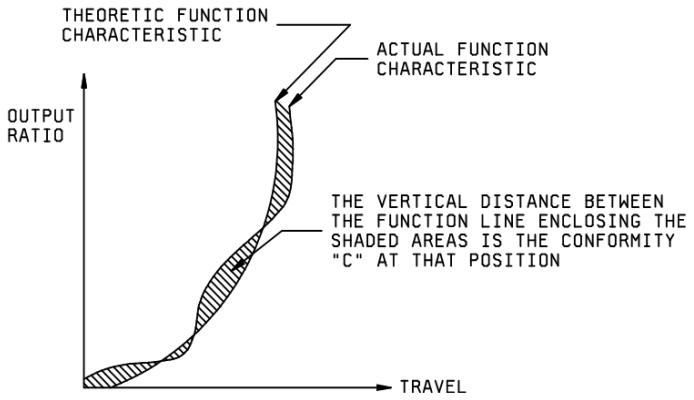



FIGURE A-14. Conformity



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as. Absolute conformity. The maximum deviation of the actual function characteristic from a fully defined theoretical function characteristic. It is expressed as a percentage of the total applied voltage and measured over the theoretical electrical travel. An index point on the actual output is required (see figure A-15).

MATHEMATICALLY: $\frac{e}{E} f\left[\frac{\theta}{\theta_T}\right] \pm C; 0 \le \theta \le \theta_T$

NOTE: The theoretical function characteristic is assumed to be a smooth curve when it can be described by a mathematical expression. When empirical data are provided, the points are assumed to be joined by straight line segments.

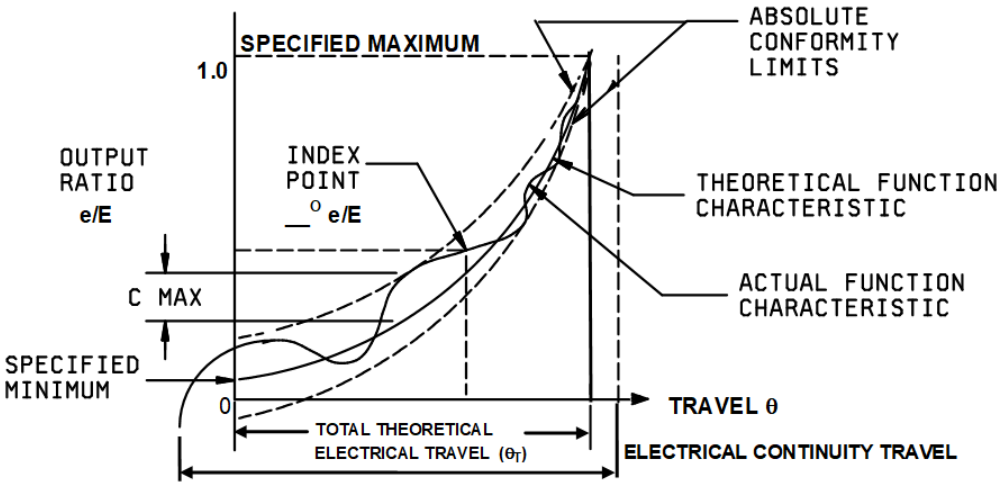



FIGURE A-15. Absolute conformity.

at. Linearity. A specific type of conformity where the theoretical function characteristic is a straight line. (see figure A-16).

MATHEMATICALLY: $\frac{e}{E} = f(\theta) \pm C = A(\theta) + B \pm C$

Where:

A = given slope

B = given intercept at $\theta$ = 0



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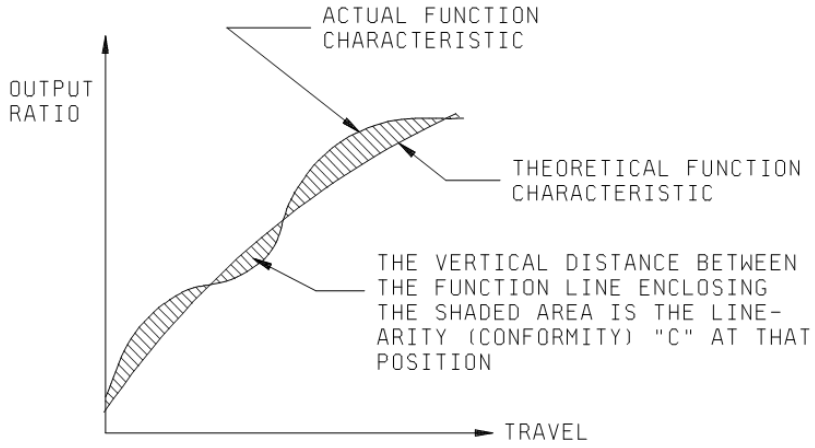



FIGURE A-16. Linearity

au. Absolute linearity. The maximum deviation of the actual function characteristic from a fully defined straight reference line. It is expressed as a percentage of the total applied voltage and measured over the theoretical electrical travel. An index point on the actual output is required. The straight reference line maybe fully defined by specifying the low and high theoretical end output ratios separated by the theoretical electrical travel. Unless otherwise specified, these end output ratios are 0.0 and 1.0, respectively (see figure A-17).

$$\text{MATHEMATICALLY: } \frac{e}{E} = A\left[\frac{\theta}{\theta_T}\right] + B \pm C$$

Where:

A = given slope

B = given intercept at $\theta = 0$.

Unless otherwise specified: A = 1; B = 0.

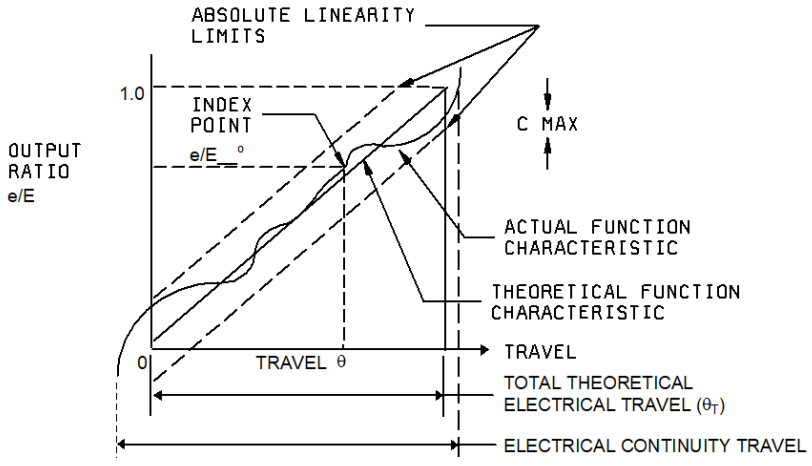



FIGURE A-17. Absolute linearity.



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av. Terminal based linearity. Terminal based linearity is the maximum deviation, expressed as a percent of the applied voltage, of the actual function characteristic from a straight reference line drawn through its specified minimum and maximum output ratios which are separated by the actual electrical travel. Unless otherwise specified, minimum and maximum output ratios are respectively zero and 100 percent of the total applied voltage (see figure A-18).

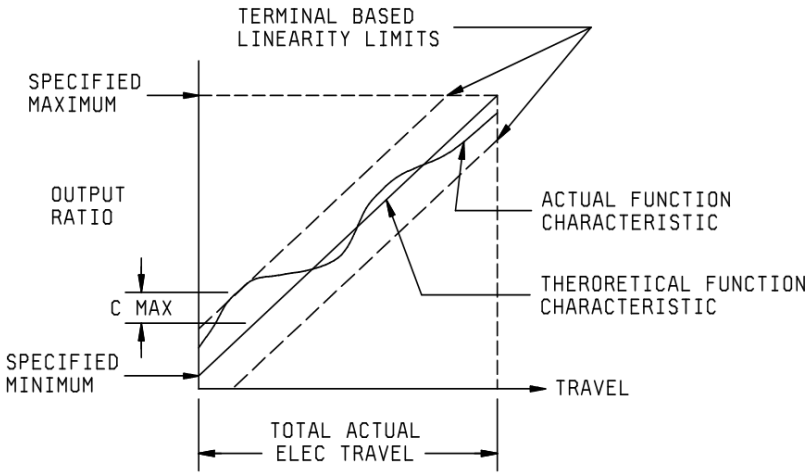



FIGURE A-18. Terminal based linearity.

aw. Zero based linearity. Zero based linearity is the maximum deviation expressed as a percentage of the total applied voltage of the actual function characteristic from a straight reference line drawn through the specified minimum output ratio, extended over the actual electrical travel and rotated to minimize the maximum deviations. Any specified end voltage requirements limits the rotation of the reference line. Unless otherwise specified, the specified minimum output will be zero (see figure A-19).

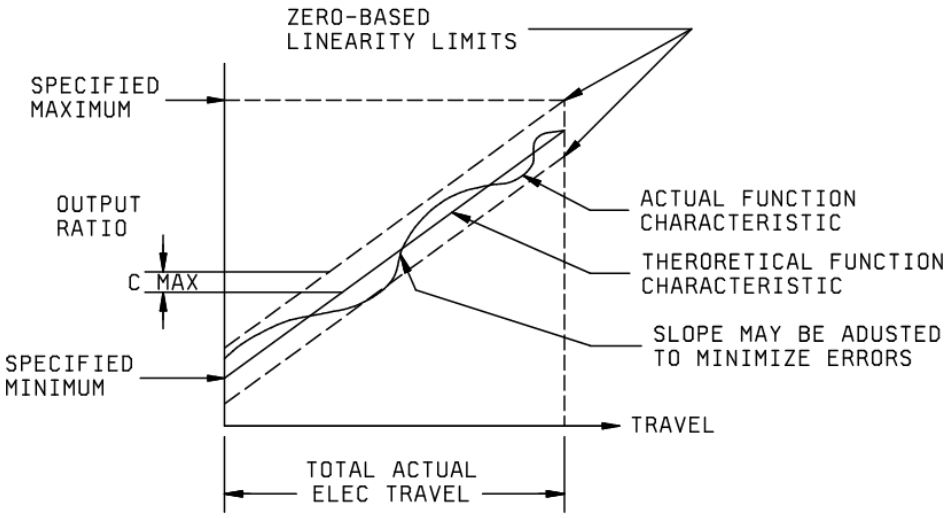



FIGURE A-19. Zero based linearity



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ax. Independent linearity. The maximum deviation of the actual function characteristics from a straight reference line with its slope and position chosen to minimize the maximum deviations. It is expressed as a percentage of the total applied voltage and is measured over the specified theoretical electrical travel. The slope of the reference line, if limited, must be separately specified. An index point on the actual output is required. Unless otherwise specified, the index point will be at $\theta$ = 180 degrees (see figure A-20).

MATHEMATICALLY: $\frac{e}{E} = P\left[\frac{\theta}{\theta_T}\right] + Q \pm C$

Where:

P = unspecified slope;

Q = unspecified intercept at $\theta$ = 0.

And both are chosen to minimize C but are limited by the end voltage requirements.

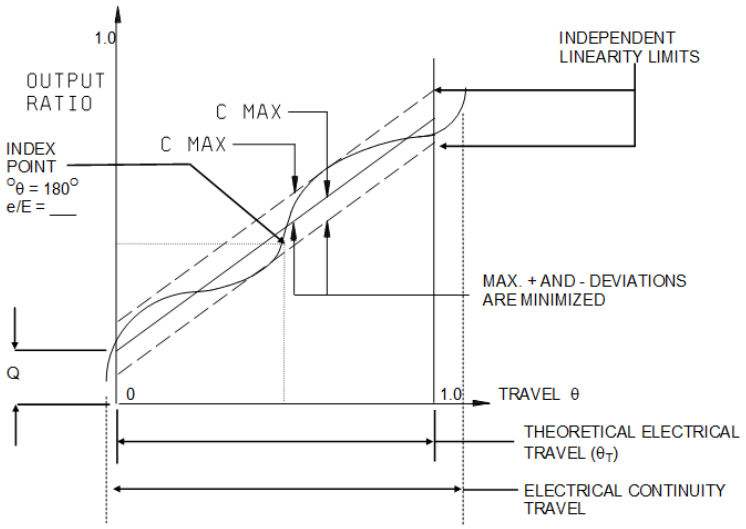



FIGURE A-20. Independent linearity.

ay. Constant limits. Permissible conformity deviations specified as a percentage of the total applied voltage (see figure A-21).

NOTE: Unless otherwise specified, all definitions in this document employ constant limits.

az. Zero-to-peak constant limits. Permissible conformity deviations specified as a percentage of zero-to-peak applied voltage (see figure A-21).

NOTE: The numerical value of zero-to-peak errors is double that of equal peak-to-peak errors, because the reference zero-to-peak applied voltage is one-half of the total (peak-to-peak) applied voltage.

ba. Proportional limits. Permissible conformity deviations specified as a percentage of the theoretical output ratio at the point of measurement (see figure A-21).

NOTE: Proportional limits may become impossibly restrictive in the vicinity of zero theoretical output and should be modified to provide a practical tolerance in that region, if the theoretical output ratio approaches zero.



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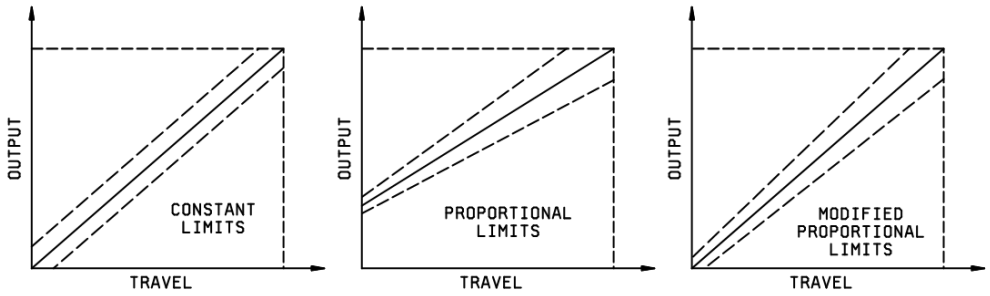



FIGURE A-21. Tolerance limits.

bb. Modified proportional limits. Any combination of constant and proportional limits (see figure A-21).

bc. Phasing point. The phasing point is a point of reference on each electrical element used to describe the relative alignment of the cups and for electrical elements of a gang with respect to the portion of the wipers on their relative electrical elements.

bd. Phasing. Phasing is the relative alignment of the cups and for electrical elements of a gang with respect to the position of the wipers on their respective electrical elements (see figure A-22).

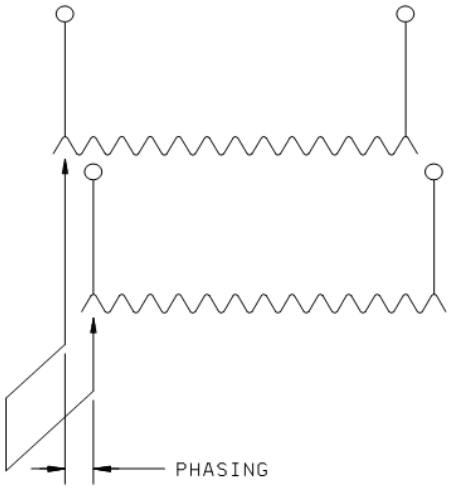



FIGURE A-22. Phasing.

be. Simultaneous conformity phasing. The relative alignment of the cups of a gang potentiometer, from a common index point, such that the output ratios of all cups fall within their respective conformity limits over the theoretical electrical travel.

bf. Voltage tracking error. The difference, at any shaft position, between the output ratios of any two commonly actuated similar electrical elements, expressed as a percentage of the single total voltage applied to them.



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bg. Output smoothness. Output smoothness is a measurement of any spurious variation in the electrical output not present in the input. It is expressed as a percentage of the total applied voltage and measured for specified travel increments over the theoretical electrical travel. Output smoothness includes effects of contact resistance variations, resolution, and other micro-nonlinearities in the output.

bh Noise. Noise is any spurious variation in the electrical output not present in the input, defined quantitatively in terms of an equivalent parasitic, transient resistance in ohms, appearing between the contact and the resistance element when the shaft is rotated or translated. The equivalent noise resistance is defined independently of the resolution, the functional characteristics, and the total travel. The magnitude of the equivalent noise resistance is the maximum departure from a specified reference line. The wiper of the potentiometer is required to be excited by a specified current and moved at a specified speed.

bi. Resolution. A measure of the sensitivity to which the output ratio of the potentiometer may be set.

bj Theoretical resolution. Theoretical resolution is used in wirewound linear potentiometers only. It is the reciprocal of the number of turns of wire in resistance winding in the actual electrical travel and is expressed as a percentage.

$$\frac{1}{N} \text{ x100} = \text{Theoretical resolution percent}$$

Where:
N = Total number of resistance wire turns

bk Travel resolution. Travel resolution is the maximum value of shaft travel in one direction per increased voltage step in any specified portion of the resistance element.

bl. Voltage resolution. The maximum incremental change in output ratio with shaft travel in one direction in any specified portion of the resistance element. (see figure A-23)

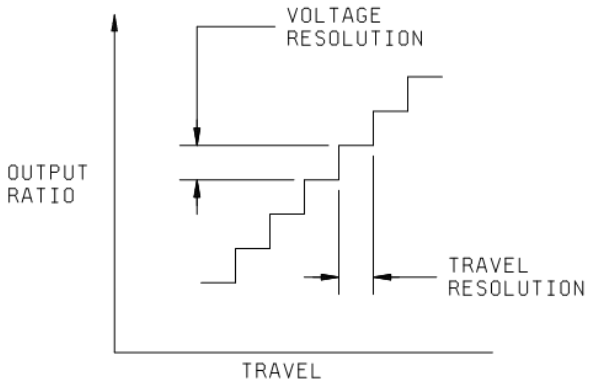



FIGURE A-23. Voltage Resolution.

bm. End voltage. End voltage is the voltage between the wiper terminal and an end terminal when the shaft is positioned at the corresponding end point. It is usually expressed as a percentage of the total applied voltage.

bn Jumpoff voltage. The jumpoff voltage is the magnitude of the first measurable voltage change as the wiper moves from the overtravel region onto the actual electrical travel region. It is usually expressed as a percentage of the total applied voltage (see figure A-24).



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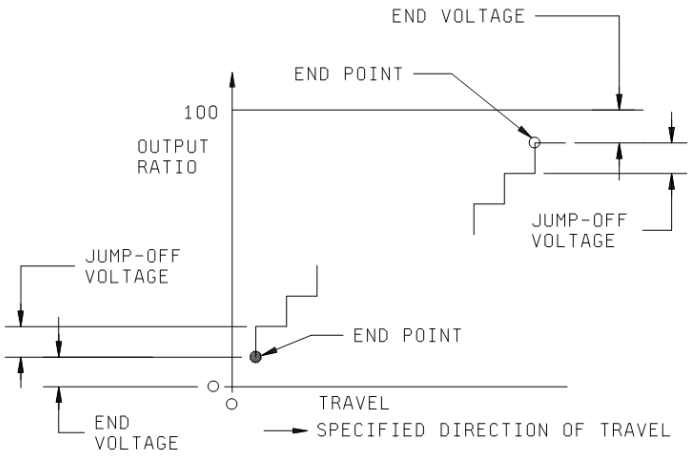



FIGURE A-24. Jumpoff voltage.

bo. Dielectric strength. Ability to withstand under prescribed conditions, a specified potential of a given characteristic between the terminals of each cup and the exposed conducting surfaces of the potentiometer, or between the terminals of each cup and the terminals of every other cup in the gang without exceeding a specified leakage current value.

bp. Insulation resistance. The resistance to a specified impressed dc voltage between the terminals of each cup and the exposed conducting surfaces of the potentiometer, or between the terminals of each cup and the terminals of every other cup in the gang, under prescribed conditions.

bq. Power rating. The maximum power that a potentiometer can dissipate under specified conditions while meeting specified performance requirements.

br. Power derating. The modification of the nominal power rating for various considerations such as load resistance, output slopes, ganging, nonstandard environmental conditions, and other factors.

bs. Life. The number of shaft revolutions or translations obtainable under specific operating conditions and within specified allowable degradation of specific characteristics.

bt. Total input impedance. The impedance between the two input terminals with open circuit between output terminals, and measured at a specified voltage and frequency with the shaft positioned to give a maximum value (see figure A-25).

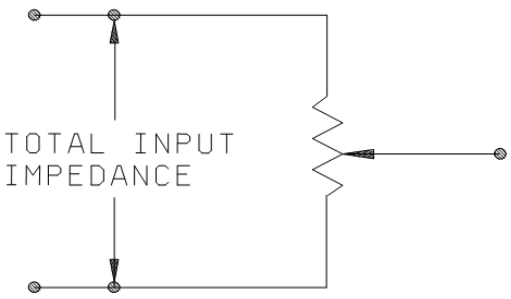



FIGURE A-25. Total input impedance.



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bu. Output impedance. Maximum impedance between slider and either end terminal with the input shorted, and measured at a specified voltage and frequency (see figure A-26).

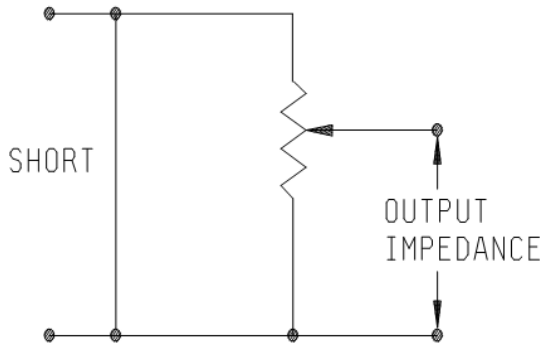



FIGURE A-26. Output impedance.

bv. Quadrature voltage. The maximum value of that portion of the output voltage which is $\pm 90^{O}$ out of time phase with the input voltage, expressed as volts-per-volt applied, measured at a specified input voltage and frequency.

bw. Phase shift. The phase difference, expressed in degrees, between the sinusoidal input and output voltages measured at a specified input voltage and frequency with the shaft at a specified position.

MATHEMATICALLY: $\Phi = \text{SIN}^{-1}\left[\frac{e_q}{e}\right] = \text{TAN}^{-1}\left[\frac{e_q}{e_1}\right]$

Where:

| | | |
|---|---|---|
| $\phi$ | = | Phase shift in degrees |
| $e_q$ | = | Quadrature voltage |
| $e_1$ | = | Inphase output voltage |
| e | = | Output voltage |

bx. Shaft runout. The eccentricity of the shaft diameter with respect to the rotational axis of the shaft, measured at a specified distance from the end of the shaft. The body of the potentiometer is held fixed, and the shaft is rotated with a specified load applied radially to the shaft. The eccentricity is expressed in inches, TIR.

by. Lateral runout. The perpendicularly of the mounting surface with respect to the rotational axis of the shaft, measured on the mounting surface at a specified distance from the outside edge of the mounting surface. The shaft is held fixed, and the body of the potentiometer is rotated with specified loads applied radially and axially to the body of the pot. The lateral runout is expressed in inches, TIR.

bz. Pilot diameter runout. The eccentricity of the pilot diameter with respect to the rotational axis of the shaft, measured on the pilot diameter. The shaft is held fixed, and the body of the potentiometer is rotated with a specified load applied radially to the body of the pot. The eccentricity is expressed in inches, TIR.

ca. Shaft radial play. The total radial excursion of the shaft, measured at a specified distance from the front surface of the unit. A specified radial load is applied alternately in opposite directions at a specified point. Shaft radial play is expressed in inches.

cb. Shaft end play. The total axial excursion of the shelf, measured at the end of the shaft with a specified axial load supplied alternately in opposite directions. Shaft end play is expressed in inches.



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cc. Starting torque. The maximum moment in the clockwise and counterclockwise directions required to initiate shaft rotation anywhere in the total mechanical travel.

cd. Running torque. The maximum moment in the clockwise and counterclockwise directions required to sustain uniform shaft rotation at a specified speed throughout the total mechanical travel.

ce. Moment of inertia. The mass moment of inertia of the rotating elements of the potentiometer about their rotational axis.

cf. Static stop strength. The maximum static load that can be applied to the shaft at each mechanical stop for a specified period of time without permanent change of the stop positions greater than specified.

cg. Dynamic stop strength. The inertia load, at a specified shaft velocity and a specified number of impacts, that can be applied to the shaft at each stop without a permanent change of the stop position greater than specified.

A.3.1.7 Special Resistors. See table A-VI.

A.3.1.7.1 MIL-PRF-29, MF, Resistor, Fixed, Meter Multiplier, External, High Voltage, Ferrule Terminal Type. These resistors are used for high-voltage, external, meter-multiplier, fixed resistors of the ferrule-terminal type for use with direct-current (dc) instruments drawing 1 milliampere at full-scale deflection.

A.3.1.7.2 MIL-PRF-23648, RTH, Resistor, Thermal (Thermistor) Insulated. These resistors exhibit a rapid change in resistance for a relative small temperature change. These resistors are used to measure temperature or to compensate for changes in temperature.

A.3.1.7.3 MIL-PRF-32192, RCTP and RCTN, Resistor Chip, Thermal. These thermistors exhibit a positive temperature coefficient (PTC) or a negative temperature coefficient (NTC) and are primarily intended for incorporation into surface mount applications. These devices are to be used for temperature control, temperature compensation, sensing, and frequency compensation over the temperature range specified.

TABLE A-VI. Special resistors selection guidance table.

| MIL-PRF-29 - Resistor, Fixed, Meter Multiplier, External, High Voltage, Ferrule Terminal Type | | | |
|---|---|---|---|
| Styles available | Resistance range (in megohms) | Voltage ratings | Maximum body size (inches) |
| MFA | 3.5, 4.0, 5.0, 6.0 | 3.5, 4.0, 5.0, 6.0 | 9.781 x 1.312 |
| MFB | 1.0, 1.5, 2.0, 2.5, 3.0, 3.5 | 1.0, 1.5, 2.0, 2.5, 3.0, 3.5 | 5.281 x 1.312 |
| MFC | 0.5, 0.8, 1.0 | 0.5, 0.8, 1.0 | 2.937 x 1.000 |
| MFD | 10 | 10 | 16.500 x 1.312 |
| MFE | 15 | 15 | 23.500 x 1.312 |
| MFF | 20 | 20 | 30.500 x 1.312 |

| MIL-PRF-23648 - Resistor, Thermal (Thermistor) Insulated | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Specification sheet MIL-PRF- | Styles available | Power rating (watts) | Thermal time constant (seconds) | Dissipation Constant (mW/°C) | Resistance tolerance (± percent) (see A.1.1.2) | Resistance Ratio 1/ | Resistance range (ohms) Min | Resistance range (ohms) max | Maximum body size (inches) and Configuration (see figure A-2) | |
| 23648/1 | RTH06 | 0.05 | 80 | 5 | F, G, J, K | A, B, C | 68 | 75k | 0.30 D. | M |
| 23648/9 | RTH22 | 0.5 | 60 | 15 | J, K | E | 10 | 39k | 0.43 x 0.16 | A |
| 23648/19 | RTH42 | 0.25 | 60 | 2.5 | | | | 10k | 0.30 x 0.11 | |
| 23648/20 | RTH44 | 0.2 | 25 | 2.0 | F, G, J, K | A, B, C | 300 | 500k | 0.25 x 0.13 x 0.135 | M |

1/ A - 19.8
B - 29.4
C - 48.7
E - 0.55



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TABLE A-VI. Special resistors selection guidance table. Continued

| MIL-PRF-32192 - Resistor, Chip, Thermal (Thermistor) | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Specification sheet MIL-PRF- | Styles available | Power rating (watts) | Thermal time constant (seconds) | Dissipation Constant (mW/$^{O}$C) | Resistance tolerance (± percent) (see A.1.1.2) | Resistance Ratio | Resistance range (ohms) Min | Resistance range (ohms) max | Maximum body size (inches) and Configuration (see figure A-2) | |
| 32192/1 | RCTP0303 | 0.125 | 30 | 1.25 | F, G, J, K | 0.53 | 10 | 10K | 0.048 x 0.048 x .082 | T |
| 32192/3 | RCTN0404 | 0.0625 | 10 | 0.625 | | Variable* | 15* | 20M* | 0.050 x 0.050 x .020 | |
| 32192/4 | RCTN0805 | 0.125 | 8 | 2.0 | | | 47* | 10M* | 0.086 x 0.057 x 0.055 | |
| 32192/5 | RCTN1206 | 0.250 | 8 | 2.0 | | | | | 0.134 x 0.068 x 0.055 | |

* Resistance ration is based on resistance range, see specification sheet for ranges.

A.3.1.7.4 Cancelled DoD Specifications. Table A-VII contains a listing of cancelled DoD specifications.

TABLE A-VII. Cancelled DoD specifications.

| Specification number | Title |
|---|---|
| MIL-DTL-7790 | Resistor, Thermocouple Lead Spool |
| | |
| MIL-PRF-122 | Resistors, Fixed, Precision, Established Reliability, General Specification for |
| MIL-PRF-23285 | Resistor, Variable, Nonwirewound, General Specification for |
| MIL-PRF-39002 | Resistors, Variable, Wirewound, Semi-Precision, General Specification for |
| | |
| MIL-R-11 | Resistors, Fixed, Composition (insulated), General Specification for |
| MIL-R-3080 | Resistors, Current-Regulating (Ballast Tubes), General Specification for |
| MIL-R-39008 | Resistors, Fixed, Composition (Insulated) Established Reliability, General Specification for |
| MIL-R-83579 | Resistors, Fixed, Wirewound (Power Type), High Reliability, General Specification for |
| MIL-R-83580 | Resistor, Fixed, Film, Chip, Established Reliability, General Specification for |
| MIL-R-87254 | Resistor, Fixed, Film, High Reliability, General Specification for |
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A.3.2 Mounting guide.

A.3.2.1 Stress mounting. Improper heat dissipation is the predominant contributing cause of failure for any resistor type; consequently, the lowest possible resistor surface temperature should be maintained. Figure A-27 illustrates the manner in which heat is dissipated from fixed resistors in free air. The intensity of radiated heat varies inversely with the square of the distance from the resistor. Maintaining maximum distance between heat generating components serves to reduces cross radiation heating effects and promotes better convection by increasing air flow. For optimum cooling without a heat sink, small resistors should have large diameter leads of minimum length terminating in tie points of sufficient mass to act as heat sinks. All resistors have a maximum surface temperature which should never be exceed. Any temperature beyond maximum can cause the resistor to malfunction. Resistors should be mounted so that there are no abnormal hot spots on the resistor surface. When mounted, resistor should not come in contact with heat insulating surfaces.

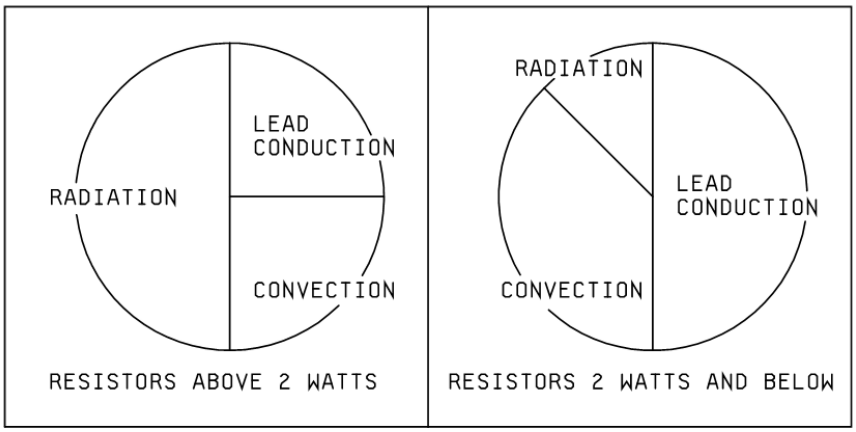



FIGURE A-27. Heat dissipation of resistors under room conditions.

A.3.2.2 Resistor mounting for vibration. Resistors should be mounted so resonance does not occur within the frequency spectrum of the vibration environment to which the resistors may be subjected. Some of the most common resistor packaging methods result in large resistor noise. Resistor mounting for vibration should provide (1) the least tension or compression between the lead and body, (2) the least excitation of the resistor in relation with any other surface, and (3) no bending or distortion of the resistor body.

A.3.2.3 Circuit packaging. Resistors that are crowed together and come into contact with each other can provide leakage paths (even well insulated parts) for external current passage. This can change the resultant resistance in the circuit. Moisture traps and dirt traps are easily formed by crowding. Moisture and dirt eventually form corrosive materials which can deteriorate the resistors and other electronic parts. Moisture can accumulate around dirt even in an atmosphere of normal humidity. Planning should be done to eliminate crowding of parts. Proper space utilization of electronic parts can reduce the package size and still provide adequate spacing of parts.



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A.3.2.4 Summary. The following is a guide for resistor mounting:

a. Maintain lead length to a minimum. The mass of the point acts as a heat sink. (NOTE: Where low temperatures are present, leads should be offset (bent slightly) to allow for thermal contraction).
b. Close tolerance and low value resistors require special precautions (such as short leads and good soldering techniques) since the resistance of the leads and the wiring may be as much as several percent of the resistance of the resistor.
c. Maintain maximum spacing between resistors.
d. For resistors mounted in series, consider the heat being conducted through the leads to the next resistor.
e. Large power units should be mounted to the chassis.
f. Do not mount high power units directly on terminal boards or printed circuits.
g. To provide for the most efficient operation and even heat distribution, power resistors should be mounted in a horizontal position.
h. Select mounting materials that will not char and can withstand strain due to expansion.
i. Consider proximity to other heat sources as well as self-heat.
j. Consider levels of shock and vibration to be encountered. Where large body mass is present, the body should be restrained from movement.

A.3.3 Effects of circuit usage. Resistors must be selected to be compatible with the conditions to which they are exposed. Numerous matters must be considered in this selection process. The most important are noted in the following.

A.3.3.1 Resistance value. This is initially determined by the circuit requirements, and may seem a trivial thing to mention. However, most resistor calculations that are made without reference to available resistors come out to a resistance value that are not standard. The design engineer should be aware of the standard resistance values that are available from manufacturers who adhere to this handbook and various Department of Defense specifications for resistors. These differ somewhat with the various types of resistors. It is a usually a fairly simple thing to bring the exact calculated value in line with a standard value. In the case where this cannot be done, a parallel or series combination of resistors can usually be used. The design engineer should also remember that the resistance value of the resistor that gets into the physical circuit will differ from the value that is stated on the circuit schematic, and that this difference will change as time goes by. The purchase tolerance of the resistor to be used will allow it to differ from the nominal stated value, depending on the type of resistor specified. Furthermore, the temperature at which the resistor works, the voltage across it, and the environment in which it lives will affects the actual value at particular times. For example, the designer should allow for a possible variation of $\pm$15 percent from the nominal value of a purchased $\pm$5 percent composition resistor, for the circuit to continue to operate satisfactorily over a very long time under moderate ambient conditions. Such a figure is a rule of thumb, based on many tests, and many resistors will remain much nearer their starting value; but if many are used, chance will ensure that some will go near this limit. A similar figure can be deduced from each variety of resistor used.



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A.3.3.1.1 Summary.

a. Select a resistor for each circuit application from the lists of standard types and values.
b. Be sure that the circuit being designed will work with any resistor whose resistance value is within the limits set by tolerance plus voltage coefficient plus temperature coefficient plus drift with time. Failure to take these precautions can possibly mean that in equipment produced in quantity for the armed services, there may be some circuits that will not work under extreme conditions.
c. Various initial tolerances are available depending on the type of resistor. It should be remembered that initial accuracies become meaningless if the inherent stability of the resistor does not support the initial accuracy.
d. During shelf life, as well as during operational life, any characteristic (such as resistance, power rating, dielectric strength, or size) of any part may change value due to stresses caused by environmental changes of temperature , humidity, pressure or vibration. Changes of characteristic caused by environmental stresses may be linear or nonlinear, reversible or nonreversible (permanent), or combinations thereof. Where a characteristic of the part undergoes a linear change during environmental stress, and the change reverses itself linearly when the environmental stress is removed so that characteristic returns to its normal value, this rate of change in characteristic value (per unit change in stress value) is designated (x) coefficient, and is usually expressed in percent or ppm/$^{O}$C.

A.3.3.2 Power rating. The minimum required power rating of a resistor is another factor that is initially set by the circuit usage, but is markedly affected by other conditions of use. As mentioned previously, the power rating is based on the hot spot temperature the resistor will withstand, while still meeting its other requirements of resistance variation, accuracy, and life. For derating information see A.3.3.2.2.

A.3.3.2.1 Self-generated heat. Self-generated-heat in a resistor is, of course, calculated as $P = I^2R$. This figure, in any circuit, must be less than the actual power rating of the resistor used. It is the usual practice to calculate this value and to use the next larger power rating available in the handbook. This calculation should, however, be considered only as a first approximation of the actual rating to be used.

A.3.3.2.2 Rating versus ambient conditions. The power rating of a resistor is based on a certain temperature rise from an ambient temperature of a certain value. If the ambient temperature is greater than this value, the amount of heat that the resistor can dissipate is correspondingly reduced, and therefore it must be derated because of temperature. The applicable section of this handbook and all of the Department of Defense specifications contain derating curves to be applied to the resistors covered.

A.3.3.2.3 Rating versus accuracy. Because of temperature coefficient of resistance that all resistors possess, a resistor which is expected to remain near its measured value under conditions of operation must remain cool. For this reason, all resistors designated as “accurate” are very much larger physically for a certain power rating than are ordinary “nonaccurate” resistors. In general, any resistor, “accurate” or not, must be derated to remain very near its original measured value when it is being operated.

A.3.3.2.4 Rating versus life. If especially long life is required of a resistor, particularly when “life” means remaining within a certain limit of resistance drift, it is usually necessary to derate the resistor, even if ambient conditions are moderate and if accuracy by itself is not important. A good rule to follow when choosing a resistor size for equipment that must operate for many thousands of hours is to derate it to one half of its nominal power rating. Thus, if the self-generated heat in the resistor is one third watt, do not use a one half watt resistor, but rather 1 watt size. This will automatically keep the resistor cooler, will reduce the long term drift, and will reduce the effect of the temperature coefficient. In equipment that need not live so long and must be small in size, this rule may be impractical, and the engineer should adjust the dependence on the rules to the circumstances at hand. A “cool” resistor will generally last longer than a “hot” one, and can absorb transient overloads that might permanently damage a “hot” resistor.



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A.3.3.2.5 Rating under pulsed conditions and under intermittent loads. When a resistor is used in circuits where power is drawn intermittently or in pulses, the actual power dissipated with safety during the pulses can sometimes be much more than the maximum rating of the resistor. For short pulses, the actual heating is determined by the duty factor and the peak power dissipated. Before approving such a resistor application, however, the engineer should be sure (1) that the maximum voltage applied to the resistor during the pulses is never greater than the permissible maximum voltage for the resistor be used, (2) that the circuit cannot fail in such a way that continuous excessive power can be through the resistor and cause it to fail also, (3) that the average power being drawn is well within the agreed on rating of the resistor, and (4) that continuous steep wavefronts applied to the resistor do not cause any unexpected troubles.

A.3.3.3 High frequency. For most resistors, the lower the resistance value, the less total impedance it exhibits at high frequency. Resistors are not generally tested for total impedance at frequencies above 120 Hz. Therefore, this characteristic is not controlled. The dominating conditions for good high frequency resistor performance are geometric considerations and minimum dielectric losses. For the best high frequency performance, the ratio of resistor length to the cross sectional area should be a maximum. Dielectric losses are kept low by proper choice of the resistor base material, and when dielectric binders are used, their total mass is kept to a minimum. The following is a discussion of the high frequency merits of these major resistor types:

a. Film type. Film type resistors have the best high frequency performance. The effective dc resistance for most resistance values remains fairly constant up to 100 MHz and decreases at higher frequencies. In general, the higher the resistance value the greater the effect of frequency.
b. Wirewound. Wirewound resistors have inductive and capacitive effects and are unsuited for use above 50kHz, even when specially wound to reduce the inductance and capacitance. Wirewound resistors usually exhibit an increase in resistance with high frequencies because of “skin” effect.

A.3.4 Effects of mechanical design and ambient conditions. Since the operation of a circuit cannot be divorced from the physical configuration it assumes when assembled, some of the points that apply herein have already been discussed. It is well, however, to check this aspect of equipment design several times, so redundancies in the following paragraphs are deliberate for the sake of emphasis.

A.3.4.1 Mechanical design of resistors. Much trouble during the life of the equipment can be eliminated if the design engineer can be sure that the resistors he is specifying for his circuits are soundly constructed and proper equipment assembly techniques are utilized. The resistor types listed in this handbook provide a great measure of this assurance and, in general, assure a uniform quality of workmanship. The areas defined in A.3.4.1.1 through A.3.4.1.7 are included as indicators of sound product construction.

A.3.4.1.1 End caps or terminations. The connection between the resistor element itself and the pigtails or leads that connect it into the circuit must be so good that no possible combination of conditions met in the proposed service can cause an intermittent connection. The Department of Defense specifications cover this point, and provide tests to check for it. When resistors are handled in automatic assembly machines, this precaution is particular important.

A.3.4.1.2 Effect of soldering. There are assembly techniques that affect resistor reliability. Resistors should never be overheated by excessive soldering iron applications, and the resistor leads should not be abraded by assembly tools. No normal soldering practice, either manual or dip soldering, should damage the resistor physically or change its resistance value appreciably.



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A.3.4.1.3 Moisture resistance. Moisture is the greatest enemy of components and electronic equipment. Usually a resistor will keep itself dry because of its own self-generated heat; this is, of course, only true when the equipment is turned on. If the equipment must stand for long periods under humid conditions without power applied, the engineer should determine whether his circuits will operate with resistance values which have changed from the "hot" condition, and whether the retrace of the resistance value during the warm up period will allow the equipment to work satisfactorily during this period. If it will not, the resistor must be adequately protected against moisture absorption. The resistor cannot be blamed for performing improperly if it is not designed for the use to which it is put. It is therefore up to the design engineer to analyze what is needed and to provide the resistor to meet these conditions. This handbook and the applicable Department of Defense specifications constitute a guide as to what various kinds of resistors will do under humid conditions.

A.3.4.1.4 Method of mounting. Large resistors that are not provided with adequate means of mounting should not be considered, under conditions of vibration or shock. Lead failure can occur, and the larger the mass supported by the leads the more probable a failure will be. Even when vibration or shock will not be a serious problem, ease of assembly and replaceability suggest that large components be mounted individually.

A.3.4.1.5 Resistor body. The body of the resistor must be sufficiently strong to withstand any handling it is likely to get. The specifications call out, through workmanship and packaging requirements, that it be shown by the manufacturer that the product will not crack, chip, or break in transit, on the shelf, or in the normal assembly process.

A.3.4.1.6 Insulation or coating. All resistors intended for use in reliable electronic equipment must be protected by an insulating coating. Sometimes this is a molded phenolic case, epoxy coating, or ceramic or glass sleeves. Wirewound power resistors use various cement and vitreous enamel coatings to protect the windings, and to insulate and provide moisture barriers. Not all of the coatings and insulations applied to commercial resistors are satisfactory for extreme variations in ambient conditions; the various Department of Defense specifications include test used to qualify the various manufacturer's products thus providing a greater confidence in the coating used.

A.3.4.1.7 Pure tin. The use of pure tin, as an underplate or final finish, is prohibited both internally and externally. Tin content of resistor components and solder will not exceed 97 percent, by mass. Tin will be alloyed with a minimum of 3 percent lead, by mass.

A.3.4.2 Effects of ambient conditions. In the establishment of rating for resistors, the design engineer has implicitly considered the mechanical design of the equipment. This may not have been realized, but it is so because the ambient conditions in which the resistor must operate determine to a large degree the power rating and mechanical construction of the resistor if long life, or any life, under extreme conditions is desired.

A.3.4.2.1 Resistor heating. A very important question in the application of resistors is how hot will they get in service. In a piece of equipment the heat in a resistor comes from several sources; namely, (1) self-generated heat, and is the thing that can be easily calculated, and (2) the heat that the resistor receives from other resistors or other heat producing components in the same immediate neighborhood by radiation, and is not so easily calculated. The important thing to remember is under these conditions each resistor will be heated more than $I^2R$ would suggest; when much heat is produced, as in stacked Wirewound resistors, the design engineer would do well not to freeze his design until he has measured a typical assembly with power on to see just how hot the resistor gets. The same thing is true of the extra heating given the resistors by convection. This is another way of saying that high ambient temperature will reduce the actual power rating of the resistor by reducing permissible temperature rise, a point that has been made several times before. The equipment designer must realize also that the heat being produced by "hot" resistors can injure other components. This is a very important point to remember; capacitors, diodes, and other resistors usually do not fail immediately when overheated. The effect of too much heat is a deteriorating one, weakening the component until at a later date it will unexpectedly fail. It is very easy to put a "heat bomb" in a piece of equipment that will not go off in normal production testing but will do its duty. It is also very easy to eliminate such troubles by strict and thoughtful attention to the problem of heating. A few rules have been given for use as guides to protect against these factors. (See A.3.3.2)



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A.3.4.2.2 High altitude. With the exception of the dielectric withstanding voltage test at reduced barometric pressure, all tests in Department of Defense specifications referenced herein are performed at ambient atmospheric pressure. This fact should be considered when the use of these resistors for high altitude conditions is contemplated.

A.3.4.2.3 Flammability. It should be noted that Department of Defense specifications referenced herein contain no requirements concerning the flammability of the materials used in the construction of these resistors. Users should take this into consideration when a particular application involves this requirement.

A.4. SUPPLEMENTAL INFORMATION

A.4.1 Reliability. The established reliability specification provides for the establishment of a failure rate figure through the single parameter of load life only. Although, in most instances, the established reliability specification provides for more frequent moisture resistance, burn in, and other types of screening tests on a 100 percent basis, the failure rate (in percent per 1,000 hours) is based only on load life test results.

A.4.2 Metric equivalent. The metric equivalents (to the nearest 0.01 mm) which are provided in the individual sections are for general information only and are based upon 1 inch = 25.4 mm.

A.4.3 International standardization agreements. Certain provisions of the specifications referenced in this handbook are subjected on international standardization agreements. When amendment, revision, or cancellation of any of these specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

A.4.4 Cross reference. A cross reference of section number, Department of Defense specification numbers, associated specification numbers, and style numbers are included for reference (see table A-VIII).

A.4.5 Tin whisker growth. The use of alloys with tin content greater than 97 percent, by mass, may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions, on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead, by mass, have shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM-B545 (Standard Specification for Electrodeposited Coatings of Tin).



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APPENDIX A

TABLE A-VIII. Detailed specification by style number.

| DoD specification | Style |
|---|---|
| MIL-PRF-22 | |
| MIL-PRF-22/1 | RP06 |
| MIL-PRF-22/2 | RP07 |
| MIL-PRF-22/3 | RP10 |
| MIL-PRF-22/4 | RP11 |
| MIL-PRF-22/5 | RP15 |
| MIL-PRF-22/6 | RP16 |
| MIL-PRF-22/7 | RP20 |
| MIL-PRF-22/8 | RP25 |
| MIL-PRF-22/9 | RP30 |
| MIL-PRF-22/10 | RP35 |
| MIL-PRF-22/11 | RP40 |
| MIL-PRF-22/12 | RP45 |
| MIL-PRF-22/14 | RP55 |
| MIL-PRF-22/15 | RP05 |
| MIL-PRF-26 | |
| MIL-PRF-26/3 | RW29<br>RW31<br>RW33<br>RW35<br>RW37<br>RW38<br>RW47 |
| MIL-PRF-26/4 | RW56 |
| MIL-PRF-29 | |
| | MFA<br>MFB<br>MFC<br>MFD<br>MFE<br>MFF |
| MIL-PRF-94 | |
| MIL-PRF-94/2 | RV5 |
| MIL-PRF-94/3 | RV6 |
| MIL-PRF-94/4 | RV2 |
| MIL-PRF-94/5 | RV4 |
| MIL-PRF-94/6 | 2RV7 |
| MIL-PRF-94/7 | RV8 |
| MIL-PRF-914 | |
| MIL-PRF-914/3 | RNS030 |
| MIL-PRF-914/4 | RNS040 |
| MIL-PRF-914/5 | RNS050 |

| DoD specification | Style |
|---|---|
| MIL-PRF-12934 | |
| MIL-PRF-12934/1 | RR0900 |
| MIL-PRF-12934/2 | RR1100 |
| MIL-PRF-12934/4 | RR2000 |
| MIL-PRF-12934/5 | RR3000 |
| MIL-PRF-12934/6 | RR1000 |
| MIL-PRF-12934/9 | RR2100 |
| MIL-PRF-12934/10 | RR3100 |
| MIL-PRF-12934/15 | RR3200 |
| MIL-PRF-12934/16 | RR3300 |
| MIL-PRF-12934/17 | RR3400 |
| MIL-PRF-12934/18 | RR3500 |
| MIL-PRF-12934/19 | RR1300 |
| MIL-PRF-12934/20 | RR1400 |
| MIL-PRF-12934/27 | RR3600 |
| MIL-PRF-12934/28 | RR3700 |
| MIL-PRF-12934/29 | RR3800 |
| MIL-PRF-12934/30 | RR3900 |
| MIL-PRF-12934/31 | RR4000 |
| MIL-PRF-12934/32 | RR4100 |
| MIL-PRF-12934/33 | RR2002 |
| MIL-PRF-12934/34 | RR1004 |
| MIL-PRF-12934/35 | 2RR2104 |
| MIL-PRF-12934/36 | RR3601 |
| MIL-PRF-12934/37 | 2RR3100 |
| MIL-PRF-18546 | |
| MIL-PRF-18546/2 | RE77<br>RE80 |
| MIL-PRF-22097 | |
| MIL-PRF-22097/2 | RJ12 |
| MIL-PRF-22684 | |
| MIL-PRF-22684/8 | RL42...TX |
| MIL-PRF-23648 | |
| MIL-PRF-23648/1 | RTH06 |
| MIL-PRF-23648/9 | RTH22 |
| MIL-PRF-23648/19 | RTH42 |
| MIL-PRF-23648/20 | RTH44 |
| MIL-PRF-27208 | |
| MIL-PRF-27208/8 | RT12 |
| MIL-PRF-27208/10 | RT26 |

| DoD specification | Style |
|---|---|
| MIL-PRF-32159 | |
| MIL-PRF-32159/1 | RCZ0502 |
| MIL-PRF-32159/2 | RCZ0505 |
| MIL-PRF-32159/3 | RCZ1005 |
| MIL-PRF-32159/4 | RCZ1505 |
| MIL-PRF-32159/5 | RCZ2208 |
| MIL-PRF-32159/6 | RCZ0705 |
| MIL-PRF-32159/7 | RCZ1206 |
| MIL-PRF-32159/8 | RCZ2010 |
| MIL-PRF-32159/9 | RCZ2512 |
| MIL-PRF-32159/10 | RCZ1010 |
| MIL-PRF-32159/11 | RCZ0402 |
| MIL-PRF-32159/12 | RCZ0603 |
| MIL-PRF-32159/13 | RCZ0302 |
| MIL-PRF-32159/14 | RCZ0201 |
| MIL-PRF-32192 | |
| MIL-PRF-32192/1 | RCTP0303 |
| MIL-PRF-32192/3 | RCTN0404 |
| MIL-PRF-32192/4 | RCTN0805 |
| MIL-PRF-32192/5 | RCTN1206 |
| MIL-PRF-32663 | |
| MIL-PRF-32663/1 | RFB0603 |
| MIL-PRF-32663/2 | RFB0805 |
| MIL-PRF-32663/3 | RFB1206 |
| MIL-PRF-32663/4 | RFB1506 |
| MIL-PRF-32663/5 | RFB2010 |
| MIL-PRF-32663/6 | RFB2512 |
| MIL-PRF-32663/7 | RFG0603 |
| MIL-PRF-32663/8 | RFG0805 |
| MIL-PRF-32663/9 | RFG1206 |
| MIL-PRF-32663/10 | RFG1506 |
| MIL-PRF-32663/11 | RFG2010 |
| MIL-PRF-32663/12 | RFG2512 |
| MIL-PRF-39005 | |
| MIL-PRF-39005/1 | RBR52 |
| MIL-PRF-39005/2 | RBR53 |
| MIL-PRF-39005/3 | RBR54 |
| MIL-PRF-39005/4 | RBR55 |
| MIL-PRF-39005/5 | RBR56 |
| MIL-PRF-39005/6 | RBR71 |
| MIL-PRF-39005/11 | RBR80 |



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TABLE A-VIII. Detailed specification number by style number - Continued.

| DoD specification | Style |
|---|---|
| MIL-PRF-39007 | |
| MIL-PRF-39007/7 | RWR78 |
| MIL-PRF-39007/8 | RWR80 |
| MIL-PRF-39007/9 | RWR81 |
| MIL-PRF-39007/10 | RWR84 |
| MIL-PRF-39007/11 | RWR89 |
| MIL-PRF-39007/12 | RWR82 |
| MIL-PRF-39009 | |
| MIL-PRF-39009/1 | RER60 |
| | RER65 |
| | RER70 |
| | RER75 |
| MIL-PRF-39009/2 | RER40 |
| | RER45 |
| | RER50 |
| | RER55 |
| MIL-PRF-39015 | |
| MIL-PRF-39015/2 | RTR22 |
| MIL-PRF-39015/3 | RTR24 |
| MIL-PRF-39017 | |
| MIL-PRF-39017/1 | RLR07 |
| MIL-PRF-39017/2 | RLR20 |
| MIL-PRF-39017/3 | RLR32 |
| MIL-PRF-39017/5 | RLR05 |
| MIL-PRF-39023 | |
| MIL-PRF-39023/1 | RQ090 |
| MIL-PRF-39023/2 | RQ110 |
| MIL-PRF-39023/3 | RQ150 |
| MIL-PRF-39023/4 | RQ200 |
| MIL-PRF-39023/5 | RQ300 |
| MIL-PRF-39023/6 | RQ100 |
| MIL-PRF-39023/7 | RQ160 |
| MIL-PRF-39023/8 | RQ210 |
| MIL-PRF-39023/9 | RQ091 |
| MIL-PRF-39023/10 | RQ051 |
| MIL-PRF-39035 | |
| MIL-PRF-39035/2 | RJR24 |
| MIL-PRF-39035/3 | RJR26 |
| MIL-PRF-39035/4 | RJR50 |
| MIL-PRF-49462 | |
| MIL-PRF-49462/3 | RHV30 |
| | RHV31 |
| | RHV32 |
| | RHV33 |
| | RHV34 |
| | RHV35 |

| DoD specification | Style |
|---|---|
| MIL-PRF-49465 | |
| MIL-PRF-49465/1 | RLV10 |
| MIL-PRF-55182 | |
| MIL-PRF-55182/1 | RNC/N/R55 |
| MIL-PRF-55182/3 | RNC/N/R60 |
| MIL-PRF-55182/5 | RNC/N/R65 |
| MIL-PRF-55182/6 | RNC/N/R70 |
| MIL-PRF-55182/7 | RNC/N/R50 |
| MIL-PRF-55182/9 | RNC/N90 |
| MIL-PRF-55182/10 | RNC/N/R75 |
| MIL-PRF-55342 | |
| MIL-PRF-55342/1 | RM0502 |
| MIL-PRF-55342/2 | RM0505 |
| MIL-PRF-55342/3 | RM1005 |
| MIL-PRF-55342/4 | RM1505 |
| MIL-PRF-55342/5 | RM2208 |
| MIL-PRF-55342/6 | RM0705 |
| MIL-PRF-55342/7 | RM1206 |
| MIL-PRF-55342/8 | RM2010 |
| MIL-PRF-55342/9 | RM2512 |
| MIL-PRF-55342/10 | RM1010 |
| MIL-PRF-55342/11 | RM0402 |
| MIL-PRF-55342/12 | RM0603 |
| MIL-PRF-55342/13 | RM0302 |
| MIL-PRF-55342/14 | RM0201 |
| MIL-PRF-83401 | |
| MIL-PRF-83401/1 | RZ010 |
| MIL-PRF-83401/2 | RZ020 |
| MIL-PRF-83401/3 | RZ030 |
| MIL-PRF-83401/4 | RZ040 |
| MIL-PRF-83401/5 | RZ050 |
| MIL-PRF-83401/6 | RZ060 |
| MIL-PRF-83401/7 | RZ070 |
| MIL-PRF-83401/8 | RZ080 |
| MIL-PRF-83401/9 | RZ090 |
| MIL-PRF-83401/10 | RZ100 |
| MIL-PRF-83401/13 | RZ130 |
| MIL-PRF-83401/14 | RZ140 |
| MIL-PRF-83401/15 | RZ150 |
| MIL-PRF-83401/18 | RZ180 |
| MIL-PRF-83401/19 | RZ190 |
| MIL-PRF-83401/21 | RZ210 |
| MIL-PRF-83401/22 | RZ220 |
| MIL-PRF-83401/23 | RZ230 |
| MIL-PRF-83401/24 | RZ240 |

| DoD Specification | Style |
|---|---|



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# APPENDIX B

## COMMERICAL ITEM DESCRIPTIONS (CIDs)

### B.1 SCOPE

B.1.1 Scope. The following appendix includes all active Commercial Item Descriptions (CIDs). These drawings are used to supplement Military Specifications.

### B.2 APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

### B.3 COMMERICAL ITEM DESCRIPTIONS (CIDs)

B.3.1 Commercial Item Descriptions (CIDs). Table B-I show a listing of Commercial Item Descriptions (CIDs) available for use.

TABLE B-I. List of Commercial Item Descriptions (CIDs).

| CID Number | Title |
|---|---|
| A-A-55088 | Resistor, Flameproof, Fusible |
| A-A-55501 | Resistor, Fixed, Zero Ohm, 1/4 Watt |
| A-A-55502 | Resistor, Fixed, Zero Ohm, 1/8 Watt |
| A-A-55512 | Resistor, Thermally Sensitive (Thermistor) |
| A-A-55517 | Resistor, Fixed, Carbon Film, General Requirements For |
| A-A-55517/1 | Resistor, Fixed, Carbon Film, 1/8 Watt |
| A-A-55517/2 | Resistor, Fixed, Carbon Film, 1/4 Watt |
| A-A-55517/3 | Resistor, Fixed, Carbon Film, 1/2 Watt |
| A-A-55517/4 | Resistor, Fixed, Carbon Film, 1 Watt |
| A-A-55517/5 | Resistor, Fixed, Carbon Film, 2 Watt |
| A-A-55534 | Resistor, Fixed, Power Type, Very Low Resistance Values, General Requirements For |
| A-A-55534/1 | Resistor, Fixed, Wirewound or Metal Element, Power Type, Style VLV1 |
| A-A-55534/5 | Resistor, Fixed, Wirewound or Metal Element, Power Type, Style VLV7 |
| A-A-55534/6 | Resistor, Fixed, Wirewound or Metal Element, Power Type, Style VLV10 |
| A-A-55562 | Resistor, Chip, Voltage Sensitive (Varistor), Metal Oxide, |
| A-A-55562/1 | Resistor, Chip, Voltage Sensitive (Varistor), Metal Oxide, Style 0603 |
| A-A-55562/2 | Resistor, Chip, Voltage Sensitive (Varistor), Metal Oxide, Style 0805 |
| A-A-55562/3 | Resistor, Chip, Voltage Sensitive (Varistor), Metal Oxide, Style 1206 |
| A-A-55562/4 | Resistor, Chip, Voltage Sensitive (Varistor), Metal Oxide, Style 1210 |
| A-A-55562/5 | Resistor, Chip, Voltage Sensitive (Varistor), Metal Oxide, Style 0402 |
| A-A-59714 | Resistor, Resettable Fuses (Non-Linear Thermistor) General Requirements for |
| A-A-59714/1 | Resistor, Resettable Fuses (Non-Linear Thermistor) Radial Leaded, Style RXE |
| A-A-59714/2 | Resistor, Resettable Fuses (Non-Linear Thermistor) Radial Leaded, Style RUE |
| A-A-59714/3 | Resistor, Resettable Fuses (Non-Linear Thermistor) Radial Leaded, Style TR |
| A-A-59715 | Resistor, Fixed and Adjustable, Wirewound, Power Type, General Requirements for |
| A-A-59715/1 | Resistor, Fixed, Wirewound, Power Type, Lead Terminal |
| A-A-59715/2 | Resistor, Adjustable, Wirewound, Power Type |
| A-A-59715/3 | Resistor, Fixed, Wirewound, Power Type |
| A-A-59715/4 | Resistor, Fixed, Wirewound, Power Type, Tab Terminals |
| A-A-59715/5 | Resistor, Fixed and Adjustable, Wirewound, Power Type |



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# APPENDIX C

# DLA LAND AND MARITIME DRAWINGS

## C.1 SCOPE

C.1.1 Scope. The following appendix includes all active DLA Land and Maritime and Defense Supply Center, Columbus Drawings. These drawings are used to supplement Military Specifications.

## C.2 APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

## C.3 DLA LAND AND MARITIME DRAWINGS

C.3.1 DLA LAND AND MARITIME Drawings. Table C-I show a listing of DLA Land and Maritime Drawings available for use.

TABLE C-I. List of DLA Land and Maritime Drawings

| Drawing Number | Title |
|---|---|
| 00001 | Resistor, Variable, Wirewound, Nonprecision, Trimmer, 3/8 Inch, Square, 1 Watt, Flexible Leads |
| 01002 | Resistor, Fixed, Chip, 1.5 Watt (MELF - 2512), Flat Ceramic Package |
| 01032 | Resistor, Fixed, Film, Chip, Beryllia Substrate, High Power, Style 1206 |
| 01033 | Resistor, Fixed, Film, Chip, Voltage Divider, Style 1206 |
| 02001 | Resistor, Fixed, Film, Precision, Chip 1/8 Watt, Style 2012 |
| 02005 | Resistors, Variable, Nonwirewound (Adjustment Type, Lead-Screw Actuated) 1/2" Rectangular, 0.3 Watts |
| 02008 | Resistor, Fixed, Film, Chip, Low and High Values, Style 1206 |
| 02010 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 1206 |
| 03003 | Resistor, Fixed, Carbon Film, High Pulse Voltage, 1/8 Watt |
| 03004 | Resistor, Fixed, Carbon Film, High Pulse Voltage, 1/4 Watt |
| 03005 | Resistor, Fixed, Carbon Film, High Pulse Voltage, 1/2 Watt |
| 03006 | Resistor, Fixed, Carbon Film, High Pulse Voltage, 1 Watt |
| 03007 | Resistor, Fixed, Carbon Film, High Pulse Voltage, 3/4 Watt |
| 03008 | Resistor, Fixed, Carbon Film, High Pulse Voltage, 2 Watt |
| 03009 | Resistor, Fixed, Carbon Film, High Pulse Voltage, 3 Watt |
| 03010 | Resistor, Fixed, Film, Chip, Surface Mounted, Ultra Precision, Style 1506 |
| 03011 | Resistor, Fixed, Film, Chip, Zero-Ohm, Style 0201 |
| 03022 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 0603 |
| 03025 | Resistor, Chip, Fixed, Film, High Voltage, Style 1206 |
| 03026 | Resistor, Chip, Fixed, Film, High Voltage, Style 2010 |
| 03027 | Resistor, Chip, Fixed, Film, High Voltage, Style 2512 |
| 04007 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 0302 |
| 04008 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 0402 |
| 04009 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 0603 |
| 04032 | Resistor, Fixed, Film, Chip, Low Values, High Power, Style 2512 |
| 05009 | Resistor, Fixed, Film, Chip (MELF), 1/4 Watt, Style 0204 |
| 06001 | Resistor, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 2010 |
| 06002 | Resistor, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 2512 |
| 06003 | Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (2 Watt) |
| 06006 | Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (3 Watt), Style 4527 |
| 06007 | Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (0.1 Watt), Style 0603 |
| 06008 | Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (0.125 Watt), Style 0805 |
| 06009 | Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (0.25 Watt), Style 1206 |
| 06010 | Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (0.5 Watt), Style 2010 |



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TABLE C-I. List of DLA Land and Maritime Drawings Continued

| Drawing Number | Title |
|---|---|
| 06011 | Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (1.0 Watt), Style 2512 |
| 06012 | Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (2.0 Watt), Style 2816 |
| 06018 | Resistor Network, Fixed, Film, Surface Mount, Voltage Divider, 3 Pin |
| 06020 | Resistor, Fixed, Bulk Metal Foil, High Precision, Surface-Mount, Molded 0.25/0.16 Watt |
| 06021 | Resistor, Fixed, Bulk Metal Foil, High Precision, Surface Mount, Molded 0.6/0.4 Watts |
| 07002 | Resistor, Fixed, Wirewound, Surface Mount, Power Type (3 Watt) |
| 07005 | Resistor, Chip, Fixed, Film, 8 Pin Array, Style 1206 |
| 07009 | Resistor, Chip, Fixed, Film, Style 0201 |
| 07010 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 0201 |
| 07011 | Resistor, Fixed, Current Sensing, Metal Strip, High Precision, Surface Mount, Style 2512 |
| 07012 | Resistor, Fixed, Current Sensing, Metal Strip, High Precision, Surface Mount, Style 3637 |
| 07024 | Resistor, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 0805 |
| 07025 | Resistor, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 1206 |
| 08003 | Resistor, Chip, Fixed, Current Sensing, Ultra Precision, Style 1625 |
| 09002 | Resistor, Fixed, Film, Precision, High Ohmic Values |
| 09004 | Resistor, Chip, Fixed, Film, Precision, Top Contact, Style 2020 |
| 09008 | Resistor, Fixed, Wirewound, (Accurate), Radial Leaded, 0.1 Watt (RBR81) |
| 09011 | Resistor Network, Fixed Film, Precision 8 Pad, Ball Grid Array |
| 09012 | Resistor Network, Fixed Film, Precision 16 Pad, Ball Grid Array |
| 09013 | Resistor Network, Fixed Film, 10 Pad, Ball Grid Array |
| 09014 | Resistor Network, Fixed Film, 32 Pad, Ball Grid Array |
| 09015 | Resistor Network, Fixed Film, 40 Pad, Ball Grid Array |
| 09016 | Resistor Network, Fixed Film, 27 Pad, Ball Grid Array |
| 09017 | Resistor Network, Fixed Film, 36 Pad, Ball Grid Array |
| 10005 | Resistor, Precision Current Sensor, 100 Watts |
| 10006 | Resistor, Current Sense, Open Air, Standard Leads, 1 Watt |
| 10007 | Resistor, Current Sense, Open Air, Standard Leads, 3 Watt |
| 10008 | Resistor, Current Sense, Open Air, Standard Leads, 5 Watt |
| 10009 | Resistor, Variable, Surface Mount, 4MM, Multiturn Trimmer, 1/4 Watt |
| 10010 | Resistor, Current Sense, Open Air, Metal Alloy Strip, 5 Watts |
| 11007 | Resistor, Chip, Fixed, Current Sense, Ultra Precision, Style 1206 |
| 11008 | Resistor, Chip, Fixed, Current Sense, Ultra Precision, Style 2010 |
| 11009 | Resistor, Chip, Fixed, Current Sense, Ultra Precision, Style 2512 |
| 12002 | Resistor, Variable, Wirewound, (Low Operating Temperature) 1.0 Watt |
| 12003 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 0502 |
| 12004 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 0505 |
| 12005 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 1005 |
| 12006 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 1505 |
| 12007 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 2208 |
| 12008 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 0705 |
| 12009 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 2010 |
| 12010 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 2512 |
| 12011 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 1010 |
| 12012 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 0402 |
| 12013 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 0302 |
| 12014 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 0201 |
| 13003 | Resistor Network, 8 Pin, Dual-in-Line Package (DIP), Multi Resistance Values |
| 13019 | Resistor, Fixed, High Surge, 1/8 Watt |
| 13020 | Resistor, Fixed, High Surge, 1/4 Watt |



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TABLE C-I. List of DLA Land and Maritime Drawings Continued

| Drawing Number | Title |
|---|---|
| 13021 | Resistor, Fixed, High Surge, 1/2 Watt |
| 13022 | Resistor, Fixed, High Surge, 1 Watt |
| 13023 | Resistor, Fixed, High Surge, 2 Watt |
| 13024 | Resistor, Fixed, High Surge, 5 Watt |
| 13025 | Resistor, Fixed, High Surge, 10 Watt |
| 14006 | Resistor, Fixed, Wirewound, (Accurate), 0.5 Watt (RBR52) |
| 14007 | Resistor, Fixed, Wirewound, (Accurate), 0.25 Watt (RBR54) |
| 14008 | Resistor, Fixed, Wirewound, (Accurate), 0.125 Watt (RBR56) |
| 15002 | Resistor, Fixed, Wirewound, (Accurate), 0.33 Watt (RBR53) |
| 15003 | Resistor, Fixed, Wirewound, (Accurate), 0.15 Watt (RBR55) |
| 15004 | Resistor, Fixed, Wirewound, (Accurate), 0.125 Watt (RBR71) |
| 15006 | Resistor, Fixed, Wirewound, (Accurate), Lug Precision, Radial Leaded, 0.5 Watt (RB17) |
| 15007 | Resistor, Fixed, Wirewound, (Accurate), Radial Leaded, 0.1 Watt (RBR80) |
| 16001 | Resistor, Fixed, Wirewound, (Accurate), 0.125 Watt (RBR75) |
| 16002 | Resistor, Fixed, Wirewound, (Accurate), Lug Precision, Radial Leaded, 0.25 Watt (RB08) |
| 17003 | Resistor, Fixed, High Resistance values, Glass Encapsulated |
| 17005 | Resistor, Fixed, Thick Film, High Voltage, High Resistance, Axial Terminal, 2 Watts |
| 17006 | Resistor, Fixed, Thick Film, High Voltage, High Resistance, Axial Terminal, 3 Watts |
| 17007 | Resistor, Fixed, Thick Film, High Voltage, High Resistance, Axial Terminal, 4 Watts |
| 17008 | Resistor, Fixed, Thick Film, High Voltage, High Resistance, Axial Terminal, 5 Watts |
| 17009 | Resistor, Fixed, Thick Film, High Voltage, High Resistance, Axial Terminal, 6 Watts |
| 18001 | Resistor, Variable, Wirewound, Semi-Precision, 1.5 Watts (RK09) - Placeholder |
| 20011 | Resistor, Thermocouple Lead Spool |
| 21001 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 1210 |
| 22001 | Resistor, Chip, Fixed, Film, 4 Terminal Current Sense, Style 2512 |
| 22002 | Resistor, Chip, Fixed, Film, 4 Terminal Current Sense, Style 2010 |
| 22003 | Resistor, Chip, Fixed, Film, 4 Terminal Current Sense, Style 1206 |
| 22004 | Resistor, Fixed, Metal Element, Power Type, (Similar to Style RLV30) |
| 22006 | Resistor, Fixed, Metal Element, Power Type, (Similar to Style RLV31) |
| 85085 | Resistor, Variable, Nonwirewound, Adjustment Type |
| 86097 | Resistor, Variable, Panel Mounted |
| 87012 | Resistor Network, Fixed, Film, Surface Mount, Gull Wing, 16 Pin |
| 87013 | Resistor Network, Fixed, Film, Surface Mount, Gull Wing, 14 Pin |
| 87014 | Resistor Network, 16 Pin, Leadless Chip Carrier |
| 87015 | Resistor Network, 28-Pin, Leadless Chip Carrier |
| 87016 | Resistor Network, 20-Pin, Leadless Chip Carrier |
| 87017 | Resistor Network, Fixed, Film, Surface Mount, 20 Pin Leadless Chip Carrier |
| 87018 | Resistor Network, 16-Pin Leadless Chip Carrier |
| 87025 | Resistor Network, 8 Pin, Dual-in-Line Package (DIP) |
| 87026 | Resistor Network, Fixed, Film, 3-Pin |
| 87030 | Resistor Network, 6-Pin, Single-in-Line Package (SIP) |
| 87031 | Resistor Network, 8-Pin, Single-in-Line Package (SIP) |
| 87032 | Resistor Network, 10-Pin, Single-in-Line Package (SIP) |
| 87045 | Resistor Network, Fixed, Film, 20 Pin, Hermetically Sealed, Dual-in-Line Package (DIP) |
| 87053 | Resistor Network, 14-Pin, Flat Pack |
| 87067 | Resistor Network, Fixed, Film, 7-Pin Sip (Low Profile), Multiple Schematics |
| 87068 | Resistor Network, Fixed, Film, 7-Pin Sip (High Profile), Multiple Schematics |
| 87071 | Resistor Network, Fixed, Film, 10-Pin Sip, Multiple Resistance Values, Multiple Schematics, (Low Profile) |
| 87072 | Resistor Network, Fixed, Film, 10-Pin Sip, Multiple Resistance Values, Multiple Schematics and Multiple Tolerances. |



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TABLE C-I. List of DLA Land and Maritime Drawings Continued

| Drawing Number | Title |
|---|---|
| 87073 | Resistor Network, Fixed, Film, Eight Pin SIP, Multiple Resistance Values, Multiple Schematics, (Low Profile) |
| 87074 | Resistor Network, Fixed, Film, Eight Pin SIP, Multiple Resistance Values, Multiple Schematics, (High Profile) |
| 87075 | Resistor, Fixed, Film, Chip, Flange Mount, Double Tab, High Power |
| 87103 | Resistor Network, 9-Pin, SIP Multiple Values, Multiple Tolerances, (Low Profile) |
| 87105 | Resistor Network, 9-Pin, SIP Multiple Values, Multiple Tolerances, (High Profile) |
| 87126 | Resistor, Variable, Nonwirewound (Adjustment Type, Lead Screw Actuated) |
| 88008 | Resistor, Tantalum Nitride, Chip |
| 88009 | Resistor, Fixed, Film, Chip, Tantalum Nitride |
| 88014 | Resistor Network, 12-Pin, Single-in-Line Package (SIP) |
| 88015 | Resistor Network, 8-Pin, Single Inline Package |
| 88016 | Resistor Network, 20 Pin Leadless Chip Carrier |
| 88018 | Resistor, Chip, Fixed, Film, 0.200 Watt, Style 0705 |
| 88020 | Resistor Network, 6-Pin, Leadless Chip Carrier |
| 88027 | Resistor, Chip, Fixed, Film, Style 0504 |
| 88030 | Resistor, Chip, Fixed, Film, Style 1005 |
| 88031 | Resistor, Chip, Fixed, Film, 0.330 Watt, Style 1505 |
| 88034 | Resistor, Fixed, Film, Chip, Low Values, High Power, Style 2512 |
| 88036 | Resistor Network, 10 Pin, Leadless Chip Carrier |
| 88040 | Resistor, Variable, Surface Mount, Nonwirewound, Trimmer |
| 89004 | Resistor-Capacitor Network, 16 Pin DIP |
| 89023 | Resistor-Capacitor Network, 16 Pin Flat Pack |
| 89039 | Resistor, Fixed, Film, Precision |
| 89040 | Resistor, Fixed, Wire Wound, Surface Mount, Power Type (2.5 Watts) |
| 89088 | Resistor, Fixed, Film, Precision, 0.1 Watt, Power Curve C |
| 89099 | Resistor, Fixed, Film, Chip, Flange Mount, Single Tab, High Power |
| 90038 | Resistor, Fixed, Film, Precision, 0.25 Watt, Power Curve C |
| 92013 | Resistor Network, 10-Pin, SIP, Extended Lead Length |
| 93030 | Resistor, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 2018 |
| 93075 | Resistor, Fixed, Wirewound, Surface Mount, Power Type (½ Watt) |
| 93076 | Resistor, Fixed, Wirewound, Surface Mount, Power Type (1 Watt) |
| 93077 | Resistor, Fixed, Wirewound, Surface Mount, Power Type (2 Watt) |
| 94012 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 0505 |
| 94013 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 1005 |
| 94014 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 2208 |
| 94015 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 0705 |
| 94016 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 1206 |
| 94017 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 2010 |
| 94018 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 2512 |
| 94019 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 1010 |
| 94025 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 0502 |
| 94026 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 1505 |
| 94047 | Resistor, Fixed, Film, Chip, (MELF), 2 Watt, Style 3610 |
| 94048 | Resistor, Fixed, Film, Chip, (MELF),1/2 Watt, Style 2010 |
| 95006 | Resistor, Fixed, Film, Chip (MELF), 1 Watt, Style 2512 |
| 95011 | Resistor, Fixed, Film, Chip (MELF), 1/8 Watt, Style 1206 |
| 96002 | Resistor, Fixed, Film, Insulated, Low Inductance |
| 97004 | Resistor, Fixed, Film (Insulated), 2 Watt |
| 97012 | Resistor, Variable, Nonwirewound, Trimmer, 3/4 Watt, Flexible Leads (Reactivation) |



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TABLE C-I. List of DLA Land and Maritime Drawings Continued

| Drawing Number | Title |
|---|---|
| 98020 | Resistor, Fixed, Film (Insulated), 1/8 Watt |
| 98021 | Resistor, Fixed, Film (Insulated), 1/2 Watt |
| 98022 | Resistor, Fixed, Film (Insulated), 1 Watt |
| 99004 | Resistor, Fixed, Film, Chip, 4 Pin Array |
| 99005 | Resistor, Fixed, Film, Chip, 8 Pin Array |
| 99006 | Resistor, Fixed, Film, Chip, 10 Pin Array |
| 99007 | Resistor, Fixed, Film, Chip, 16 Pin Array |
| 99011 | Resistor, Fixed, Film (Insulated), ¼ Watt |
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C.3.1.1 Resistor, Chip. Table C-II show a breakdown listing of Chip Resistors

TABLE C-II. Resistor, Chip.

| Drawing number | Title |
|---|---|
| 01002 | Resistor, Fixed, Chip, 1.5 Watt (MELF - 2512), Flat Ceramic Package |
| 01032 | Resistor, Fixed, Film, Chip, Voltage Divider, Style 1206 |
| 01033 | Resistor, Fixed, Film, Chip, Voltage Divider, Style 1206 |
| 02008 | Resistor, Fixed, Film, Chip, Low and High Values, Style 1206 |
| 02010 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 1206 |
| 03010 | Resistor, Fixed, Film, Chip, Surface Mounted, Ultra Precision, Style 1506 |
| 03011 | Resistor, Fixed, Film, Chip, Zero-Ohm, Style 0201 |
| 03022 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 0603 |
| 03025 | Resistor, Chip, Fixed, Film, High Voltage, Style 1206 |
| 03026 | Resistor, Chip, Fixed, Film, High Voltage, Style 2010 |
| 03027 | Resistor, Chip, Fixed, Film, High Voltage, Style 2512 |
| 04007 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 0302 |
| 04008 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 0402 |
| 04009 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 0603 |
| 04032 | Resistor, Fixed, Film, Chip, Low Values, High Power, Style 2512 |
| 05009 | Resistor, Fixed, Film, Chip (MELF), 1/4 Watt, Style 0204 |
| 06001 | Resistor, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 2010 |
| 06002 | Resistor, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 2512 |
| 06003 | Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (2 Watt) |
| 06006 | Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (3 Watt), Style 4527 |
| 06007 | Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (0.1 Watt), Style 0603 |
| 06008 | Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (0.125 Watt), Style 0805 |
| 06009 | Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (0.25 Watt), Style 1206 |
| 06010 | Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (0.5 Watt), Style 2010 |
| 06011 | Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (1.0 Watt), Style 2512 |
| 06012 | Resistor, Chip, Fixed, Power Metal Strip, Surface Mount, Low Value (2.0 Watt), Style 2816 |
| 07005 | Resistor, Chip, Fixed, Film, 8 Pin Array, Style 1206 |
| 07009 | Resistor, Chip, Fixed, Film, Style 0201 |
| 07010 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 0201 |
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TABLE C-II. Resistor, Chip. Continued

| Drawing number | Title |
|---|---|
| 07024 | Resistor, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 0805 |
| 07025 | Resistor, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 1206 |
| 08003 | Resistor, Chip, Fixed, Current Sensing, Ultra Precision, Style 1625 |
| 09004 | Resistor, Chip, Fixed, Film, Precision, Top Contact, Style 2020 |
| 11007 | Resistor, Chip, Fixed, Current Sense, Ultra Precision, Style 1206 |
| 11008 | Resistor, Chip, Fixed, Current Sense, Ultra Precision, Style 2010 |
| 11009 | Resistor, Chip, Fixed, Current Sense, Ultra Precision, Style 2512 |
| 12003 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 0502 |
| 12004 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 0505 |
| 12005 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 1005 |
| 12006 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 1505 |
| 12007 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 2208 |
| 12008 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 0705 |
| 12009 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 2010 |
| 12010 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 2512 |
| 12011 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 1010 |
| 12012 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 0402 |
| 12013 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 0302 |
| 12014 | Resistor, Chip, Fixed, Film, Values Less Than 1 Ohm, Style 0201 |
| 21001 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 1210 |
| 22001 | Resistor, Chip, Fixed, Film, 4 Terminal Current Sense, Style 2512 |
| 22002 | Resistor, Chip, Fixed, Film, 4 Terminal Current Sense, Style 2010 |
| 22003 | Resistor, Chip, Fixed, Film, 4 Terminal Current Sense, Style 1206 |
| 87075 | Resistor, Fixed, Film, Chip, Flange Mount, Double Tab, High Power |
| 88008 | Resistor, Tantalum Nitride, Chip |
| 88009 | Resistor, Fixed, Film, Chip, Tantalum Nitride |
| 88018 | Resistor, Chip, Fixed, Film, 0.200 Watt, Style 0705 |
| 88027 | Resistor, Chip, Fixed, Film, Style 0504 |
| 88030 | Resistor, Chip, Fixed, Film, Style 1005 |
| 88031 | Resistor, Chip, Fixed, Film, 0.330 Watt, Style 1505 |
| 88034 | Resistor, Fixed, Film, Chip, Low Values, High Power, Style 2512 |
| 89099 | Resistor, Fixed, Film, Chip, Flange Mount, Single Tab, High Power |
| 93030 | Resistors, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 2018 |
| 94011 | Resistor, Chip, Fixed, Film, Zero-Ohm, Style RM1206 |
| 94012 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 0505 |
| 94013 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 1005 |
| 94014 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 2208 |
| 94015 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 0705 |
| 94016 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 1206 |
| 94017 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 2010 |
| 94018 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 2512 |
| 94019 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 1010 |
| 94025 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 0502 |
| 94026 | Resistor, Chip, Fixed, Film, Moisture Resistant, Military and Space Level, Style 1505 |
| 94047 | Resistor, Fixed, Film, Chip, (MELF), 2 Watt, Style 3610 |
| 94048 | Resistor, Fixed, Film, Chip (MELF), 1/2 Watt, Style 2010 |
| 95006 | Resistor, Fixed, Film, Chip (MELF), 1 Watt, Style 2512 |
| 95011 | Resistor, Fixed, Film, Chip (MELF), 1/8 Watt, Style 1206 |



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TABLE C-II. Resistor, Chip. Continued

| Drawing number | Title |
|---|---|
| 99004 | Resistor, Fixed, Film, Chip, 4 Pin Array |
| 99005 | Resistor, Fixed, Film, Chip, 8 Pin Array |
| 99006 | Resistor, Fixed, Film, Chip, 10 Pin Array |
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C.3.1.2 Resistor, Network. Table C-III show a breakdown listing of Network Resistors.

TABLE C-III. Resistor, Network.

| Drawing number | Title |
|---|---|
| 06018 | Resistor Network, Fixed, Film, Surface Mount, Voltage Divider, 3 Pin |
| 09011 | Resistor Network, Fixed Film, Precision 8 Pad, Ball Grid Array |
| 09012 | Resistor Network, Fixed Film, Precision 16 Pad, Ball Grid Array |
| 09013 | Resistor Network, Fixed Film, 10 Pad, Ball Grid Array |
| 09014 | Resistor Network, Fixed Film, 32 Pad, Ball Grid Array |
| 09015 | Resistor Network, Fixed Film, 40 Pad, Ball Grid Array |
| 09016 | Resistor Network, Fixed Film, 27 Pad, Ball Grid Array |
| 09017 | Resistor Network, Fixed Film, 36 Pad, Ball Grid Array |
| 13003 | Resistor Network, 8 Pin, Dual-in-Line Package (DIP), Multi Resistance Values |
| 87012 | Resistor Network, Fixed, Film, Surface Mount, Gull Wing, 16 Pin |
| 87013 | Resistor Network, Fixed, Film, Surface Mount, Gull Wing, 14 Pin |
| 87014 | Resistor Network, 16 Pin, Leadless Chip Carrier |
| 87015 | Resistor Network, 28-Pin, Leadless Chip Carrier |
| 87016 | Resistor Network, 20-Pin, Leadless Chip Carrier |
| 87017 | Resistor Network, Fixed, Film, Surface Mount, 20 Pin Leadless Chip Carrier |
| 87018 | Resistor Network, 16-Pin Leadless Chip Carrier |
| 87025 | Resistor Network, 8 Pin, Dual-in-Line Package (DIP) |
| 87026 | Resistor Network, Fixed, Film, 3-Pin |
| 87030 | Resistor, Network, 6-Pin, Single-in-Line Package (SIP) |
| 87031 | Resistor, Network, 8-Pin, Single-in-Line Package (SIP) |
| 87032 | Resistor Network, 10-Pin, Single-in-Line Package (SIP) |
| 87033 | Resistor Network, 10-Pin, Single-in-Line Package (SIP) |
| 87045 | Resistor Network, Fixed, Film, 20 Pin, Hermetically Sealed, Dual-in-Line Package (DIP) |
| 87053 | Resistor, Network, 14-Pin, Flat Pack |
| 87067 | Resistor Network, Fixed, Film, 7-Pin Sip (Low Profile), Multiple Schematics |
| 87068 | Resistor Network, Fixed, Film, 7-Pin Sip (High Profile), Multiple Schematics |
| 87071 | Resistor Network, Fixed, Film, 10-Pin Sip, Multiple Resistance Values, Multiple Schematics, (Low Profile) |
| 87072 | Resistor Network, Fixed, Film, 10-Pin Sip, Multiple Resistance Values, Multiple Schematics and Multiple Tolerances |
| 87073 | Resistor Network, Fixed, Film, Eight Pin SIP, Multiple Resistance Values, Multiple Schematics, (Low Profile) |
| 87074 | Resistor Network, Fixed, Film, Eight Pin SIP, Multiple Resistance Values, Multiple Schematics, (High Profile) |
| 87103 | Resistor, Network, 9-Pin, SIP Multiple Values, Multiple Tolerances, (Low Profile) |
| 87105 | Resistor, Network, 9-Pin, SIP Multiple Values, Multiple Tolerances, (High Profile) |
| 88014 | Resistor, Network, 12-Pin, Single-in-Line Package (SIP) |
| 88015 | Resistor, Network, 8-Pin, Single Inline Package |
| 88016 | Resistor Network, 20 Pin Leadless Chip Carrier |



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TABLE C-III. Resistor, Network. Continued

| | |
|---|---|
| 88020 | Resistor Network, 6-Pin, Leadless Chip Carrier |
| 88036 | Resistor Network, 10 Pin, Leadless Chip Carrier |
| 89004 | Resistor-Capacitor Network, 16 Pin DIP |
| 89023 | Resistor-Capacitor Network, 16 Pin Flat Pack |
| 92013 | Resistor Network, 10-Pin, SIP, Extended Lead Length |
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C.3.1.3 Resistor, Current Sensing. Table C-IV show a breakdown listing of Current Sensing Resistors.

TABLE C-IV. Resistor, Current Sensing.

| Drawing number | Title |
|---|---|
| 07011 | Resistor, Fixed, Current Sensing, Metal Strip, High Precision, Surface Mount, Style 2512 |
| 07012 | Resistor, Fixed, Current Sensing, Metal Strip, High Precision, Surface Mount, Style 3637 |
| 10005 | Resistor, Precision Current Sensor, 100 Watts |
| 10006 | Resistor, Current Sense, Open Air, Standard Leads, 1 Watt |
| 10007 | Resistor, Current Sense, Open Air, Standard Leads, 3 Watt |
| 10008 | Resistor, Current Sense, Open Air, Standard Leads, 5 Watt |
| 10010 | Resistor, Current Sense, Open Air, Metal Alloy Strip, 5 Watts |
| | |
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C.3.1.4 Resistor, Carbon Film. Table C-V show a breakdown listing of Carbon Film Resistors.

TABLE C-V. Resistor, Carbon Film.

| Drawing number | Title |
|---|---|
| 03003 | Resistor, Fixed, Carbon Film, High Pulse Voltage, 1/8 Watt |
| 03004 | Resistor, Fixed, Carbon Film, High Pulse Voltage, 1/4 Watt |
| 03005 | Resistor, Fixed, Carbon Film, High Pulse Voltage, 1/2 Watt |
| 03006 | Resistor, Fixed, Carbon Film, High Pulse Voltage, 1 Watt |
| 03007 | Resistor, Fixed, Carbon Film, High Pulse Voltage, 3/4 Watt |
| 03008 | Resistor, Fixed, Carbon Film, High Pulse Voltage, 2 Watt |
| 03009 | Resistor, Fixed, Carbon Film, High Pulse Voltage, 3 Watt |
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C.3.1.5 Resistor, Fixed. Table C-VI show a breakdown listing of Fixed Resistors.

TABLE C-VI. Resistor, Fixed.

| Drawing number | Title |
|---|---|
| 13019 | Resistor, Fixed, Film, Precision, Chip 1/8 Watt, Style 2012 |
| 13020 | Resistor, Fixed, Film, Radial Lead, 2.25 Watts (Up to 20 Watts with Heatsink) |
| 13021 | Resistor, Fixed, Film, Radial Lead, 1 Watt (Up to 35 Watts with Heatsink) |
| 13022 | Resistor, Fixed, Film, Radial Lead, 1 Watt (Up to 50 Watts with Heatsink) |
| 13023 | Resistor, Fixed, Film, Radial Lead, 3 Watts (Up to 100 Watts with Heatsink) |
| 13024 | Resistor, Fixed, Film, Precision, High Ohmic Values |
| 13025 | Resistor, Fixed, High Surge, 1/8 Watt |
| 17003 | Resistor, Fixed, High Surge, 1/4 Watt |
| 17005 | Resistor, Fixed, High Surge, 1/2 Watt |
| 17006 | Resistor, Fixed, High Surge, 1 Watt |
| 17007 | Resistor, Fixed, High Surge, 2 Watt |
| 17008 | Resistor, Fixed, High Surge, 5 Watt |
| 17009 | Resistor, Fixed, High Surge, 10 Watt |
| 13019 | Resistor, Fixed, High Resistance Values, Glass Encapsulated |
| 13020 | Resistor, Fixed, Thick Film, High Voltage, High Resistance Axial Terminal, 2 Watts |
| 13021 | Resistor, Fixed, Thick Film, High Voltage, High Resistance Axial Terminal, 3 Watts |
| 13022 | Resistor, Fixed, Thick Film, High Voltage, High Resistance Axial Terminal, 4 Watts |
| 13023 | Resistor, Fixed, Thick Film, High Voltage, High Resistance Axial Terminal, 5 Watts |
| 13024 | Resistor, Fixed, Thick Film, High Voltage, High Resistance Axial Terminal, 6 Watts |
| 22006 | Resistor, Fixed, Metal Element, Power Type, (Similar to Style RLV31) |
| 88032 | Resistor, Fixed, Film, Zero-Ohm, Chip, Style RM0502 |
| 89039 | Resistor, Fixed, Film, Precision |
| 89088 | Resistor, Fixed, Film, Precision, 0.1 Watt, Power Curve C |
| 90038 | Resistor, Fixed, Film, Precision, 0.25 Watt, Power Curve C |
| 96002 | Resistor, Fixed, Film, Insulated, Low Inductance |
| 97004 | Resistor, Fixed, Film (Insulated), 2 Watt |
| 97010 | Resistor, Fixed, Film, Precision, 0.3 Watt, Power Curve A |
| 97011 | Resistor, Fixed, Film, Precision, 0.3 Watt, Power Curve B |
| 97004 | Resistor, Fixed, Film, Precision, 0.15 Watt, Power Curve B |
| 98020 | Resistor, Fixed, Film (Insulated), 1/8 Watt |
| 98021 | Resistor, Fixed, Film (Insulated), 1/2 Watt |
| 98022 | Resistor, Fixed, Film (Insulated), 1 Watt |
| 99011 | Resistor, Fixed, Film (Insulated), ¼ Watt |
| | |
| 07002 | Resistor, Fixed, Wirewound, Surface Mount, Power Type (3 Watt) |
| 09008 | Resistor, Fixed, Wirewound, (Accurate), Radial Leaded, 0.1 Watt (RBR81) |
| 14006 | Resistor, Fixed, Wirewound, (Accurate), 0.5 Watt (RBR52) |
| 14007 | Resistor, Fixed, Wirewound, (Accurate), 0.25 Watt (RBR54) |
| 14008 | Resistor, Fixed, Wirewound, (Accurate), 0.125 Watt (RBR56) |
| 15002 | Resistor, Fixed, Wirewound, (Accurate), 0.33 Watt (RBR53) |
| 15003 | Resistor, Fixed, Wirewound, (Accurate), 0.15 Watt (RBR55) |
| 15004 | Resistor, Fixed, Wirewound, (Accurate), 0.125 Watt (RBR71) |
| 15006 | Resistor, Fixed, Wirewound, (Accurate), Lug Precision, Radial Leaded, 0.5 Watt (RB17) |
| 15007 | Resistor, Fixed, Wirewound, (Accurate), Radial Leaded, 0.1 Watt (RBR80) |
| 16001 | Resistor, Fixed, Wirewound, (Accurate), 0.125 Watt (RBR75) |
| 16002 | Resistor, Fixed, Wirewound, (Accurate), Lug Precision, Radial Leaded, 0.25 Watt (RB08) |



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TABLE C-VI. Resistor, Fixed. Continued

| Drawing number | Title |
|---|---|
| 89040 | Resistor, Fixed, Wirewound, Surface Mount, Power Type (2.5 Watts) |
| 93075 | Resistor, Fixed, Wirewound, Surface Mount, Power Type (½ Watt) |
| 93076 | Resistor, Fixed, Wirewound, Surface Mount, Power Type (1 Watt) |
| 93077 | Resistor, Fixed, Wirewound, Surface Mount, Power Type (2 Watt) |
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C.3.1.6 Resistor, Variable. Table C-VII show a breakdown listing of Variable Resistors.

TABLE C-VII. Resistor, Variable.

| Drawing number | Title |
|---|---|
| 00001 | Resistor, Variable, Wirewound, Nonprecision, Trimmer, 3/8 Inch, Square, 1 Watt, Flexible Leads |
| 02005 | Resistor, Variable, Nonwirewound (Adjustment Type, Lead-Screw Actuated) 1/2" Rectangular, 0.3 Watts |
| 10009 | Resistor, Variable, Surface Mount, 4MM, Multiturn Trimmer, 1/4 Watt |
| 18001 | Resistor, Variable, Wirewound, Semi-Precision, 1.5 Watts (RK09) |
| 85085 | Resistor, Variable, Nonwirewound, Adjustment Type |
| 86097 | Resistor, Variable, Panel Mounted |
| 87126 | Resistor, Variable, Nonwirewound (Adjustment Type, Lead Screw Actuated) |
| 88040 | Resistor, Variable, Surface Mount, Nonwirewound, Trimmer |
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C.3.1.7 Resistor, Thermal. Table C-VIII show a breakdown listing of Thermal Resistors.

TABLE C-VIII. Resistor, Thermal.

| Drawing number | Title |
|---|---|
| 20011 | Resistor, Thermocouple Lead Spool |
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C.3.1.8 Resistor, Bulk Metal. Table C-IX show a breakdown listing of Bulk Metal Resistors.

TABLE C-IX. Resistor, Bulk Metal.

| Drawing number | Title |
|---|---|
| 06001 | Resistor, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 2010 |
| 06002 | Resistor, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 2512 |
| 07024 | Resistor, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 0805 |
| 07025 | Resistor, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 1206 |
| 93030 | Resistor, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 2018 |
| | |
| 06020 | Resistor, Fixed, Bulk Metal Foil, High Precision, Surface-Mount, Molded 0.25/0.16 Watt |
| 06021 | Resistor, Fixed, Bulk Metal Foil, High Precision, Surface Mount, Molded 0.6/0.4 Watts |
| | |
| | |
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C.3.1.9 Cancelled Documents. Table C-X show cancelled drawing that have replacements

TABLE C-X. Cancelled drawings.

| Drawing number | Title |
|---|---|
| 02009 | Resistor, Fixed, Film, Flip Chip, Ultra Precision, Style 1206 |
| 04025 | Resistor, Chip, Fixed, Film, Surface Mount, 5 Watts (Up to 25 Watts with Heatsink) |
| 05015 | Resistor, Chip, Fixed, Film, Surface Mount, 2.25 Watts (Up to 20 Watts with Heatsink) |
| 05016 | Resistor, Fixed, Film, Radial Lead, 2.25 Watts (Up to 20 Watts with Heatsink) |
| 07017 | Resistor, Fixed, Film, Radial Lead, 1 Watt (Up to 35 Watts with Heatsink) |
| 07018 | Resistor, Fixed, Film, Radial Lead, 1 Watt (Up to 50 Watts with Heatsink) |
| 07019 | Resistor, Fixed, Film, Radial Lead, 3 Watts (Up to 100 Watts with Heatsink) |
| 84146 | Resistor, Antenna Termination Kit, Receive 2-30 MHz |
| 85083 | Resistor, Fixed, Chip |
| 85095 | Resistor, Coil Strip |
| 86005 | Resistor, Variable, 25 Turn, Multiple Tap |
| 87009 | Resistor, Fixed, Zero Ohm, 1/4 Watt |
| 87010 | Resistor, Fixed, Zero Ohm, 1/8 Watt |
| 87027 | Resistor Network, Fixed, Film, 3-Pin |
| 87033 | Resistor Network, 10-Pin, Single-in-Line Package (SIP) |
| 87037 | Resistor Chip, 1 Watt (MELF) |
| 87063 | Resistor, Voltage Sensitive (Varistor) |
| 87064 | Resistor Network, 16-Pin, Surface Mount |
| 87078 | Resistor, Hot Molded, Panel, Variable, Dual Section (Cancellation Notice) |
| 88039 | Resistor, Variable, Non-Wirewound, Trimmer, 4mm, J-Hook Leads |
| 88057 | Resistor, Voltage Sensitive (Varistor) |
| 88063 | Resistor, Voltage Sensitive (Varistor), Base Mount Metal-Oxide |
| 89005 | Resistor, Variable, 25 Turn, Multiple Tap |
| 89097 | Resistor Network, Fixed, Film, 3 Pin |
| 90027 | Resistor, Variable, Nonwirewound, Trimmer, 4mm, Gull-Wing Leads |
| 90039 | Thermistor, Thermally Sensitive Resistor |
| 90065 | Resistor, Voltage Sensitive (Varistor) |
| 90096 | Resistor, Voltage Sensitive, Chip, Metal Oxide Varistor |



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TABLE C-X. Cancelled drawings. Continued

| Drawing number | Title |
|---|---|
| 91017 | Resistor, Variable, Nonwirewound, Trimmer 1/4 Watt |
| 90047 | Resistor, Chip, Fixed, Zero-Ohm, Style RM2208 |
| 90048 | Resistor, Chip, Fixed, Film, Zero-Ohm, Style 0705 |
| | |
| | |
| 93078 | Resistor, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 2110 |
| 93079 | Resistor, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 1505 |
| 94047 | Resistor, Fixed, Film, Chip, (MELF), 2 Watt, Style 3610 |
| 94048 | Resistor, Fixed, Film, Chip (MELF), 1/2 Watt, Style 2010 |
| 99007 | Resistor, Fixed, Film, Chip, 16 Pin Array |
| | |
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| | |

* DLA Land and Maritime Drawing



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APPENDIX D

INACTIVE FOR NEW DESIGN DoD SPECIFICATIONS,
COMMERCIAL ITEM DESCRIPTIONS (CIDs) and DLA LAND AND MARITIME DRAWINGS

D.1 SCOPE

D.1.1 Scope. The following appendix includes all inactive for new design DoD Specifications, Commercial Item Descriptions (CIDs) and DLA LAND AND MARITIME Drawings. These specifications are used for replacement purposes only.

D.2 APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

D.3 INACTIVE DoD SPECIFICATIONS, COMMERCIAL ITEM DESCRIPTIONS (CIDs) AND DLA LAND AND MARITIME DRAWINGS.

D.3.1 INACTIVE DoD SPECIFICATIONS.

D.3.1.1 Inactive DoD Specifications. Table D-I contains a listing of Inactive for new design DoD specifications.

TABLE D-I. Inactive DoD specifications.

| Specification number | Title |
|---|---|
| MIL-PRF-19 | Resistor, Variable, Wirewound, Low Operating Temperature |
| MIL-PRF-19/2 | Resistor, Variable, Wirewound (Low Operating Temperature), Style RA20 |
| MIL-PRF-19/3 | Resistor, Variable, Wirewound (Low Operating Temperature), Style RA30 |
| | |
| MIL-PRF-26/2 | Resistor, Fixed, Wirewound (Power Type), Styles RW20, RW21, RW22, RW23, and RW24 |
| MIL-PRF-26/4 1/ | Resistor, Fixed, Wirewound (Power Type), Styles RW55, RW56, RW67, RW68, and RW69 |
| MIL-PRF-26/5 | Resistor, Fixed, Wirewound (Power Type), Styles RW70, RW74, RW78, and RW79 |
| MIL-PRF-26/6 | Resistor, Fixed, Wirewound (Power Type), Styles RW80 and RW81 |
| | |
| MIL-PRF-93 | Resistor, Fixed, Wirewound (Accurate), General Specification for |
| MIL- PRF -93/10 | Resistor, Fixed, Wirewound (Accurate), Style RB52 |
| MIL- PRF -93/11 | Resistor, Fixed, Wirewound (Accurate), Style RB53 |
| MIL- PRF -93/12 | Resistor, Fixed, Wirewound (Accurate), Style RB54 |
| MIL- PRF -93/13 | Resistor, Fixed, Wirewound (Accurate), Style RB55 |
| MIL- PRF -93/14 | Resistor, Fixed, Wirewound (Accurate), Style RB56 |
| MIL- PRF -93/19 | Resistor, Fixed, Wirewound (Accurate), Style RB70 |
| MIL- PRF -93/20 | Resistor, Fixed, Wirewound (Accurate), Style RB71 |
| | |
| MIL-DTL-6749 | Resistor, Variable, Wirewound, Nonprecision, Aircraft Power |
| | |
| MIL-R-10509 | Resistor, Fixed, Film (High Stability), General Specification For |
| MIL-R-10509/1 | Resistor, Fixed, Film (High Stability), Style RN60 |
| MIL-R-10509/2 | Resistor, Fixed, Film (High Stability), Style RN65 |
| MIL-R-10509/3 | Resistor, Fixed, Film (High Stability), Style RN70 |
| MIL-R-10509/7 | Resistor, Fixed, Film (High Stability), Style RN55 |
| MIL-R-10509/8 | Resistor, Fixed, Film (High Stability), Style RN50 |
| | |
| MIL-R-18546/1 | Resistor, Fixed, Wirewound (Power Type, Chassis Mounted), Styles RE60, RE65, RE70, and RE75 |
| | |
| MIL-R-19365 | Resistor, Adjustable, Wirewound, Power |
| | |
| MIL-PRF-22097/1 | Resistor, Variable, Nonwirewound (Adjustment Type, Lead-Screw Actuated), Style RJ11 |
| MIL-PRF-22097/3 | Resistor, Variable, Nonwirewound (Adjustment Type, Lead-Screw Actuated), Style RJ22 |
| MIL-PRF-22097/4 | |
| MIL-PRF-22097/5 | Resistor, Variable, Nonwirewound (Adjustment Type, Lead-Screw Actuated), Style RJ26 |
| MIL-PRF-22097/6 | Resistor, Variable, Nonwirewound (Adjustment Type, Single Turn), Style RJ50 |

1/ MIL-PRF-26/4 – Only styles RW55, RW67, RW68, and RW69 are “Inactive for New Design.”



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TABLE D-I. Inactive DoD specifications. - Continued

| Specification number | Title |
|---|---|
| MIL-PRF-22684/1 | Resistor, Fixed, Film, Insulated, Style RL07 |
| MIL-PRF-22684/2 | Resistor, Fixed, Film, Insulated, Style RL20 |
| MIL-PRF-22684/3 | Resistor, Fixed, Film, Insulated, Style RL32 |
| MIL-PRF-22684/4 | Resistor, Fixed, Film, Insulated, Style RL42 |
| | |
| MIL-PRF-23648/2 | Resistor, Thermal (Thermistor), Insulated, Negative Temperature Coefficient, Style RTH08 |
| MIL-PRF-23648/3 | Resistor, Thermal (Thermistor), Insulated, Negative Temperature Coefficient, Style RTH10 |
| | |
| MIL-PRF-27208/4 | Resistor, Variable, Wirewound (Adjustment Type, Lead- Screw Actuated) Style RT22 |
| MIL-PRF-27208/9 | |
| | |
| MIL-PRF-32192/2 | Resistor, Chip, Thermal (Thermistor), Insulated Positive Temperature Coefficient Style RCTP0805 |
| | |
| MIL-PRF-39005/7 | Resistor, Fixed, Wirewound (Accurate), Nonestablished Reliability, and Established Reliability, Style RBR57 |
| MIL-PRF-39005/8 | Resistor, Fixed, Wirewound (Accurate), Nonestablished Reliability, and Established Reliability, Style RBR74 |
| MIL-PRF-39005/9 | Resistor, Fixed, Wirewound (Accurate), Nonestablished Reliability, and Established Reliability, Style RBR75 |
| | |
| MIL-PRF-39007/5 | Resistor, Fixed, Wirewound (Power Type), Nonestablished Reliability, Established Reliability, and Space Level, Style RWR71 |
| MIL-PRF-39007/6 | Resistor, Fixed, Wire-Wound (Power Type), Nonestablished Reliability, Established Reliability, and Space Level, Style RWR74 |
| | |
| MIL-PRF-39015/1 | Resistor, Variable, Wirewound (Lead Screw Actuated), Nonestablished Reliability and Established Reliability, Style RTR12 |
| | |
| MIL-PRF-39035/1 | Resistor, Variable, Nonwirewound (Adjustment Type, Lead-Screw Actuated), Nonestablished Reliability, and Established Reliability Style RJR12 |
| | |
| MIL-PRF-49465/6 | Resistor, Fixed, Metal Element, Power Type, Style RLV30 |
| MIL-PRF-49465/7 | Resistor, Fixed, Metal Element, Power Type, Style RLV31 |
| | |
| MIL-PRF-55182/2 | Resistor, Fixed, Film, Nonestablished Reliability, Established Reliability, and Space Level, Style RNR57 |
| MIL-PRF-55182/11 | Resistor, Fixed, Film, Established Reliability Style RNR51 |
| MIL-PRF-55182/12 | Resistor, Fixed, Film, Established Reliability Style RNR56 |
| | |
| MIL-PRF-83530 | |
| MIL-PRF-83530/1 | Resistors, Voltage Sensitive (Varistors) Style RVS10 |
| | |
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D.3.2 INACTIVE COMMERCIAL ITEM DESCRIPTIONS (CIDs)

D.3.2.1 Inactive Commercial Item Descriptions (CIDs). Table D-II contains a listing of Inactive Commercial Item Descriptions (CIDs).

TABLE D-II. Inactive Commercial Item Descriptions (CIDs)

| Drawing Number | Title |
|---|---|
| A-A-55534/2 | Resistor, Fixed, Wirewound or Metal Element, Power Type, Style VLV1206 |
| A-A-55534/3 | Resistor, Fixed, Wirewound or Metal Element, Power Type, Style VLV3 |
| A-A-55534/4 | Resistor, Fixed, Wirewound or Metal Element, Power Type, Style VLV5 |
| A-A-55534/7 | Resistor, Fixed, Wirewound or Metal Element, Power Type, Style VLV2010 |
| A-A-55534/8 | Resistor, Fixed, Wirewound or Metal Element, Power Type, Style VLV2512 |
| A-A-55534/9 | Resistor, Fixed, Wirewound or Metal Element, Power Type, Style VLV2 |
| A-A-55564 | Resistor, Voltage Sensitive (Varistor), General Specification for |
| A-A-55564/1 | Resistor, Voltage Sensitive (Varistor), Base Mount |
| A-A-55564/2 | Resistor, Voltage Sensitive (Varistor), Metal Oxide, High Energy |
| A-A-55564/3 | Resistor, Voltage Sensitive (Varistor), Metal Oxide, Radial Lead |
| | |
| | |
| | |

D.3.3 INACTIVE DLA LAND AND MARITIME DRAWINGS

D.3.3.1 Inactive DLA Land and Maritime Drawings. Table D-II contains a listing of Inactive DLA Land and Maritime Drawings with supersession data.

TABLE D-III. Inactive DLA LAND AND MARITIME Drawings.

| Drawing Number | Title | Superseding document |
|---|---|---|
| 03002 | Resistor, Chip, Fixed, Film, Zero-Ohm, Style 0505 | MIL-PRF-32159/2 |
| 03012 | Resistor, Chip, Fixed, Film, Zero-Ohm, Style 0302 | MIL-PRF-32159/13 |
| 03013 | Resistor, Chip, Fixed, Film, Zero-Ohm, Style 0603 | MIL-PRF-32159/12 |
| 03014 | Resistor, Chip, Fixed, Film, Zero-Ohm, Style 0402 | MIL-PRF-32159/11 |
| 03015 | Resistor, Chip, Fixed, Film, Zero-Ohm, Style 2010 | MIL-PRF-32159/8 |
| 03016 | Resistor, Chip, Fixed, Film, Zero-Ohm, Style 2512 | MIL-PRF-32159/9 |
| 03017 | Resistor, Thermal, (Thermistor), Die Chip, Positive Temperature Coefficient (PTC) | MIL-PRF-32159/1 |
| 03018 | Resistor, Thermal, (Thermistor), Die Chip, Negative Temperature Coefficient (NTC), Style 0404 | MIL-PRF-32159/3 |
| 87011 | Resistor, Chip, Fixed, Zero-Ohm, Style 1010 | MIL-PRF-32159/10 |
| 88032 | Resistor, Chip, Fixed, Film, Zero-Ohm, Style RM0502 | MIL-PRF-32159/1 |
| 88033 | Resistor, Chip, Fixed, Film, Style 1010 | MIL-PRF-32159/10 |
| 90047 | Resistor, Chip, Fixed, Zero-Ohm, Style RM2208 | MIL-PRF-32159/5 |
| 90048 | Resistor, Chip, Fixed, Film, Zero-Ohm, Style 0705 | MIL-PRF-32159/6 |
| 90049 | Resistor, Chip, Fixed, Film, Zero-Ohm, Style 1005 | MIL-PRF-32159/3 |
| 90092 | Resistor, Chip, Fixed, Film, Zero-Ohm, Style RM1505 | MIL-PRF-32159/4 |
| 93073 | Resistor, Chip, Thermally Sensitive, (Thermistor), NTC, Style 0805 | MIL-PRF-32159/4 |
| 93074 | Resistor, Chip, Thermally Sensitive, (Thermistor), NTC, Style 1206 | MIL-PRF-32159/5 |

* DLA Land and Maritime Drawing



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TABLE D-III. Inactive DLA LAND AND MARITIME Drawings. Continued

| Drawing Number | Title | Superseding document |
|---|---|---|
| 93078 | Resistor, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 2110 | 06001* |
| 93079 | Resistor, Chip, Fixed, Bulk Metal Foil, Ultra Precision, Style 1505 | 03010* |
| 94011 | Resistor, Chip, Fixed, Film, Zero-Ohm, Style RM1206 | MIL-PRF-32159/7 |
| 95013 | Resistor, Chip, Fixed, Film, Style 0302 | MIL-PRF-32159/13 |
| 95014 | Resistor, Chip, Fixed, Film, Style 0402 | MIL-PRF-32159/11 |
| 97008 | Resistor, Chip, Fixed, 1/4 Watt, MELF | 05009* |
| 97009 | Resistor, Fixed, Film, Precision, 0.3 Watt, Power Curve A | 89039* |
| 97010 | Resistor, Fixed, Film, Precision, 0.3 Watt, Power Curve B | 89039* |
| 97011 | Resistor, Fixed, Film, Precision, 0.15 Watt, Power Curve B | 89039* |
| 98018 | Resistor, Chip, Thermally Sensitive (Thermistor), Positive Temperature Coefficient (PTC), Style 0805 | MIL-PRF-32192/2 |
| 99001 | Resistor, Fixed, Ceramic Composition (Insulated) | |
| 99007 | Resistor, Chip, Fixed, Film, 16 Pin Array | No replacement |

* DLA Land and Maritime Drawing



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# APPENDIX E

## REPLACEMENT PARTS FOR MIL-R-11 and MIL-R-39008

### E.1 SCOPE

E.1.1 Scope. The military specifications MIL-R-11 (Resistor, Fixed, Composition, Insulated) and MIL-R-39008 (Resistor, Fixed, Composition, Insulated) Established Reliability, General Specification For)have been cancelled. There are no direct replacements available. The following suggested substitutes do not meet all of the requirements of MIL-R-11 and MIL-R-39008.

### E.2 APPLICABLE DOCUMENTS. This section is not applicable to this section.

### E.3 SUGGESTED REPLACEMENTS

E.3.1 Replacement for general applications. The following specifications are considered acceptable substitutes for general applications of the MIL-R-39008 and MIL-R-11.

Department of Defense Specifications:

| | | |
|---|---|---|
| MIL-PRF-22684/8 | - | Resistor, Fixed, Film, Insulated, Style RL42….TX. |
| MIL-PRF-39017/1 | - | Resistor, Fixed, Film, Insulated, Nonestablished Reliability, and Established Reliability, Style RLR07. |
| MIL-PRF-39017/2 | - | Resistor, Fixed, Film, Insulated, Nonestablished Reliability, and Established Reliability, Style RLR20. |
| MIL-PRF-39017/3 | - | Resistor, Fixed, Film, Insulated, Nonestablished Reliability, and Established Reliability, Style RLR32. |
| MIL-PRF-39017/5 | - | Resistor, Fixed, Film, Insulated, Nonestablished Reliability, and Established Reliability, Style RLR05. |

DLA LAND AND MARITIME drawings:

| | | |
|---|---|---|
| 97004 | - | Resistor, Fixed, Film, Insulated, 2 Watt |
| 98020 | - | Resistor, Fixed, Film, Insulated, 1/8 Watt |
| 98021 | - | Resistor, Fixed, Film, Insulated, 1/2 Watt |
| 98022 | - | Resistor, Fixed, Film, Insulated, 1 Watt |
| 99011 | - | Resistor, Fixed, Film, Insulated, 0.25 Watt |

Designers are CAUTIONED on using these resistors in pulse applications. See E.4.1

E.3.2 Replacement for pulse applications. The following specifications are considered acceptable substitutes for pulse applications for MIL-R-11 and MIL-R-39008.

Designers are CAUTIONED on using these resistors in high power pulse applications. Since they have not been qualified nor tested for such applications, damage and premature failure are possible. These resistors only see a one-time pulse when Short-time overload testing is performed.



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# APPENDIX E

## REPLACEMENT PARTS FOR MIL-R-39008 AND MIL-R-11

Designers MAY CONSIDER using the following DLA Land and Maritime drawings for high power pulse applications.

DLA LAND AND MARITIME drawings:

| | | |
|---|---|---|
| 03003 | - | Resistor, Fixed, Carbon Film, High Pulse Voltage, 1/8 Watt |
| 03004 | - | Resistor, Fixed, Carbon Film, High Pulse Voltage, 1/4 Watt |
| 03005 | - | Resistor, Fixed, Carbon Film, High Pulse Voltage, 1/2 Watt |
| 03006 | - | Resistor, Fixed, Carbon Film, High Pulse Voltage, 1 Watt |
| 03007 | - | Resistor, Fixed, Carbon Film, High Pulse Voltage, 3/4 Watt |
| 03008 | - | Resistor, Fixed, Carbon Film, High Pulse Voltage, 2 Watt |
| 03009 | - | Resistor, Fixed, Carbon Film, High Pulse Voltage, 3 Watt |

NOTE: These alternative resistors do not have the geometry (form, fit) of the MIL-PRF-39017 resistors, nor are they subject to the same qualification/verification and periodic Group C inspection requirements as the RLR style resistors. Additionally, Group B for the DLA Land and Maritime drawings parts may be satisfied by providing generic data.

### E.4 ADDITIONAL INFORMATION

E.4.1 Fixed film resistors. Designers are CAUTIONED on using these resistors in pulse applications. These resistors only see a one-time pulse when Short-time overload testing is performed. These resistors are subject to damage and premature failures when they see excessive pulsing.



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Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Review activities:
Army - AR, AT, AV, CR4, MI
Navy - AS, CG, MC, OS, SH
Air Force - 19

Preparing activity:
DLA - CC

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NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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